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Understanding The Virtue of Complexity



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Abstract

Recent papers have challenged certain aspects of the “virtue of complexity” described by [Kelly et al. \(2024b\)](#) (KMZ) and related work. These challenges ultimately have little bearing on the theoretical arguments or empirical findings of KMZ. They do, however, provide a valuable opportunity to better understand the nuanced behavior of complex models. In addition to responding to recent challenges, we provide detailed discussions of how complex models learn in small samples, the roles of “nominal” and “effective” complexity, the unique effects of implicit regularization, and the importance of limits to learning. We then present new empirical and theoretical analyses that expand on KMZ. Finally, we introduce and demonstrate the virtue of ensemble complexity.

Keywords: Portfolio choice, machine learning, random matrix theory, benign overfit, overparameterization

JEL: C3, C58, C61, G11, G12, G14

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1 Introduction

“It is an empirical fact that more is better in modern machine learning. State-of-the-art models are commonly trained with as many parameters and for as many iterations as compute budgets allow, often with little regularization. This ethos of enormous, underregularized models contrasts sharply with the received wisdom of classical statistics.” (Simon et al., 2024)

The above quote is based on findings in the natural and applied sciences. Evidence is now accumulating that the same is true in finance, economics, and other social sciences.¹

Kelly et al. (2024b) (henceforth KMZ) and its extensions² analyze, both theoretically and empirically, the role of heavily parameterized models for describing asset returns. Their conclusion is simple: Unusually large asset pricing models are surprisingly successful out-of-sample.

In the wake of its publication, some authors have challenged the findings of KMZ. A common pattern in these papers is misapprehension of (admittedly complicated) properties of complex statistical models. By articulating their concerns, these papers provide a valuable opportunity to expand our understanding of complexity in finance. We use this opportunity to shed further light on the virtue of complexity (or “VoC”). Our contribution takes three forms. First, we provide an expanded discussion of the intuition behind the subtle forces at play in KMZ and related papers. Second, we extend the empirical analysis of KMZ to illustrate how genuine learning occurs in complex models and further demonstrate the robustness of KMZ’s finding. Third, we generalize the theory of KMZ to more thoroughly understand the conditions that give rise to virtue of complexity.

The paper is organized as follows. Section 2 provides a conceptual overview of what complex model do and do not accomplish. First, the virtue of complexity is not a statement

¹See Athey (2018), Grimmer et al. (2021), and Kelly and Xiu (2023), among many others.

²Didisheim et al. (2024), Kelly et al. (2023), and Kelly et al. (2025a).

that complex models recover the true model. Complex models face *limits to learning* that impair their ability to recover the truth.³ The world of complex models is fundamentally different than the traditional econometric setting that empiricists often assume. When using a parsimonious model (and assuming that it is correctly specified), we rest in the comfort that reasonably large sample sizes ensure the estimated model hones in on the truth with reasonable precision. In machine learning, we acknowledge the reality that we do not know the true model, so correct specification is off the table. As an alternative we turn to non-parametric approximation. Non-parametric models face a tension: The more parameters in the non-parametric model, the better its capacity to approximate the truth. But with more parameters, the more thinly the data is spread to estimate those parameters, the less precise the estimates. Traditional econometric theory (built on laws of large numbers and central limit theorems) breaks down when the parameters outnumber the data. For this reason, the virtue of complexity is not a claim that complex models can recover the true model. Instead, it is a statement that large approximating models get closer to the truth than small approximating models. In other words, the benefits of large approximation capacity more than make up for the costs of estimating so many parameters.

We emphasize the importance of apples-to-apples comparisons. We take a view that *all* empirical models are misspecified approximating models. An econometrician may specify a parsimonious model based on first principles (i.e., theory, prior, or intuition) but we work under the assumption that this results in a misspecified model. It is impossible to know how badly misspecified a human-designed model is. The advantage of KMZ’s empirical approach is that it allows us to compare models solely on their complexity while holding the inputs to all models fixed. In particular, KMZ’s smaller random feature models are nested in their larger models in order to isolate the key statistical tradeoffs. Larger models always face a more onerous learning curve because they have to estimate more parameters. But because the random feature models are nested, we know that larger model have smaller

³See [Didisheim et al. \(2024\)](#) and [Chen et al. \(2025a\)](#).

specification error. Because of this, KMZ can show their main point about the virtue of complexity: The benefits of better approximation dominate the costs of estimating more parameters. Comparing non-nested models (e.g., linear versus nonlinear) ruins our ability to rank models’ specification error. For this reason, one cannot “test” the virtue of complexity by comparing a simple linear model with say a complex nonlinear model.

Section 2 also explains the role that sample size plays in the virtue of complexity. KMZ’s theory of complex model behavior is asymptotic in nature, building on random matrix limit theory when $P, T \rightarrow \infty$ and $P/T \rightarrow c > 0$. The condition $c > 0$ means that, at its core, it is a theory of statistical learning in data scarce conditions. Just like traditional econometric theory (with P fixed so $c \rightarrow 0$) relies on large T asymptotics to describe the behavior of estimators, we would never say that a t -statistic is only applicable in the large T limit. The entire motivation for asymptotic theory is to characterize estimators in finite samples. The same logic applies to the theory of complex models.

We demonstrate that the limiting results of KMZ accurately characterize complex model behavior when T is finite and often even when T is very small. We also show that complex models display genuine nonlinear learning in surprisingly small training samples. This is an exciting finding because small T models with high complexity are not just an intellectual curiosity. Instead they are the norm in time series machine learning models for finance and economics, as economic time series suffer from rather extreme data scarcity. This is compounded by the vexing problem of structural change in economic data. We believe that the ability of complex models to function well in small samples will make them indispensable for the time series toolkit moving forward.

Third, we resolve confusion around the term “complexity.” At the heart of this confusion is the simultaneous use of large parameters counts *along with* parameter shrinkage in machine learning models. Does shrinkage undermine the virtue of complexity? Should we think of complexity in terms of parameter counts or should it be adjusted for shrinkage? To

answer this, we refine the discussion in terms of “nominal complexity,” which is the KMZ definition based on raw parameter count, and “effective complexity” which accounts for model bias (for example through shrinkage). We show that the virtue of complexity holds even after accounting for shrinkage, bias, or any other notion of effective parameterization. We demonstrate this by comparing models with the same effective complexity and showing that, among these, the models with the largest nominal complexity tend to dominate. Regardless, KMZ is not interested in characterizing complexity in a semantic sense. The sole objective of KMZ is to understand the determinants of out-of-sample model performance. Whatever its name, the theory of KMZ clearly demonstrates that the ratio $c = P/T$ is central for understanding the behavior of machine learning models.

Challenges of KMZ have focused on narrow aspects of their empirics such as very short training windows or the use of random Fourier feature (RFF) models. These challenges miss the point that KMZ’s empirical analysis is a deliberately simple proof-of-concept for their main contribution, which is theoretical. KMZ *do not* advocate RFF as their preferred machine learning model for return prediction. Their claim *is not* that machine learning works best with small samples. Their point is plainly that unusually large asset pricing models are surprisingly successful out-of-sample. KMZ’s experiments are designed to convey this point clearly and concisely.

In Section 3, we extend the empirical analysis of KMZ to establish the following facts.⁴ The KMZ predictive results are not spurious or mechanical effects of small samples and persistent predictors, as claimed by Nagel (2025). First, KMZ themselves show that the same VoC patterns emerge with large samples. And when we purge return predictors of their time series persistence, the same VoC patterns again emerge. We then perform placebo tests which, according to Nagel (2025), should generate a spurious VoC. The placebo results produce no such spurious behavior.

⁴In addition to the analysis in Section 3, Kelly et al. (2023) provide an expansive analysis of complex prediction models for a wide variety of asset returns, including equity factors, international equities indices, currencies, commodities, and fixed income.

The quest to understand complexity was born from the real-world success of deep neural networks. Only after seeing that larger networks were more successful in application after application did statisticians begin using random feature regression as a toy model for deep learning. To reiterate, we (and KMZ) are not interested in RFF per se. We are interested in the viability of complexity in financial modeling. To this end we show that, like RFF, deep neural networks deliver powerful out-of-sample performance. Furthermore, VoC again emerges as we gradually increase the deep network’s size.⁵

We then investigate the claim that KMZ’s empirical findings are driven by a mechanical momentum effect. Recalling that KMZ already perform an extensive comparison between their complex model and momentum, we analyze and decompose the various construction choices made by Nagel in his preferred momentum benchmark. Our new tests clearly rule out any mechanical momentum effect as the driver of KMZ’s empirical results. Instead, Nagel’s derivation of an interpretable predictor from the KMZ model is a useful demonstration of how machine learning can be leveraged to infer more tractable economic representations. His effort is also related to so-called “distillation” of large models into more compact models with ideally little loss in performance ([Hinton et al., 2015](#)).

Our last empirical contribution in Section 3 is to describe and document a new dimension of VoC. There are many varieties of complex models for return prediction. We show that the virtue of complexity emerges among models with different architectures, objective functions, data inputs, and so on. The different complex models we study all have large capacity for approximating unknown prediction functions, yet we find that these models produce rather different and complementary trading strategies. The mean-variance efficient portfolio of complex strategies puts weight on strategies derived from many different complex models. In other words, even individual complex models are not complex enough—they are dominated by an even larger model that is a combination of multiple complex models. We refer to this

⁵The same point is made with transformer asset pricing models by [Kelly et al. \(2025a\)](#).

form of complexity-on-top-of-complexity as the “virtue of ensemble complexity.” We then rationalize how this phenomenon emerges.

Sections 2 and 3 address the main comments of Nagel (2025). In Section 4 we discuss some of Nagel’s peripheral points and also address comments raised by Buncic (2025), Cartea et al. (2025), Fallahgoul (2025), and Berk (2023).

In section 5 we extend the theory of KMZ to more general data environments. We show that the shape of “VoC curves” that relate a model’s out-of-sample performance to its complexity depends on two key properties of the data generating process (DGP), which we refer to as “concentration” and “alignment.” Concentration refers to the distribution of eigenvalues among the predictor covariance matrix. A concentrated eigenvalue distribution has a few dominant components indicating substantial redundancy among predictors, but when covariance is disperse among many eigenvalues it means that many predictors contribute differentiated information to the predictor set. The less concentrated the eigenvalue distribution, the more model performance increases with complexity. Alignment refers to the dependence between predictor betas and eigenvalues. If predictors associated with large eigenvalues tend to have the largest betas, then the generating process has a high degree of alignment. If alignment is low (or perhaps negative), then even minor eigenvalues are important for prediction. The less alignment the more prominent the virtue of complexity. The upward sloping Sharpe ratio curves in KMZ’s empirics (and also in Kelly et al., 2023; Didisheim et al., 2024) are consistent with low concentration and low alignment in return prediction data sets.

At last, Section 6 concludes by outlining what we see as important directions for future work on complex financial models.

2 Can Complex Models Learn From Small Samples?

Nagel’s (2025) first central claim asserts that it is implausible for complex models to learn

from small numbers of observations, therefore the success of complex models in predicting returns in small samples must be spurious. In his words:

“The effective complexity of this kernel ridgeless regression is low because it is capped by the number of observations in the training window. In the case $T = 12$, the effective number of parameters is 12, orders of magnitude lower than the number of RFF when P is in the thousands. Predictive success in the large- P , small- T case therefore cannot be a consequence of virtue of complexity.”

This claim boils down to an assertion. Nagel (2025) provides no theoretical basis for this claim and instead attempts to support his claim with empirical analysis. In Section 3 we argue that his empirical analysis does not substantiate his claim. In this section, we provide theoretical arguments along with a concrete example to show that complex models indeed “learn” predictive relationships even when T is small.

2.1 “Nominal” Complexity Versus “Effective” Complexity

Nagel’s claim has two logical elements: (1) Effective model complexity is constrained by sample size, and (2) There can be no meaningful learning achieved by models with low effective complexity. Putting these two together, Nagel concludes that the small- T results of KMZ must be spurious. We show that part (2) of this claim is incorrect and this can be seen immediately from the theoretical analysis of KMZ. We have no issue with part (1), but we also show that this point has no bearing on the results of KMZ.

The first step toward resolution is to avoid conflating different notions of complexity. In KMZ, there is a single definition of complexity. It is the ratio of estimable parameters of an empirical model to the number of observations used to train that model, $c = P/T$. One may dislike the *name* that KMZ use for this ratio, but its definition is precise.⁶ Disagreements

⁶Note that this definition of complexity comes from the ML literature. A few prominent examples: Hastie et al. (2009), p.37, define complexity as “the number of basis functions” (P in our setting), while Hastie et al. (2022) refer to c as the “complexity of the feature space,” and Belkin et al. (2019) refer to P as “model complexity.”

about whether this ratio is a good descriptor of model complexity is a separate and wholly semantic issue. KMZ is not interested in characterizing the complexity of a model in a semantic sense. **The sole objective of KMZ is to understand the determinants of out-of-sample model performance.** Whatever its name, KMZ show that the ratio $c = P/T$ is critical for understanding out-of-sample model performance. For clarity, we will refer to the KMZ definition as “nominal complexity” where necessary.

Nagel introduces the term “effective complexity” though never formally defines it. He references a measure of effective number of parameters (Williams and Rasmussen, 2006). There are many ways to measure effective number of parameters (or, as sometimes studied, effective degrees of freedom) in machine learning models.⁷ More importantly, neither does Nagel provide an explanation for why effective complexity matters for understanding model behavior. The failure to understand the impact of effective complexity on model behavior gives rise to the flaw in part (2) of his claim, and ultimately to Nagel’s erroneous conclusion about the virtue of complexity.

Our discussion here does not rely on any particular notion of effective parameter count, so we will generally refer to “effective complexity” (which we denote ec) as the ratio of effective number of parameters (however measured) to the number training observations. In Nagel’s preferred definition and in the special case of ridgeless kernel regression, the effective parameter count is T , in which case effective complexity is fixed at exactly one ($ec = 1$) for any KMZ model with nominal complexity $c \geq 1$. This is a key point. We show that there can be many models with the same effective complexity yet that have very different out-of-sample behaviors. Then, among those models with the same ec , differences in their out-of-sample performance are well explained by differences in their nominal complexity.

⁷See Abbas et al. (2021), Rosset and Tibshirani (2020), and Patil et al. (2024), for example. Statistical learning theory provides several formal ways to quantify model complexity, such as the Vapnik–Chervonenkis dimension, Rademacher complexity, covering numbers (metric entropy), and related measures like Gaussian complexity. Although it is believed that the semantic meaning of statistical complexity should involve some sort of parameter norm, parameter count still plays a key role, and it does not seem possible to get rid of it (see, e.g., Bartlett et al., 2017).

It turns out that the theory of KMZ already captures the impact of both nominal complexity and effective complexity on model behavior.⁸ While the mathematical background of KMZ may be challenging, the intuition can be illustrated clearly in a simple example with ordinary least squares regression. Consider a small sample of $T = 10$ observations from a data generating process in which the dependent variable Y_t is driven by 100 iid predictors X_t :

$$Y_t = \beta' X_t + \epsilon_t, \quad t = 1, \dots, T. \quad (1)$$

Imagine that the form of the DGP is known to the researcher, but that she knows nothing at all about β (i.e., she has an uninformative prior).⁹ The objective is to make the best possible out-of-sample predictions for Y given X . What degree of model complexity should the researcher use, and what is the model’s ultimate performance?

We focus on linear models and ridgeless regression for concreteness.¹⁰ We explore how empirical model behavior changes as we vary its nominal complexity from as few as one parameter ($c = 1/10$) to as many as the full 100 parameters in the true model ($c = 10$). Following KMZ, we analyze how model complexity impacts the behavior of key model attributes: The norm of the estimated predictive coefficient and the out-of-sample expected return, volatility, and Sharpe ratio of the market timing strategy. These quantities tell us everything we need to know to understand model behavior (including effective complexity).

The expected return plot in the upper right of Figure 1 describes the model’s bias in simulated data.¹¹ The smallest model ($P = 1$) is badly biased—this specification is

⁸Though KMZ use different terminology, as we make clear below.

⁹This is the exact setting of KMZ, with the exception that in this example we show behaviors even in finite dimensions, without relying on large P and large T RMT as in KMZ.

¹⁰Our main point can be illustrated in more complicated settings using e.g. neural networks and other shrinkage schemes.

¹¹Our simulation of monthly returns in equation (1) fixes annualized residual return variance at 20%. Predictors X_t are drawn as iid standard normal variables and β is drawn as iid normal with variance that fixes the population R^2 at 0.2. We repeat our $T = 10$ and $P = 1, \dots, 100$ regression analysis for ten million simulations and report average out-of-sample performance across simulations.

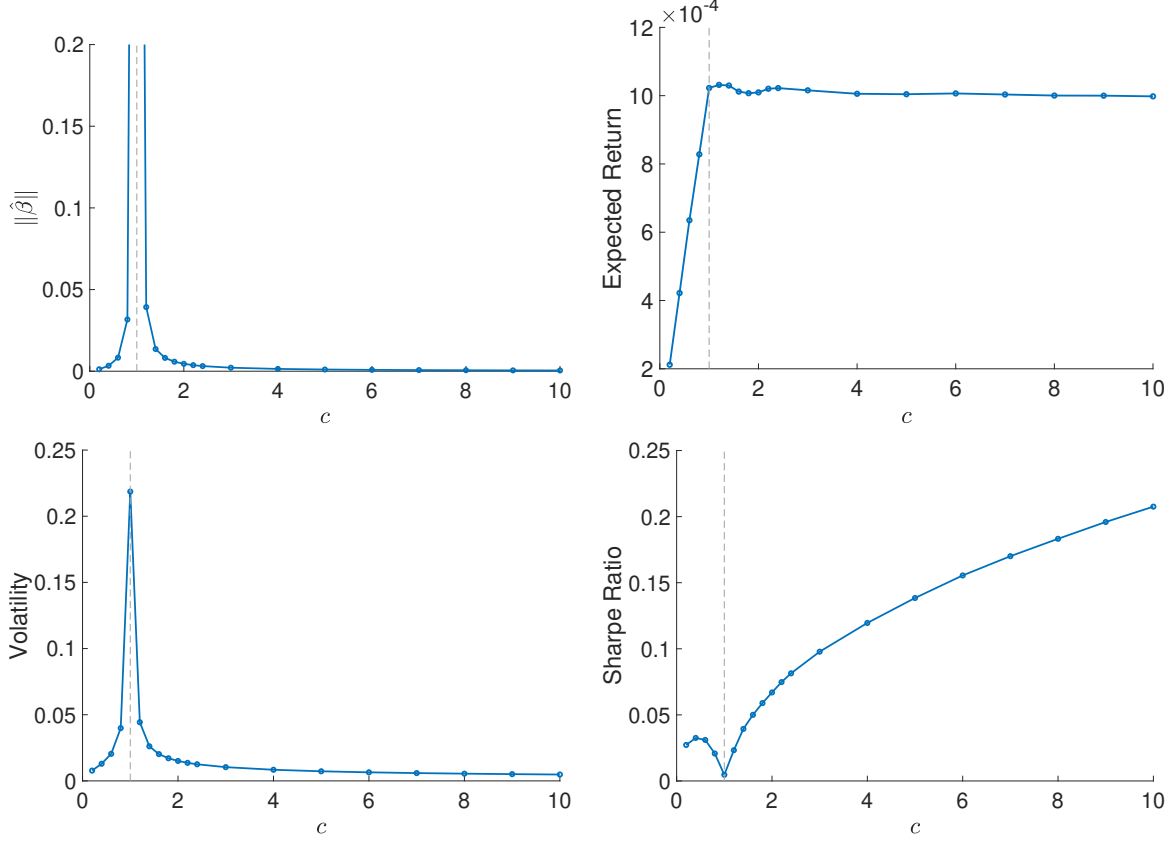


Figure 1: Illustrating Virtue of Complexity in a Simulation

Note. The figure reports simulated out-of-sample behavior of linear regression models with varying degrees of complexity c (shown on the x-axis). Data are simulated from (1) with a sample size of $T = 10$ and the models with $P = 1, \dots, 100$ regressors are trained using ridgeless regression. The top left panel shows the norm of the beta coefficient estimate, the top right shows the out-of-sample mean return of the market timing strategy, the bottom left shows the strategy’s volatility, and the bottom right shows its out-of-sample Sharpe ratio.

constrained to use too few parameters, so its expected return is close to zero and far from the expected return of the DGP.¹² Adding predictors gradually reduces bias (improves the model’s expected return) until we reach $P = T$, at which point the bias-reducing effect of adding predictors abruptly stops. At this point, the model is saturated—all models with $P \geq T$ exactly fit (or “interpolate”) the training data, and there is no further bias

¹²The infeasible return of the true timing strategy is 0.01 in our simulated model. The infeasible Sharpe ratio is 1.4, versus 0.2 for the feasible timing strategy with high complexity.

reduction from additional parameters beyond $P = T$.¹³ This suggests one potential way to conceptualize “effective parameterization”: It is the saturation point, i.e. the minimum number of parameters necessary to minimize bias. In the current linear example, the saturation point (or effective number of parameters) is $\min(P, T)$, which is essentially the notion of effective parameterization referenced by Nagel.¹⁴ In other words, effective complexity be understood in terms of bias, and in the KMZ theory effective complexity is pinned down by the expected return. In this example, there is “virtue of complexity”—Sharpe ratio increases with P even when $P > T$. But effective complexity is fixed for $P > T$. So where does the virtue come from?

Bias is an incomplete descriptor of model behavior, so if effective complexity is intrinsically related to bias then it too is only a partial description of model behavior. So next consider the model’s out-of-sample volatility as shown in Figure 1. As we begin adding parameters, volatility rises. This is because scarce data is spread more and more thinly to estimate the growing number of parameters. Eventually we hit $P = T$, where we learn one parameter value from each data point, and we interpolate the training data. Volatility is enormous at this point, and it seems that it will only get worse if consider even larger models with $P > T$. But volatility surprisingly retreats as P grows beyond T . The reason for this is *implicit regularization*. When $P \leq T$, the least squares solution is unique. Once P exceeds T , there are infinite least squares solutions that all interpolate the training data. The ridgeless estimator selects one of these solutions—the β that exactly fits the training data with the smallest ℓ_2 norm (i.e., with the smallest variance). There are other least squares interpolators

¹³To understand the flattening of the bias, let $c = P/T$ and let P_1 be the number of signals used in the regression. The ratio of the empirical model complexity to the true model complexity is $q = P_1/P$. When $c \cdot q < 1$, ridgeless regression is unbiased and the expected return is proportional to the size of the β vector, which is linear in P_1 : $\beta(P_1)' \Psi_{1,1} \beta(P_1)$. However, once we cross the $P_1 = T$ boundary, ridgeless regression becomes biased due to implicit regularization. As we increase P_1 the size $\beta(P_1)' \Psi_{1,1} \beta(P_1)$ keeps increasing linearly in q . However, the bias is also increasing: With larger P_1 , ridgeless finds a β estimate with an ever smaller norm. Interestingly, when $\Psi = I$ and β is drawn iid, the two effects exactly cancel each other.

¹⁴For nonlinear machine learning models the saturation point may exceed $\min(P, T)$, and may even be infinite.

out there, but they all have larger variance than the ridgeless solution. By refining the set of solutions to be the one with minimum variance, we are imposing a form of shrinkage. It is not explicit ridge shrinkage because $z \rightarrow 0$, and we are *still* interpolating the training data. Instead, the shrinkage is implicit and it is there only because nominal complexity is high. As we push P higher and higher, the space of “perfect fit” solutions expands, so we can find least squares interpolators with ever smaller norm. Hence the continual decline in β norm and in turn the decline in trading strategy volatility.

Should we care if shrinkage is implicit or explicit? Either way, shrinkage constrains the β solution and controls variance, right? The critical difference is that while explicit shrinkage lowers variance, it also increases bias. In contrast, implicit shrinkage lowers variance *without sacrificing bias*. There is no way for explicit ridge shrinkage to accomplish this feat, it is a special property that can only be achieved by complexity-induced implicit shrinkage. Figure 1 demonstrates how implicit shrinkage in ridgeless regression maintains a fixed bias as P grows beyond T , yet volatility continues to fall. This delicate balance is what gives rise to the virtue of complexity. This also clearly illustrates that complex models continue to learn when P grows beyond T . The reduction of variance without a concomitant increase in bias is evidence that models with higher nominal complexity learn more efficiently, even though their effective complexity isn’t changing! Adding parameters allows us to learn models that deliver the same expected return but with more precision. It is immediately clear that effective complexity is inadequate for characterizing model performance. All models with $P > T$ have the same effective complexity, but differences in their out-of-sample performance continue to be explained by differences in their nominal complexity.

There are really two candidate definitions of “effective complexity” already in the theory of KMZ. As noted, one is the expected return, as this captures model bias. The other is the beta norm, which is in essence a shrinkage-adjusted parameter count. Neither of these is, on their own, sufficient for describing the behavior of a model. This is not surprising. In order

to characterize out-of-sample performance of any model, we need to understand both its bias and its variance. This is exactly what KMZ delivers, a characterization of bias, variance, and how these two interact to determine the out-of-sample Sharpe ratio of a model. This is another way to understand that the semantic definition of complexity is neither here nor there for understanding machine learning models. What is critical is that we understand the model’s tradeoff of bias and variance. This is what the KMZ theory delivers, and it so happens that $c = P/T$ is the key driver of this characterization.

Interestingly, the machine learning literature has demonstrated that low effective complexity is critical for successful out-of-sample model behavior (see, e.g., [Abbas et al., 2021](#); [Rosset and Tibshirani, 2020](#); [Patil et al., 2024](#); [Bartlett et al., 2017](#)). But there are different ways to build models with low effective complexity, and they are not all created equal. One way is to constrain the model specification, for example by specifying a model with a small number of raw parameters, which hardwires $c = ec$ (relatedly, one can use heavy *explicit* regularization to control ec). A different way is by using models with high nominal complexity, which inevitably leads to low effective complexity due to *implicit* regularization. KMZ demonstrates that the latter approach is more effective.

There is another important reason to think about models in terms of their nominal complexity. The nominal parameterization, P , determines a model’s ability to interpolate data. Empirical evidence throughout the machine learning literature suggests that interpolation capacity is a critical property for building models that “generalize” well (i.e., are successful out-of-sample). In other words, it appears that *both* low effective complexity and high nominal complexity are necessary ingredients for optimal predictions. It is not entirely understood why this is. [Simon et al. \(2024\)](#) argue that interpolation is valuable because it allows models to capture behaviors associated with low variance principal components of the data. We often think of interpolation as “overfitting,” but the widespread empirical benefits of interpolation might better be viewed as evidence that simple models pervasively *underfit*

data. The world is a complex place and our data sets are often too small to accurately inform our models about all nuances of the data generating process. This is why interpolation is so beneficial in the simulation example above! The DGP is complex relative to the size of the data set, and even if we train a model 100 parameters from 10 data points, we are still underfitting. We don’t have enough data to overfit; there is much more to learn about the DGP despite the fact that we have exactly fit the training data.

We close this discussion noting that our simulation above is an overly simplistic example, but its power lies in its transparency and tractability. The lessons learned from this simple example capture important properties observed in the Byzantine world of deep neural networks, and this example also produces patterns that match the empirical patterns in returns data. Of course, changing assumptions in the example above (such as the iid nature of the predictors) can alter the bias variance tradeoff and can produce interesting variation in the behavior of complex models. We address this point in the theory of Section 5.

2.2 Complexity and Small Samples

We next turn to the topic of complexity in small training samples. Nagel (2025) asserts that it is implausible for heavily parameterized models to learn meaningful predictive relationships in data scarce environments. Nagel questions the applicability of KMZ’s theory (which relies in part on $T \rightarrow \infty$ asymptotics) for understanding the behavior of small- T environments. As described in the introduction, the theory of model complexity is the theory of statistical learning in data scarce conditions.

Incredulity regarding the success of complex models in small data sets may stem from fuzziness about what “success” means. The theory results of KMZ show that, holding the number of training observations fixed at T , out-of-sample model performance tends to improve as the number of model parameters grows. In other words, success of complex models needs to be understood in *relative* terms. KMZ demonstrate the outperformance of

complex models when compared apples-to-apples with simpler models that have access to the same information set and have similar functional forms, but use fewer parameters.

To understand complexity, we must appreciate that complex models face challenges in *absolute* terms. Complex models suffer from an inability to recover the “true” model. Imagine fixing the number of parameters in your model to some large number P . Now consider the behavior of this model as you gradually increase complexity by *reducing* the number of observations T . Reducing the amount of training data diminishes model performance. When $c > 0$, standard asymptotic results (laws of large numbers and central limit theorems) break down. This is because there are too few data points to ensure that parameter estimates converge on their true values. The lack of data introduces “limits to learning,” leading to complex model performance that is inferior to the infeasible true model performance. But this is where the relative performance benefits of complexity come in. A complex model cannot recover the truth, but it gets closer to it than simple models.

In this section, we provide more intuition on the background theory from KMZ and its extension in [Didisheim et al. \(2024\)](#) (DKKM henceforth). We elaborate on DKKM’s concept of limits to learning, which is responsible for the behavior of complex models in small samples. We also provide simulation evidence that the KMZ limit theory is an accurate description of model behavior even in samples as small as a few training observations.

2.2.1 Limits to Learning

Low complexity ($c \approx 0$) describes settings with many more observations than parameters to estimate. This is the purview of traditional econometrics. In these conditions, the law of large numbers kicks in, and appropriate estimators tend to recover the true model parameters.¹⁵ Importantly, when $c \approx 0$, the in-sample performance of a model accurately portrays its expected out-of-sample performance.

¹⁵This statement is true if the model is correctly specified. If the model is misspecified, estimates tend toward the nearest “pseudo-true” parameters (see e.g. [White, 1982](#)).

In high complexity settings ($c \gg 0$), parameterization is large relative to the number of observations. This is the machine learning case. Here, the law of large numbers breaks down. “Best” estimators fail to converge on the actual parameters because there is not enough data to go around. As a result, complexity creates two wedges in model performance.

The first wedge is “overfit.” Naturally, heavy parameterization mechanically inflates in-sample performance. We define overfit as the difference between the in-sample performance (in terms of SDF Sharpe ratio or pricing errors) of the trained model versus the performance of the true but infeasible model:

$$\text{Overfit} = \text{In-sample Performance} - \text{True Performance}.$$

Note that overfit is more about a dearth of training observations than excessive parameters—it exists even for correctly specified models with no “excess” parameters (as in the example of Section 2.1).¹⁶

Second, high complexity creates a wedge between the true model and the expected out-of-sample performance of the trained model. The failure of the law of large numbers when $c > 0$ means that, even in the limit, there remains a gap between the trained model and the truth. We call this wedge “limits to learning,” as it describes how far the trained model is from the best possible model in terms of expected performance.

$$\text{Limits to Learning} = \text{True Predictability} - \text{Out-of-sample Performance}.$$

¹⁶It is intuitive that complexity exaggerates in-sample performance, but by how much? By taking into account the details of the model (its specification and the number of parameters) and the training data conditions (the amount of training data and its covariance structure), the theory of DKKM infers expected overfit via random matrix theory. A fascinating implication is that we can, in turn, infer the amount of predictability in the true underlying model and, along with it, the expected performance of a trading strategy based on the true model (see [Chen et al., 2025a,b](#)). This provides an answer to economic questions like “if an investor had full information about the prediction environment, including the true model structure and parameter values, how much predictability would she see in financial markets, and how profitable would this information be to trade on?”

Like overfit, this wedge is due to the statistical cost of estimating many parameters when training data is relatively scarce.

Finally, we define the overall complexity wedge as the difference in in-sample model performance versus expected out-of-sample performance, which is determined by the amount of overfitting plus the limits to learning:

$$\begin{aligned}\text{Complexity Wedge} &= \text{In-sample Performance} - \text{Out-of-sample Performance} \\ &= \text{Overfit} + \text{Limits to Learning}.\end{aligned}$$

To summarize, when the model is correctly specified and $c = 0$, the law of large numbers eliminates these wedges and in-sample, out-of-sample, and true predictability are equalized. When $c > 0$, complexity causes the trained model’s in-sample predictability to exceed that of the true model, which in turn exceeds the trained model’s expected out-of-sample predictability. The sum of overfit and limits to learning equals the difference between the in-sample and out-of-sample performance.¹⁷ In addition to these wedges, if the model is *misspecified*, there is another gap that emerges due to specification error. When $c = 0$, then in-sample and out-of-sample performance remain equalized, but because of specification error both fall short of the true model.

Figure 2 illustrates these concepts of KMZ and DKKM in a stylized example. Consider two hypothetical empirical methods (bars represent in-sample and out-of-sample model performance). The left side of the figure depicts the “traditional approach” to econometric modeling in which the researcher specifies a deliberately simplified model in the name of parsimony. Restricting the specification so $P/T \approx 0$ ensures that model estimates

¹⁷Complexity wedges can be partially mitigated by ridge shrinkage. Positive shrinkage can reduce the extent of overfit and improve the limits to learning, but as long as complexity is greater than zero, both wedges are positive. Much of the work in machine learning goes into identifying the extent of shrinkage that maximizes a model’s expected out-of-sample performance, typically using data-driven methods like cross-validation. Another contribution of our theoretical approach is that, for a given set of training conditions, it identifies the optimal degree of shrinkage to maximize expected out-of-sample model performance.

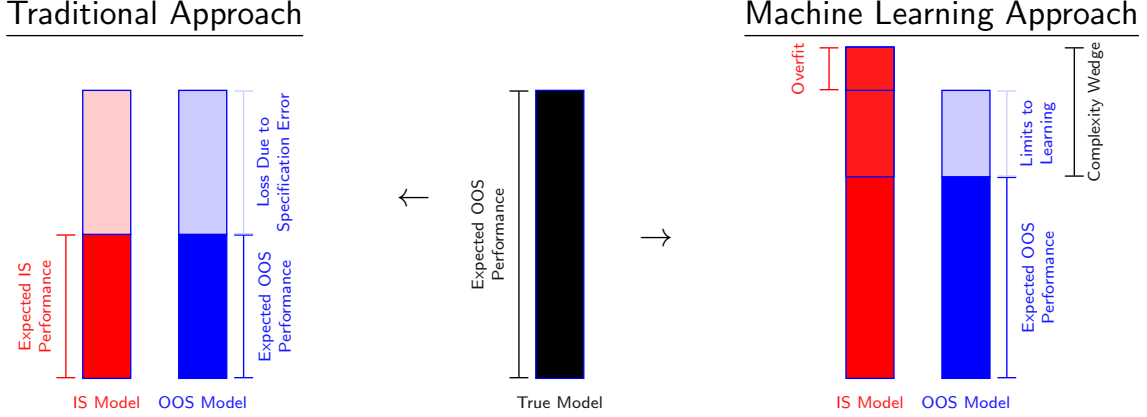


Figure 2: Limits to Learning, Overfit, and the Complexity Wedge

Note. Illustration of limits to learning, overfit, and the complexity wedge. The left side shows in-sample (IS) and out-of-sample (OOS) model performance in the “traditional approach” to econometric modeling in which the researcher specifies a deliberately simplified model. The right side shows the “machine learning approach” in which the researcher uses a complex non-parametric model. The center bar illustrates the frame of reference by showing the infeasible true model.

converge and therefore the in-sample performance is indicative of expected out-of-sample performance. The equivalence of in-sample and expected out-of-sample performance is the virtue of parsimony and arises from the low variance of a model whose parameter estimates have converged due to the law of large numbers. But, because the model is simple, it is a coarse approximation of the true model thus it converges to an incorrect model, and as a result its performance suffers. Its limited out-of-sample performance is due to specification error arising from the model’s simplicity.¹⁸

The right side of Figure 2 illustrates the “machine learning approach” in which the researcher uses a complex non-parametric model. Because the model is flexible and heavily parameterized, it has the capacity to approximate the true model with high fidelity ($P \rightarrow \infty$ mitigates the specification error). However, due to realistic data limitations, the model is complex (P/T is large). This results in overfit. But this “overfit” typically *improves* out-

¹⁸If the researcher *knows* the correct simple model specification, then of course using this model is first best and recovers the truth. But this is a heroic and unrealistic assumption. We acknowledge that the researcher might get lucky with their simple specification, but we don’t expect researchers to be lucky on average.

of-sample performance (recall the intuition of the example in Section 2.1—if the data set is small and the true model is complex, then the model may see exaggerated in-sample performance, it is still underfitting the data, especially if you use anything less than all of the predictors). Despite its high capacity to approximate the true model, its out-of-sample performance is below that of the true model due to limits to learning. The “virtue of complexity” references the fact that, although complex model performance is limited by data constraints, the benefits of its improved approximation outweigh the simple model’s benefit of low variance, thanks in large part to the effectiveness of implicit shrinkage.

2.2.2 Complex Models Excel in Surprisingly Small Samples

To appreciate limits to learning, let’s see it at work. In this section, we report the behavior of complex models in small samples via simulation. In real-world data analyzed by KMZ, no one knows the ground truth that generated the data, so it is impossible to definitively know how much learning is happening. In simulations, we know the generating process and therefore can make concrete statements about the extent of learning by comparing how close an empirical model comes to the truth.

For our ground truth, we use a nonlinear prediction function

$$Y_t = \beta \frac{\sum_{i=1}^3 X_{i,t}}{1 + \sum_{i=1}^3 X_{i,t}^2} + \epsilon_t,$$

where the noise ϵ_t is standard normal, and the coefficient β is fixed to ensure the true R^2 of the model is 0.1.¹⁹ To make the predictor data similar to typical economic predictors, we generate persistent variables: $X_{i,t}$, $i=1,\dots,3$ are AR(1) processes with an AR coefficient of 0.5 and iid standard normal innovations.

To approximate this nonlinear prediction function, we use the same RFF methodology

¹⁹While the true R^2 of the model is 0.1, the empirical R^2 will be substantially smaller due to limits to learning.

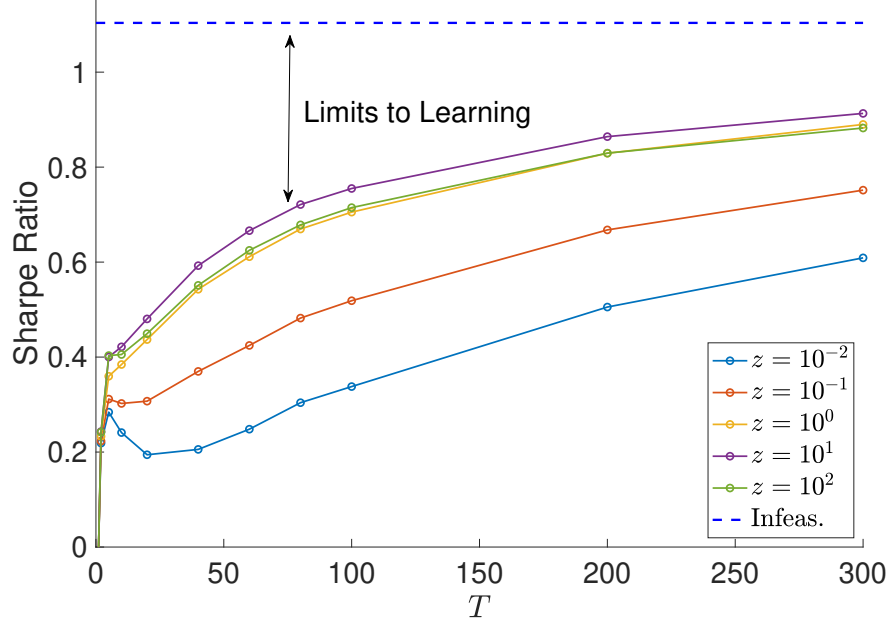


Figure 3: Limits to Learning in Small Samples

Note. Illustration of limits to learning in simulated samples. The empirical model uses RFF with a fixed number of parameters $P = 100$. The curves report out-of-sample model performance in training sample of size T (ranging from 1 to 300 observations) and for various degrees of explicit ridge shrinkage, z . The dashed horizontal line shows the infeasible best model performance, and the vertical distance between empirical models and the infeasible best model describes limits to learning.

used by KMZ. We train an RFF model fixing $P = 100$ in a sample of size T , ranging from a single observation, $T = 1$, training samples as large as $T = 300$ observations. For each trained model, we make an out-of-sample forecast, μ_T . As in KMZ, we construct a “market timing strategy” whose realized out-of-sample return is $\mu_T \cdot Y_{t+1}$. We repeat this procedure for one million simulated data sets, then calculate the out-of-sample timing strategy Sharpe ratio averaged over all simulations. We repeat this entire procedure for each training sample size T .

Figure 3 reports the Sharpe ratio of the trading strategy as we vary the size of the training sample holding the number of model parameters fixed. The dashed horizontal line shows the infeasible best Sharpe ratio attainable with full knowledge of the DGP. The vertical distance between the dashed line and the empirical performance curve describes the limits

to learning. Even when the sample is relatively large at $T = 300$, model complexity is nontrivial at $c = 1/3$. Clearly, the empirical model learns a substantial amount about the DGP, though with $c = 1/3$ the data is still too scarce for the traditional law of large numbers to fully kick in and the empirical model fails to match the true model. As the sample gets smaller, limits to learning become more pronounced.

What is fascinating about this figure is that extremely complex models learn meaningful predictive relationships even when T is miniscule. With $T = 10$, complexity is $c = 10$, yet still the model achieves an out-of-sample Sharpe ratio of 0.4, versus the infeasible true Sharpe ratio of about 1.1. With $T = 2$ ($c = 50$) the Sharpe ratio is about 0.2!²⁰

Figure 3 holds P fixed and varies T , essentially plotting one point on the VoC curve for each hypothetical value of T . In Figure 4, we show more detail by plotting entire VoC curves (i.e., a range of model sizes P) for each sample size T . This figure demonstrates that the main result of KMZ—namely, that complex models dominate simple model in apples-to-apples comparison—holds in small samples. Recall that the VoC curves in KMZ are based on random matrix theory in the $P, T \rightarrow \infty$ limit. In contrast, the results in Figure 4 are exact *finite sample* results. Evidently, the asymptotic theory of KMZ faithfully describes the virtue of complexity even in samples with as few as $T = 10$ training observations.

The simulation evidence above demonstrates that complex models can learn meaningful nonlinear predictive relationships even in small samples. As shown by KMZ, complex models outperform simple models, and this is true even when T is small. Limits to learning tells us that the extent of this outperformance is dictated by the amount of data available for training. It is hardly surprising that complex models deteriorate as T shrinks, but Nagel’s assertion of no learning when T is small is clearly incorrect. Rather than dismissing the virtue of complexity in small T , an interesting and open research direction is quantifying the extent of complex learning as a function of (finite) T under general conditions (the results of Figure 3 are from simulations of one particular DGP).

²⁰This is significant with a p -value of less than 0.001 in our simulations.

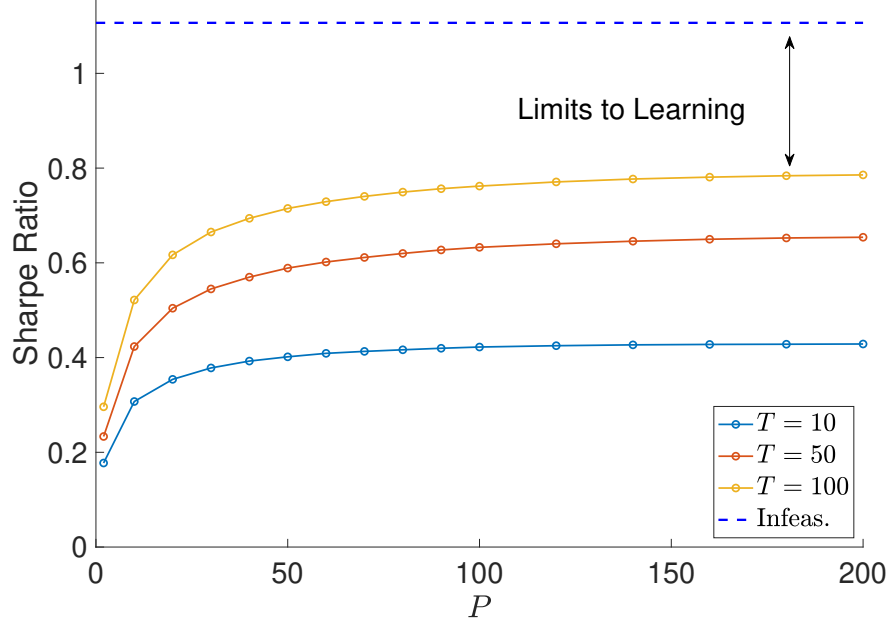


Figure 4: Virtue of Complexity in Small Samples

Note. Illustration of virtue of complexity in simulated samples. Empirical models uses RFF with varying numbers of parameters P (ranging from 2 to 200 parameters), holding sample size fixed at $T = 10, 50$, or 100 (shown with different curves). Explicit ridge shrinkage is fixed at $z = 10$. The dashed horizontal line shows the infeasible best model performance.

By focusing his critique on $T = 12$, Nagel zeroes in on a problem where limits to learning are severe, and thus where the virtue of complexity is known to be weakest. The point of KMZ’s deliberately conservative $T = 12$ analysis is to show that complexity is valuable even in such extreme conditions. Indeed, the virtue of complexity is impressive even in this challenging environment (as further demonstrated in Section 3).

2.2.3 Summary

Effective parameterization of a complex model is inadequate for understanding model behavior. KMZ describes complex model behavior in terms of bias and variance and from this provides a complete description of complex model behavior (including its effect on Sharpe Ratio, a novel and challenging result in statistical complexity theory). The machine learning literature already shows that low effective complexity is a prerequisite for successful out-of-

sample performance. KMZ theory shows that low effective complexity achieved through *high nominal complexity* (and its concomitant implicit shrinkage) is more effective than forcing low effective complexity with tightly constrained specifications.

The virtue of complexity is a relative statement—complex models dominate simpler models on an apples-to-apples basis. But complex models face limits to learning in absolute terms that prevent them from recovering the true model. Limits to learning is a critical concept for understanding complexity. It reveals there is always learning in complex models, even when T is small. But the larger the sample the greater the learning. We demonstrate that the asymptotic theory of KMZ provides an accurate description of complex model behavior in small samples. And we show that, despite limits to learning, complex models are capable of learning meaningful nonlinear predictive associations in small samples.

The benefits of complex models in small samples is especially valuable in a time series discipline like asset pricing, which faces natural constraints on sample sizes. These benefits are compounded further by the nonstationarity that likely underlies financial and economic data.

3 Empirical Evidence for the Virtue of Complexity

Nagel’s second central claim is that KMZ’s empirical findings fail to demonstrate that high complexity models learn meaningful predictive associations. In his words:

“I find that the high-complexity ridgeless regression with RFF in fact does not learn to extract useful predictive information from the K predictors. Instead, the market-timing strategy based on the high-complexity ridgeless regression forecasts from short training windows reduces to a certain volatility-timed momentum strategy.”

Nagel’s empirical arguments are focused on a subset of the results in KMZ that use training samples of $T = 12$.

In this section we extend the empirical analysis of KMZ, demonstrating that the KMZ model learns genuine and robust predictive relationships (even when $T = 12$).

3.1 Genuine Learning in Small Samples

Nagel argues that if the regressors are persistent and the training sample is small, then RFF will mechanically recover a momentum-like strategy. Perhaps the clearest way to evaluate this claim is to generate “placebo” predictors that have the same distributional properties as the Goyal-Welch predictors (especially their persistence) but that are independent of returns by construction. If Nagel’s argument is correct that persistence and small samples give rise to predictability even when there is no true predictability in the data, then we should see the placebo predictors spuriously demonstrating a virtue of complexity.

The placebo analysis proceeds as follows. Let R_t to be the genuine historical market return data and X_t be the (15×1) vector of genuine Goyal-Welch predictors used by KMZ. For each predictor i we estimate the autoregression

$$X_{i,t} = \rho_{i,0} + \rho_{i,1}X_{i,t-1} + u_{i,t}. \quad (2)$$

We then generate a placebo version of this predictor as

$$\check{X}_{i,t} = \hat{\rho}_{i,0} + \hat{\rho}_{i,1}\check{X}_{i,t-1} + \check{u}_{i,t} \quad (3)$$

where $\hat{\rho}_{i,0}$ and $\hat{\rho}_{i,1}$ are the coefficients estimated from equation (2). The innovations in (3) are generated from a multivariate normal distribution that preserves the covariance structure of the predictors, $\check{u}_t \sim \text{iid}N(0, \hat{\Sigma})$, where $\hat{\Sigma}$ is the sample covariance of the estimated regression residuals in equation (2). The placebo data are initiated at $\check{X}_{i,0} = \hat{\rho}_{i,0}$ for each i . Finally, we produce a P -dimensional set of random Fourier features $\check{S}_t$ from the placebo Goyal-Welch data $\check{X}_t$. We forecast the genuine market return with placebo predictors in ridge regressions

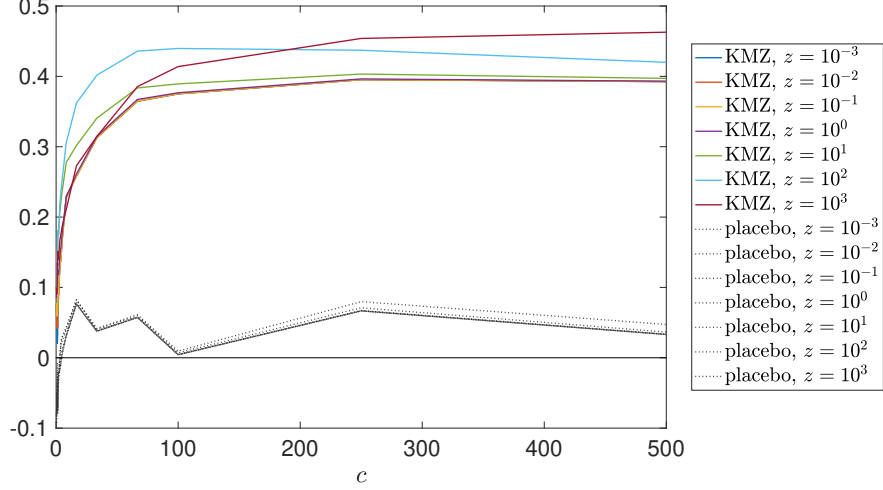


Figure 5: Virtue of Complexity Placebo Test

Note. The figure reports true empirical Sharpe ratios in complex models (shown in solid lines with color) and placebo Sharpe ratio (black dotted lines). The placebo strategy is trained using real returns data, but with predictor data that is simulated to have the same persistence and variance as the true predictors but is independent of the return. The placebo Sharpe ratios demonstrate that the virtue of complexity is not a spurious result of small samples and persistent predictors (as claimed by Nagel).

of the form

$$R_{t+1} = \check{S}_t^t \beta + \epsilon_{t+1}.$$

In other words, all aspects of our placebo test are identical to the $T = 12$ analysis in KMZ with the exception of substituting placebo predictors in place of the true predictors.

Figure 5 reports results. The colored curves are the original KMZ results using genuine data. The dotted lines are placebo results. We find almost no spurious predictability in the placebo test, despite the fact that $T = 12$ and 9 of the 15 predictors have an AR(1) coefficient over 0.96 (3 of those are over 0.99). We can conclude from this test that the high complexity KMZ model indeed learns about the predictive relationship between the Goyal-Welch variables and returns (but of course the learned relationship is imperfect due to limits to learning).

Another way to see genuine learning in the KMZ model is to study how performance

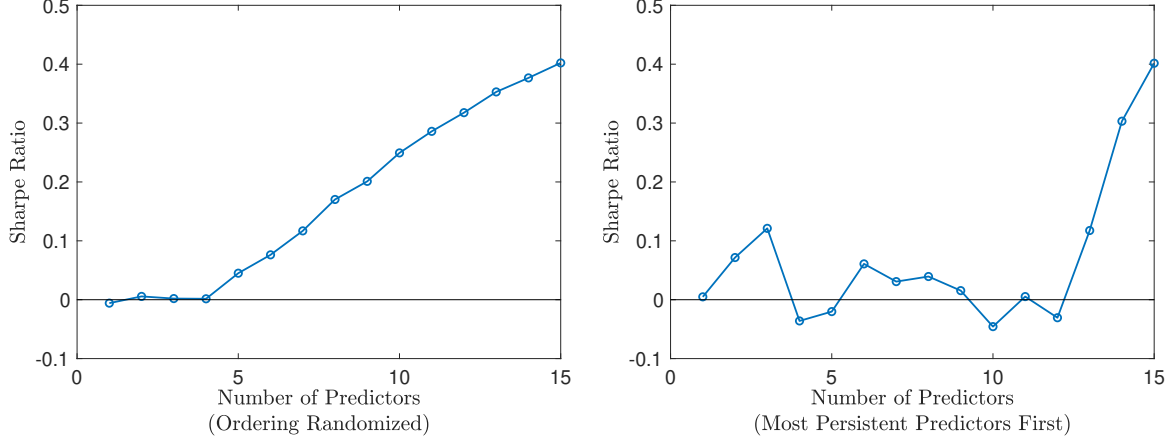


Figure 6: Virtue of Complexity With Varying Numbers of Predictors

Note. The figure reports out-of-sample model performance of the highest complexity KMZ model ($P = 12,000$) with various subset of Goyal-Welch predictors using $T = 12$ month training windows. The left panel includes one to 15 predictors and randomizes the order in which predictors are included. The right panel introduces predictors starting with the most persistent and works its way down to the least persistent.

changes as we vary the set of Goyal-Welch predictors used in the model. Figure 6 reports out-of-sample model performance of the highest complexity model ($P = 12,000$) when we use as few as a single predictor and then gradually add predictors until we get back to the full 15-predictor model. In the left panel we report results when we randomize the order in which predictors are included (we do this repeatedly and average over orderings), while the right panel introduces predictors starting with the most persistent and works its way down to the least persistent. We continue to focus on the $T = 12$ setting with all aspects of the analysis identical to that of KMZ with the exception of varying the number of raw predictors.

There are two main conclusions from this figure. First, if the KMZ results are spurious and are due to persistent predictors and small T , then this should manifest when we use just a few predictors. Instead, we see that adding predictors gradually increases out-of-sample model performance. Second, we see evidence of nonlinear predictions. Since the predictors are randomly ordered, the convexity in the left panel suggests that the complex

model benefits from interactions among predictors. Third, the right panel shows that the biggest gains come from the *least persistent* predictors, contrary to Nagel’s claim that our findings must be driven by persistent predictors.²¹

3.2 More Than Momentum

Nagel (2025) claims that he can “*approximate KMZ’s market timing strategy with a very simple one: A momentum strategy that has weights that decay with lag length ... combined with a volatility-timing scaling factor that increases the magnitude of weights on past returns if the volatility of the 15 predictor variables is low.*” He then proceeds to benchmark KMZ empirical analysis against this replicating model.

It’s unclear how “very simple” this momentum strategy is (we discuss this more below), but first there is a more important conceptual point. We should think of Nagel’s momentum strategy not as a benchmark for KMZ, but as an approximation of it, reverse-engineered from their model specification and empirical results. Nagel is clear about the steps he goes through to derive his alternative momentum strategy from the KMZ model. If this were a pre-existing strategy (like the standard time series momentum as in [Moskowitz et al. \(2012\)](#) or volatility-managed strategy of [Moreira and Muir \(2017\)](#)), or if it were an a priori theoretical choice, it might be a blow to the empirical claims of KMZ. Instead, Nagel walks through his steps of constructing an approximator of KMZ.²² This is a useful demonstration of how machine learning can be leveraged to produce more tractable representations of predictive models. Indeed, this is related to the concept of “distillation” in the machine learning literature.

What makes less sense, however, is using alpha tests to compare the high complexity model to its distilled approximation. Are we surprised that the models are similar? Is it sensible to denounce a complex machine learning model on the grounds that it performs

²¹This last fact is not a surprise; it is already shown in a different way by KMZ (see their Figures 11 and IA5).

²²As an aside, the fact that Nagel can derive any reasonable recreation of the RFF model illustrates the benefit of working first with RFF before delving into more complicated models like deep networks trained with SGD or other methods, where such a derivation would be next to impossible.

similarly to a human-specified model when the human-specified model is inspired by the findings of the ML model? One first needs the complex model in order to “distill” it into something interpretable. While machine learning distillation is typically a process of a “student” model training on the larger “teacher” model, the process taken by Nagel is human-oriented and time-intensive. In another dataset with another DGP, the findings of the complex model will again be learned in seconds, but its reverse-engineered approximation must be pursued anew.²³

At a much more basic level, momentum must be present in the data if returns are predictable and expected returns (i.e., “predictors”) are persistent (this point is emphasized by Kelly et al., 2021). Showing that there is a form of momentum in the data (and an arcane one at that) cannot be seen as evidence either for or against learning from another prediction model. Our placebo test makes this clear. Seeing momentum in the data simply shows that a sufficient condition for a particular form of predictability is satisfied.

3.2.1 A Closer Look and Economic Implications

The specific momentum model chosen by Nagel to approximate the KMZ model is:

$$\mu_t^{DW/PV} = \frac{1}{\hat{\sigma}_{x,t}^2} \times \sum_{k=0}^{11} \frac{12-k}{78} Y_{t-k}, \quad (4)$$

where $\hat{\sigma}_{x,t}^2$ is the average trailing 12-month variance of the predictors.²⁴ The realized return of the associated trading strategy is $Y_{t+1}\mu_t^{DW/PV}$.

There are a couple of important preliminary points to note. First, the returns Y_t and

²³For example, in our results below, we report various machine learning models trained on data orthogonalized to the original KMZ dataset that also demonstrate VoC but that is not approximated with Nagel’s momentum scheme.

²⁴In detail,

$$\hat{\sigma}_{x,t}^2 = \frac{1}{15} \sum_{i=1}^{15} \left[\frac{1}{12} \sum_{k=0}^{11} \left(X_{i,t-k} - \left(\frac{1}{12} \sum_{k=0}^{11} X_{i,t-k} \right) \right)^2 \right].$$

the predictors X_t are already standardized by their individual trailing variance. This is a design choice from KMZ to reduce the influence of heteroskedasticity and to ensure that the benefits of “volatility-management” (Moreira and Muir, 2017) are accounted for *before* the complex model is applied (KMZ always compare their strategies to the simple volatility-managed strategy, which is a buy-and-hold investment in the volatility-standardized return, Y_t). Nagel’s momentum strategy therefore has three levels of volatility standardization influencing its performance: The standardization already built-in to Y_t , followed by the standardization of predictor variance $\hat{\sigma}_{x,t}^2$, which is itself derived from X_t ’s that are already variance standardized using a longer trailing windows. Thus, as alluded to above, the extent to which Nagel’s momentum strategy is “a very simple one” is in the eye of the beholder.

Next, we decompose the behavior of Nagel’s momentum strategy. For this we study four strategies based on the following signals:

$$\mu_t^{\text{EW}} = \frac{1}{12} \sum_{k=0}^{11} Y_{t-k} \quad (5)$$

$$\mu_t^{\text{DW}} = \sum_{k=0}^{11} \frac{12-k}{78} Y_{t-k} \quad (6)$$

$$\mu_t^{\text{PV}} = \frac{1}{\hat{\sigma}_{x,t}^2} \quad (7)$$

$$\mu_t^{\text{EW/PV}} = \frac{1}{\hat{\sigma}_{x,t}^2} \times \frac{1}{12} \sum_{k=0}^{11} Y_{t-k}. \quad (8)$$

μ_t^{EW} is a standard equal weighted average of the trailing 12-month returns, commonly known in the literature as “time series momentum” (Moskowitz et al., 2012). μ_t^{DW} is the declining-weight average of past 12-month returns that constitutes the pure momentum component of Nagel’s strategy. Third, μ_t^{PV} uses *only* the predictor variance component of Nagel’s strategy. $\mu_t^{\text{EW/PV}}$ is the product of equations (5) and (7), the same Nagel’s strategy but without declining momentum weights. And finally, Nagel’s momentum in (4) is the product of equations (6) and (7).

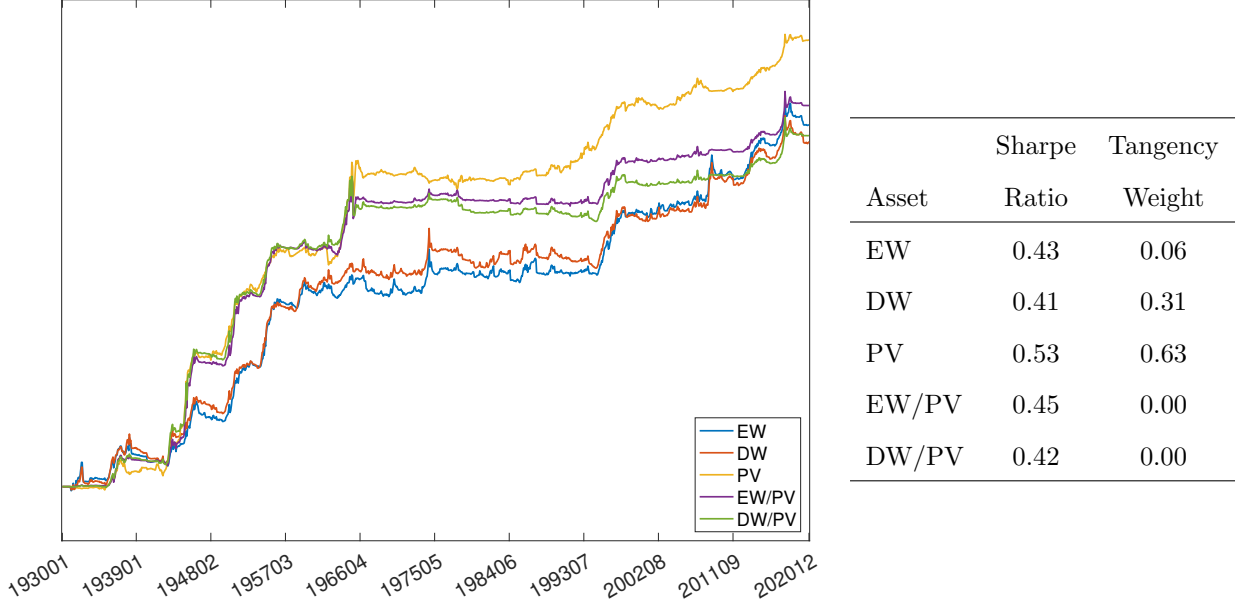


Figure 7: What Drives Nagel’s “Mechanical” Momentum Strategy?

Note. This figure decomposes the performance of Nagel’s momentum strategy into five related sub-strategies described in equations (4) through (8). The figure shows cumulative returns of each strategy and the table shows Sharpe ratio and full sample tangency portfolio weights (imposing a no-shorting constraint and normalizing all strategies to the same volatility).

Figure 7 breaks out the performance of the signals in equations (4) through (8). EW and DW are nearly identical in performance (and are 93% correlated). These are therefore already accounted for in KMZ (who show that the complex model has large and significant alpha over EW in their Section VII).

A bigger insight from this breakout is that PV is a quite different, and substantially stronger, performer than any form of momentum (its correlation is 43% and 35% with EW and DW respectively). Scaling momentum by predictor volatility doesn’t accomplish much. Among the five strategies, the PV strategy is the strongest by a large margin. A long-only tangency portfolio of these five strategies puts 63% of its weight on PV and spreads the rest between EW and DW momentum. Nagel’s scaled momentum strategy gets 0% allocation. Furthermore, KMZ has a large and significant alpha to PV (7% per annum on a 10% volatility scale with a t -statistic of 2.2).

The success of PV has a deeper implication. It suggests that Nagel distills an economically interpretable strategy from KMZ’s complex model. The Goyal-Welch predictors are largely macroeconomic in nature. The success of the PV strategy implies that a portfolio managed on the basis *macroeconomic volatility* is utility-enhancing (above and beyond the benefits of [Moreira and Muir, 2017](#), market-volatility management). KMZ results hint at this finding. They show that the complex model earns its alpha by taking positive market risk during economic expansions and taking risk off heading into recessions. Nagel’s derivation provides an interpretation of this trading pattern. The model detects that rising macroeconomic volatility associates with declining market returns. This is an educational example of how machine learning can be a means to an economic end.

3.3 The Many Varieties of Complex Models

In this section we demonstrate the virtue of complexity in an entirely different way than KMZ. There are many varieties of complex models for return prediction. Complex models can differ by their model architecture, by their tuning parameters, by their data inputs, by their objective functions, and so on. We show that the virtue of complexity holds in a wide variety of complex models. We also show that trading strategies from different complex models demonstrate surprisingly different and complementary behaviors. The mean-variance efficient portfolio puts weight on many strategies derived from different complex models. In other words, even individual complex models not complex enough—they are dominated by an even larger model that is a combination of many complex models. We refer to this form of complexity-on-top-of-complexity as the “virtue of ensemble complexity.” We then rationalize how this phenomenon emerges.

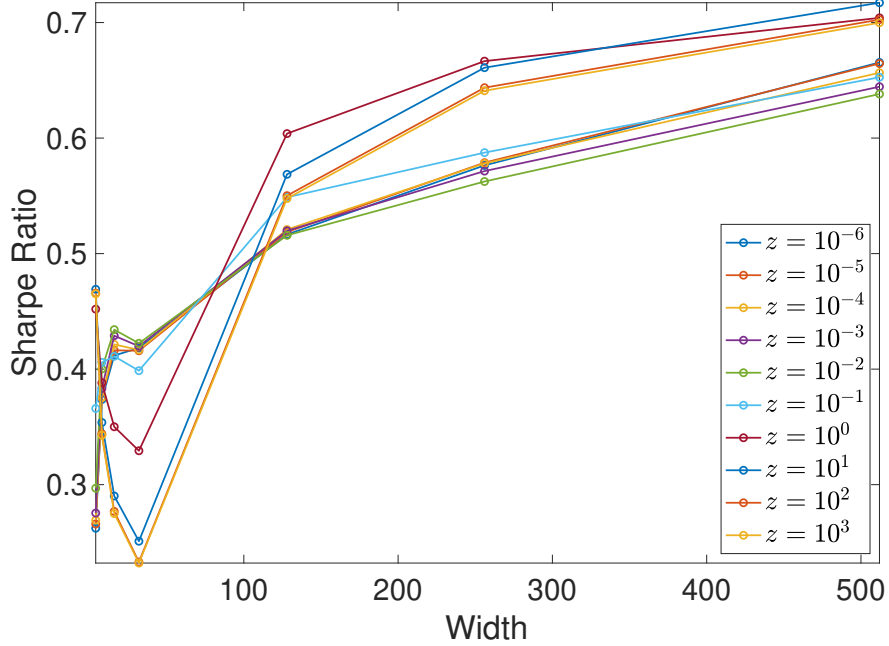


Figure 8: KMZ Analysis With Deep Neural Networks

Note. The figure shows out-of-sample Sharpe ratio of feed-forward neural networks with four hidden layers. The x-axis reports the width of each network, ranging from 4 to 512 neurons in each hidden layer.

3.3.1 Deep Networks

The quest to understand complexity was born from the real-world success of deep neural networks. Only after seeing that increasingly large networks dominated in application after application did statisticians begin using random feature regression as a toy model for deep learning. It is an important reminder that we (and KMZ) are not interested in RFF per se. We are interested in understanding the viability for complex models for financial modeling, and RFF is merely a convenient vehicle for analyzing empirical counterparts to theoretical comparative statics. KMZ is clear on this point.

Given that deep neural networks are the main attraction in machine learning, we now demonstrate that 1) deep neural networks can deliver more powerful out-of-sample trading strategy performance than RFF and 2) there is a virtue of complexity in network size. We build a deep neural network with a standard architecture. It is a feedforward network with

Goyal-Welch predictors as inputs, it uses four hidden layers with width of K neurons in each layer, and the activation function for all neurons is hyperbolic tangent.²⁵ These models have $16K + 3K^2$ weight parameters ($15K$ for the input layer, K^2 for the fully connected hidden layers, and K for the output layer, plus a bias term in each layer). The linear output layer is ridge-regularized with a penalty parameter z (following our ridge-regularized RFF). The entire network is trained with stochastic gradient descent (including the input weights, which is different from the randomized input weights of RFF). We train on an expanding training window beginning with a window of 36 months in the first training sample.

The results are shown in Figure 8. The plot shows how the deep network’s performance changes as we vary the hidden layers’ width. We consider widths of $K = 4, 8, 16, 32, 64, 128, 256$, and 512 . The first conclusion of the figure is that Sharpe ratios in some cases exceed 0.70, well above any of KMZ’s RFF specifications or any momentum strategy. The second conclusion is that the virtue of complexity holds even in deep networks: Holding depth fixed, wider models dominate narrower models (this is the deep net analogue of increases the number of terms in RFF regressions).

3.3.2 Alternative Objective Functions

KMZ use a two-step approach. They first train a time series return prediction function and then use its predictions to build a market timing strategy. In contrast, DKKM propose a one-step process to train the timing strategy directly. Applying DKKM’s insight to the KMZ problem, we replace the least squares prediction function

$$Y_t = f(X_{t-1}) + \epsilon_t$$

²⁵We use four layers to contain the computational costs, but it is straightforward to extend this analysis to much richer models (see [Kelly et al., 2025a](#)).

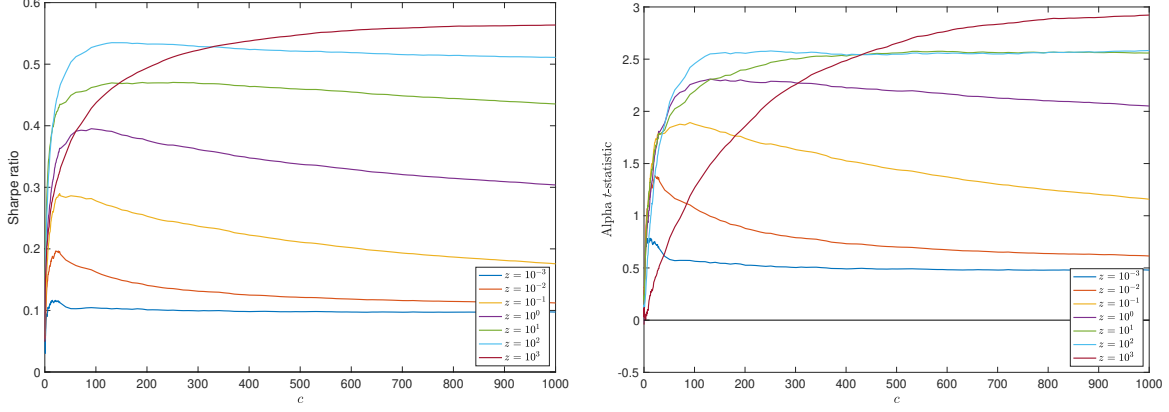


Figure 9: KMZ Analysis With Sharpe Ratio Objective

Note. The left panel plots the complex MSRR model’s out-of-sample Sharpe ratio as a function of complexity. The right panel shows the alpha t -statistic of this model versus the original KMZ model with least squares objective.

with a maximum Sharpe ratio regression (MSRR)²⁶

$$1 = w(X_{t-1})Y_t + \epsilon_t. \quad (9)$$

As discussed in [Kelly and Xiu \(2023\)](#), MSRR direct optimizes the Sharpe ratio objective, rather than the indirect approach of first optimizing a mean squared prediction error objective whose solution is then plugged into the timing portfolio. Intuitively, the objective of (9) is to recover a functional form for $w()$ such that the timing returns $w(X_{t-1})Y_t$ behave as closely as possible to a positive constant (and thus has the highest possible Sharpe ratio). For interested readers, DKKM elaborate on this idea in detail.

We approximate the weight function $w()$ using RFF. More specifically, let

$$w(X_t) = \sum_{i=1}^{P/2} (\beta_{i,1} \sin(\gamma \omega'_i X_t) + \beta_{i,2} \cos(\gamma \omega'_i X_t)), \quad \omega_i \sim iidN(0, I).$$

²⁶See [Britten-Jones \(1999\)](#) and [Kelly and Xiu \(2023\)](#).

As a result, (9) can be written as a P -dimensional linear regression

$$1 = \sum_{j=1}^P b_j S_{j,t-1} Y_t + \epsilon_t \quad (10)$$

where each term j is a random nonlinear transformation of the original features X with sine or cosine activation.²⁷

The left panel in Figure 9 demonstrates that, when the model object is changed from least squares to Sharpe ratio maximization, we again find that large P models achieve high Sharpe ratio (especially with appropriate shrinkage). More strikingly, the MSRR models behave differently than KMZ’s least squares model. The right panel in Figure 9 reports the alpha t -statistic of MSRR models against the highest complexity KMZ model. MSRR models have significant alpha versus KMZ (especially with appropriate shrinkage).

Next, we extend the MSRR framework further and replace the RFF formulation of $w()$ with a four hidden-layer neural network—the MSRR analogue of the neural networks reported in Section 3.3.1. The results are shown in Figure 10. MSRR with deep nets dominate MSRR-RFF and deliver alpha t -statistics as high as 5.0 versus the original KMZ model.

3.3.3 Alternative Inputs

In the previous sections, we altered the model structure of KMZ but held the underlying data fixed. Now, we show that the virtue of complexity appears when using alternative model

²⁷As Kelly et al. (2024a) show, this is a novel and finance-specific kernel method. In the limit of a large number of random features, this MSRR-RFF model converges to a “portfolio kernel” given by

$$R_{T+1}^{MSRR} = K_R((x_T, R_{T+1}), (X, R))(zI + K_R((X, R), (X, R))/T)^{-1} \mathbf{1} T^{-1}, \quad (11)$$

with

$$K_R((x, r), (\tilde{x}, \tilde{r})) = r\tilde{r}K(x, \tilde{x}). \quad (12)$$

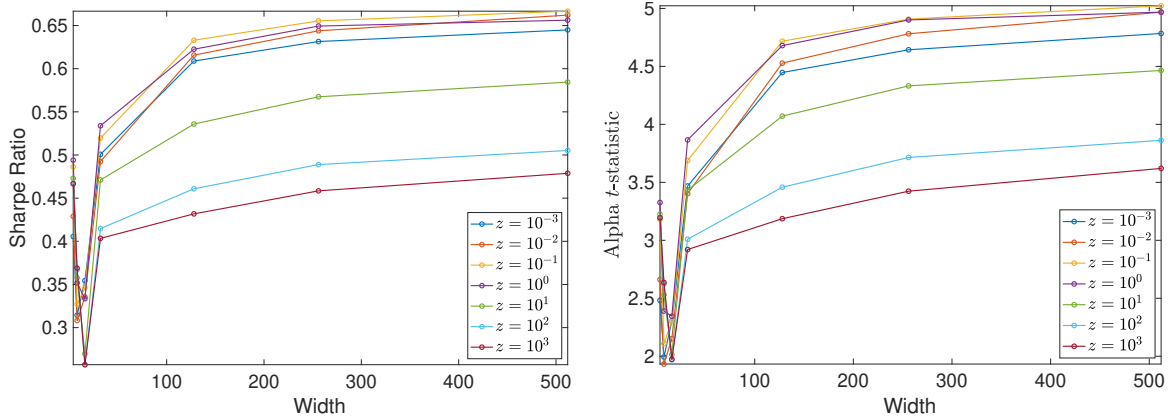


Figure 10: Deep Neural Network With Sharpe Ratio Objective

Note. The left panel plots the deep neural network MSRR model’s out-of-sample Sharpe ratio as a function of network width (the number of neurons in each hidden layer). The right panel shows the alpha t -statistic of this model versus the original KMZ model.

inputs. In particular, we pre-process the data to eliminate the conditions that Nagel argues give rise to a spurious virtue of complexity, and yet we continue to find the same qualitative results as KMZ.

First, instead of forecasting Y_t , we orthogonalize it versus Nagel’s momentum strategy in a full-sample regression:

$$\hat{\epsilon}_t = Y_t - \hat{\beta} \left(Y_t \mu_{t-1}^{\text{DW/PV}} \right). \quad (13)$$

We then repeat the analysis of KMZ using the residualized return $\hat{\epsilon}_t$ as the forecast target.

The results are shown in the left panel of Figure 11. We continue to find that complex models dominate simple models when forecasting residual returns. Naturally, the magnitude of the outperformance is smaller as we have purged the return of some of its predictable content, but the upward sloping curves have the same qualitative shape found in KMZ.

Next, we go back to the original returns Y_t , but now replace each predictor $X_{i,t}$ with its

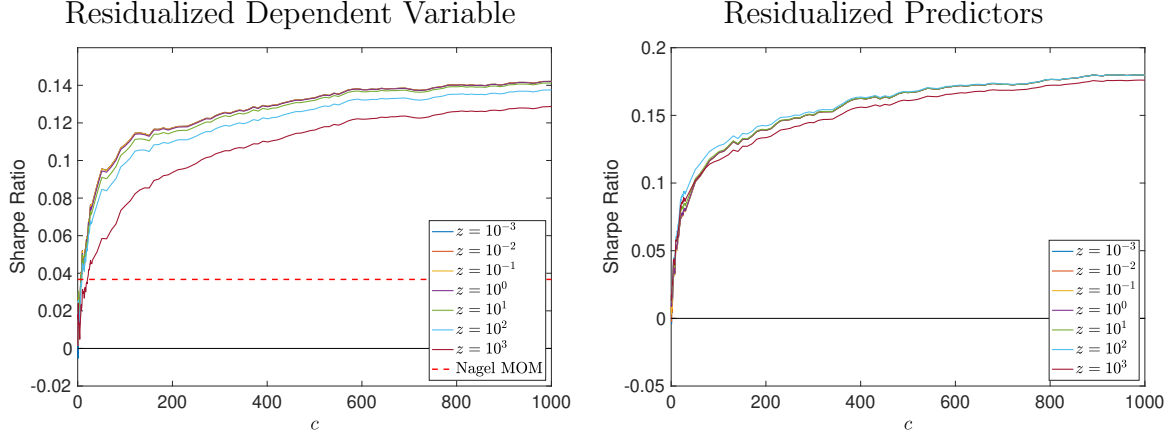


Figure 11: KMZ Analysis With Modified Data Inputs

Note. The figure repeats the analysis of KMZ with $T = 12$ but alters the data inputs. The left panel uses the same predictors of KMZ but uses returns that are orthogonalized to Nagel’s momentum strategy. The right panel uses the same returns of KMZ, but replaces predictors with their AR(1) innovations.

AR(1) innovation estimated from the full sample:²⁸

$$\hat{u}_{i,t} = X_{i,t} - \hat{\rho}_{i,0} - \hat{\rho}_{i,1}X_{i,t-1}. \quad (14)$$

Nagel argues that the virtue of complexity is a spurious symptom of persistent predictors. We see on the right of Figure 11 that virtue of complexity arises even when predictors are stripped of their persistence. As in the preceding example, the magnitude of the complex model outperformance is smaller as we have purged part of the useful information in predictors.

In Figure 12 we train the deep network specification of 3.3.1 using the residualized returns of (13) (shown in the left panel) and the residualized predictors of (14) (shown in the right panel). Again, we see the virtue of complexity in deep nets even with alternative data inputs.

3.3.4 Non-Gaussian Random Features

Nagel (2025) argues that when predictors are persistent and training windows are small,

²⁸We do not modify the two predictors whose estimated AR(1) coefficient below 0.2.

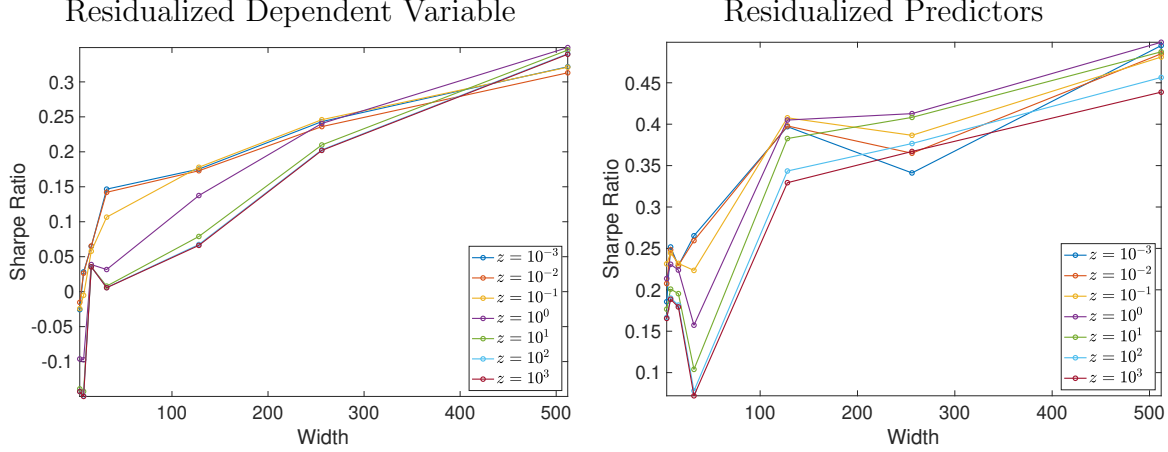


Figure 12: Deep Neural Networks With Modified Data Inputs

Note. The figure repeats the analysis of Figure 11 using alternative data inputs, but replaces the RFF functional form with the deep neural model of Section 3.3.1.

return forecasts tend to look like a combination past returns with positive weights. This is a feature of the Gaussian kernel that underlies random *Fourier* features. But it is not a property of random features models more generally, especially for kernels that are non-smooth at zero.²⁹

In light of this, we study random binning features (RBF), which can be thought of as a random feature analogue of tree-based models.³⁰ Like a tree model, the RBF feature map is non-differentiable, thus its behavior can differ rather drastically from RFF. Figure 13 finds Sharpe ratios increasing in RBF complexity (left panel) and demonstrates that the RBF model produces differentiated behavior and is significantly additive to KMZ’s RFF model.

²⁹The intuition is the following. Let $K(x, \tilde{x}) = f(\|x - \tilde{x}\|^2)$ and suppose that $\tilde{x} = x + \delta$, where δ is small. Then,

$$K(x, \tilde{x}) = f(\|\delta\|^2) \approx f(0) + f'(0)\|\delta\|^2. \quad (15)$$

For many kernels, such as the commonly used Laplace kernel, $f'(0) = \infty$. Hence, even small changes in x produce large responses in K . See, [Cheng and Singer \(2013\)](#); [Do and Vu \(2013\)](#) and [Favero et al. \(2021\)](#); [Cagnetta et al. \(2023\)](#) who also argue that the success of modern convolutional nets is partially driven by this property that non-smooth kernels are able to discriminate outcomes based on small variations in the input data.

³⁰See [Rahimi and Recht \(2007\)](#) for an overview of the algorithm. Our specific implementation fixes $T = 12$ and varies the parameter R from 1 to 50, which governs the parameterization of the RBF model.

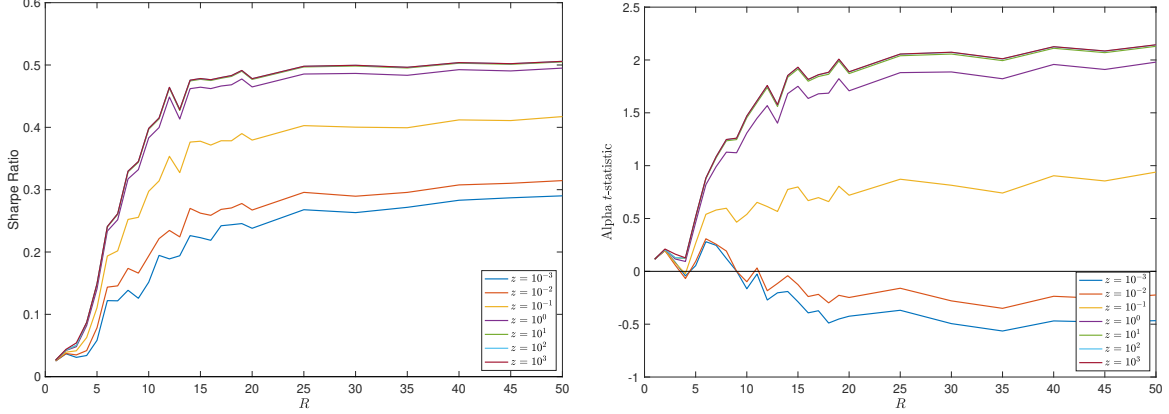


Figure 13: KMZ Analysis With Random Binning Features

Note. The left panel plots the performance of the complex RBF model’s out-of-sample Sharpe ratio as a function of complexity. The right panel shows the alpha t -statistic of this model versus the original KMZ model using RFF.

3.3.5 Semi-parametric Momentum

Nagel argues that the complex model of KMZ merely recovers a sophisticated form of momentum. In this section we argue that if one wanted to explicitly devise a momentum strategy, there would *still* be a virtue to building momentum with a complex model.

Defining a momentum strategy as a weighted average of recent 12 months returns,

$$\mu_t^{\text{MOM}} = w_{1,t}Y_t + w_{2,t}Y_{t-1} \dots + w_{12,t}Y_{t-11}, \quad (16)$$

we design a semi-parametric momentum model that learns weights on lagged returns as a function of KMZ’s usual Goyal-Welch predictor variables, X_t :

$$w_{j,t} = \sum_{i=1}^{P/24} (\beta_{j,i,1} \sin(\gamma \omega'_i X_t) + \beta_{j,i,2} \cos(\gamma \omega'_i X_t)), \quad \omega_i \sim iidN(0, I). \quad (17)$$

Substituting these weights into equation (16), we can estimate the semi-parametric momentum model as a random feature ridge regression where the regressors are lags of returns

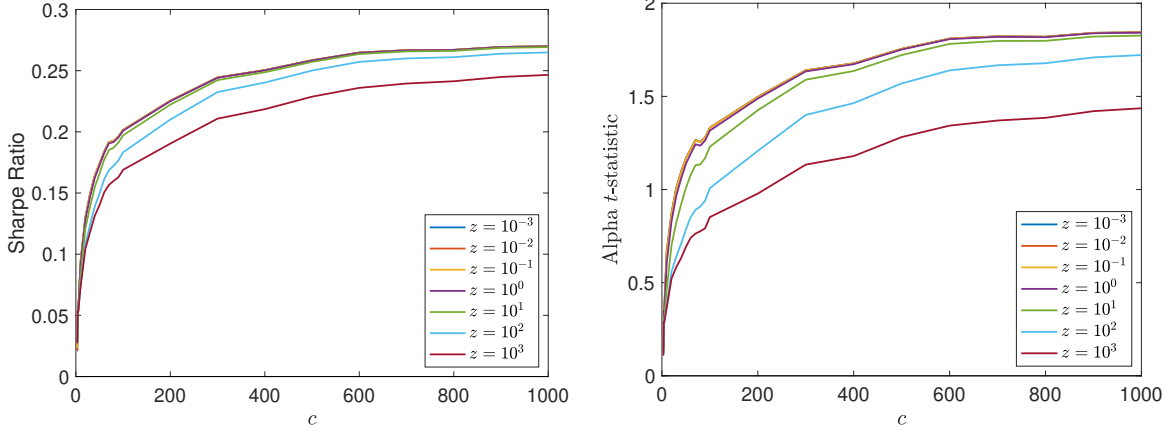


Figure 14: Semi-parametric Momentum and Complexity

Note. The left panel plots the semi-parametric momentum model’s out-of-sample Sharpe ratio as a function of complexity. The right panel shows the alpha t -statistic of this model versus Nagel’s momentum strategy.

with each RFF. This produces P regression coefficients: $(\beta_{j,i,1}, \beta_{j,i,2})$ for $j = 1, \dots, 12$ and $i = 1, \dots, P/24$. The out-of-sample performance of this strategy as a function of model complexity is shown in the left panel of Figure 14. The virtue of complexity clearly emerges even in the confines of momentum modeling.

For better or worse, the random features formulation in equation (17) does not ensure that momentum weights on lagged returns are all positive. Since Nagel emphasizes the positivity of weights in his momentum strategy, we also consider a deep “neural momentum” architecture:

$$w_{j,t} = \frac{\exp\left(\beta'_j X_t^{(2)}\right)}{\sum_{i=1}^{12} \exp\left(\beta'_i X_t^{(2)}\right)}, j = 1, \dots, 12, \quad (18)$$

where

$$X_t^{(l)} = \tanh(W^{(l)} X_t^{(l-1)}), l = 1, \dots, 4 \text{ and } X_t^{(0)} = X_t.$$

This is a four-layer neural network for momentum weights that enforces non-negativity of

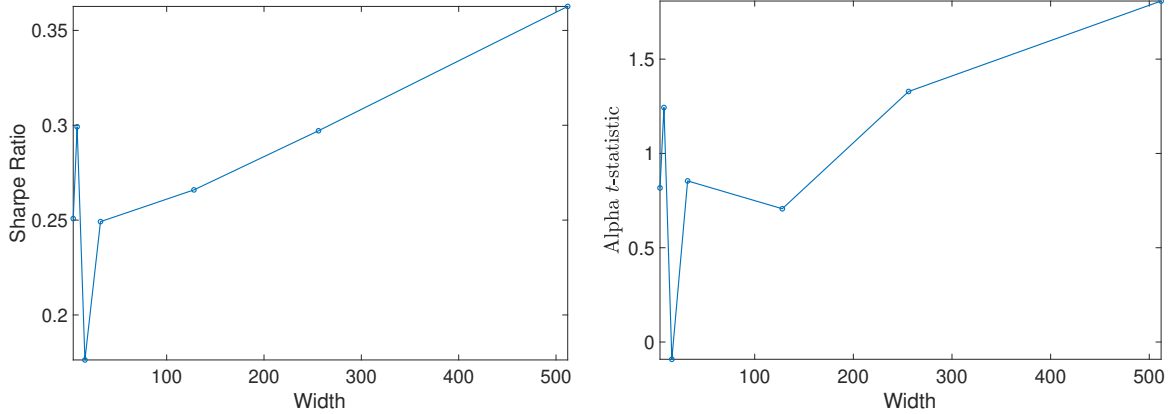


Figure 15: Neural Momentum

Note. The left panel plots the neural momentum model’s out-of-sample Sharpe ratio as a function of network width (the number of neurons in its hidden layer). The right panel shows the alpha t -statistic of this model versus Nagel’s momentum strategy.

weights with a softmax output layer. The original Goyal-Welch predictors are passed through four hidden layers of K neurons, with input coefficient matrices $W^{(l)}$ ($K \times 15$) and hyperbolic tangent activation function. The output from the last hidden layer is then mapped into momentum weights $w_{j,t}$ on the j^{th} most recent return. We alter the complexity of the model by varying the number of neurons $K = 4, 16, 32, 64, 128, 256$, or 512.

For this specification we again find performance increasing with complexity, though the alpha versus KMZ is only marginally significant.

3.4 The Virtue of Ensemble Complexity

The evidence presented for the model specifications in Section 3.3 indicates that the virtue of complexity emerges in a wide variety of models. Panel A of Table 1 shows that trading strategies from these models are substantially different than one another. Their average pairwise correlation is only 0.29.

Even more interestingly, these models are generally additive to one another. In Panels B and C of Table 1 we report pairwise alpha tests of each row model versus each column

Table 1: Comparison of High Complexity Models

Panel A reports correlation of timing strategy returns for the models presented in Section 3.3. Panels B and C report alphas and their t -statistics from pairwise tests of the row model versus the column model. “Mkt.” is the volatility managed market return. “Resid.” refers to models forecasting market return orthogonalized the Nagel’s momentum strategy. “Pred. AR” refers to models using AR(1) innovations in Goyal-Welch predictors.

	Mkt.	KMZ	Nagel Mom.	Deep NN	MSRR RFF	MSRR NN	Resid. MSRR	Pred. AR MSRR	Resid. NN	Pred. AR NN	RBF	RFF Mom.	Neural Mom.
Panel A: Correlation													
Mkt.	1.00	0.36	0.35	0.52	0.37	0.54	0.07	0.23	-0.12	0.25	0.41	0.15	-0.02
KMZ		1.00	0.64	0.34	0.82	0.31	0.55	0.47	0.03	0.18	0.68	0.53	0.38
Nagel Mom.			1.00	0.30	0.63	0.25	0.15	0.54	0.07	0.21	0.82	0.23	0.48
Deep NN				1.00	0.26	0.75	0.11	0.18	0.49	0.27	0.40	0.22	0.10
MSRR-RFF					1.00	0.25	0.47	0.44	0.00	0.19	0.64	0.41	0.34
MSRR NN						1.00	0.12	0.13	0.40	0.25	0.37	0.17	0.09
Resid. MSRR							1.00	0.06	0.06	0.05	0.32	0.32	0.29
Pred. AR MSRR								1.00	0.04	0.15	0.43	0.12	0.20
Resid. NN									1.00	0.03	0.07	0.01	0.12
Pred. AR NN										1.00	0.25	0.05	0.03
RBF											1.00	0.33	0.67
RFF Mom.												1.00	0.21
Neural Mom.													1.00
Panel B: Annualized Alpha (row model vs. column model)													
Mkt.	-	0.04	0.04	0.02	0.03	0.02	0.05	0.05	0.06	0.04	0.03	0.05	0.05
KMZ		0.03	-	0.02	0.00	0.03	0.04	0.04	0.05	0.04	0.01	0.04	0.04
Nagel Mom.			0.02	0.02	0.01	0.03	0.04	0.03	0.04	0.03	0.00	0.04	0.02
Deep NN				-	0.05	0.02	0.07	0.07	0.06	0.06	0.05	0.06	0.07
MSRR-RFF				0.04	-	0.04	0.05	0.05	0.06	0.05	0.02	0.05	0.05
MSRR NN				0.04	0.05	-	0.06	0.06	0.06	0.06	0.05	0.06	0.06
Resid. MSRR			0.01	0.01	-0.01	0.01	-	0.01	0.01	0.01	0.00	0.01	0.00
Pred. AR MSRR			0.00	0.01	-0.01	0.01	0.02	-	0.02	0.01	0.00	0.02	0.01
Resid. Mkt. NN			0.02	-0.01	0.02	0.00	0.02	0.02	-	0.02	0.02	0.02	0.02
Pred. AR NN			0.03	0.02	0.03	0.02	0.04	0.03	0.04	-	0.02	0.04	0.04
RBF			0.02	0.02	0.01	0.03	0.05	0.04	0.05	0.04	-	0.04	0.03
RFF Mom.			0.02	0.01	0.00	0.01	0.02	0.02	0.03	0.02	0.01	-	0.02
Neural Mom.			0.02	0.03	0.02	0.03	0.03	0.03	0.03	0.04	0.00	0.03	-
Panel C: Alpha t -statistic (row model vs. column model)													
Mkt.	-	3.61	3.88	1.90	3.26	1.96	4.97	4.76	5.36	4.31	3.34	4.72	5.09
KMZ		-	2.64	2.42	0.25	2.73	4.63	4.23	4.49	3.98	1.75	3.86	3.52
Nagel Mom.			-	2.02	0.68	2.45	3.80	3.56	3.79	3.26	0.04	3.47	2.61
Deep NN			5.58	-	5.26	2.78	6.44	6.32	6.23	5.80	5.00	6.15	6.20
MSRR-RFF			3.76	3.74	-	3.89	5.43	5.13	5.38	4.78	2.98	4.81	4.47
MSRR NN			4.10	5.01	4.97	-	6.11	6.04	5.78	5.50	4.75	5.88	5.91
Resid. MSRR			0.90	-1.50	-1.43	0.52	-	1.16	1.12	1.07	-0.29	0.52	0.28
Pred. AR MSRR			0.58	-0.44	-0.68	0.94	1.67	-	1.66	1.23	-0.31	1.46	1.09
Resid. Mkt. NN			2.88	2.11	2.26	-0.22	2.19	2.20	-	2.15	1.91	2.24	1.85
Pred. AR NN			2.25	2.63	2.42	1.89	3.38	3.21	3.37	-	2.30	3.33	3.33
RBF			2.93	2.25	1.69	2.63	4.61	4.45	4.62	4.02	-	4.21	3.35
RFF Mom.			1.64	-0.01	0.24	1.34	2.12	2.21	2.39	2.25	0.89	-	1.73
Neural Mom.			3.46	1.79	1.63	2.80	3.16	3.11	3.13	3.28	0.23	2.95	-
Panel D: Tangency Portfolio Weights (no shorting constraint)													
Tangency	0.08	0.00	0.00	0.26	0.23	0.17	0.00	0.00	0.00	0.10	0.00	0.00	0.16

model. The average pairwise alpha is 2.9% per annum (on a 10% volatility scale) and the average t -statistic is 2.8. The alpha is positive for 94% of the pairs, and the alpha has a t -statistic of at least 2.0 for 67% of pairs.

To compare models jointly rather than pairwise, we report the no-shorting tangency portfolio of all the machine learning models studied in Section 3.3 (all strategies are scaled to the same sample volatility). We find that the tangency portfolio puts positive weight on six of the 13 models, shown in Panel D of Table 1. Individually, each of these models is highly complex. The tangency portfolio is an ensemble model. The combined parameterization is the sum of the parameterizations of all models with positive weights. Evidently, the virtue of complexity documented for each individual specification in Section 3.3 is an understatement, as model performance is further enhanced by expanding complexity via ensembling.

The fact that there is no clearly dominant machine learning specification appears puzzling at first brush. The explanation of this phenomenon is rooted in limits to learning. To aid our discussion, Figure 16 imagines a 2-dimensional space of models (represented by the plane), and rising from the plane is a surface that describes models’ expected out-of-sample performance. The highest point on the surface corresponds to the “true” model. Consider specifying a model and initializing it with a parameter guess—this is represented as a point on this plane, say point A . At its initial parameter guess, the model is weak, but as we work through its gradient descent, model A learns from the training data and migrates in the direction of the true model. In traditional asymptotic theory, when the training data set is large and the model has no specification error, this process works its way to the top of the hill to recover the truth. When data is scarce relative to the number of parameters, there is no such convergence. The model improves through training, but it falls short of the infeasible best model due to limits to learning. The training process runs out of fuel (i.e., data) before reaching the peak.

Next, imagine initializing other models, say at points B and C , and performing this

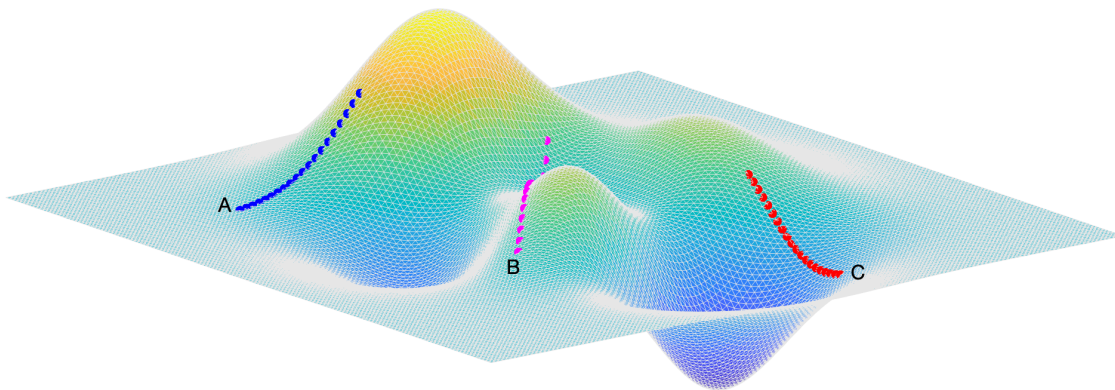


Figure 16: Limits To Learning and Ensembles

Note. Conceptual illustration of limits to learning with multiple models.

same training procedure. Again, these models migrate toward the truth, but due to limits to learning they too fall short. Interestingly, A , B , and C all arrive at models with reasonable out-of-sample performance, but they are all different! Naturally, these models will be correlated with each other; after all, they were trained on the same data. But because they began with different “perspectives” on the data, they learned different things from the data, and in essence specialized in the perspectives that they were each exposed to.³¹

To make this effect of limits to learning relatable we have illustrated it in a low-dimensional setting. In reality, machine learning models work through this training process in an extraordinarily high-dimensional space. Even when all models are universal approximators (such as neural networks) and have the capacity to represent the true model if given sufficient training data, different approximators learn different things from the same data. In fact, it is the heavy parameterization of universal approximators that give rise to complexity (data scarcity relative to the number of parameters), which in turn constrains learning and produces model specialization. In the end, no one model rules them all, and there are benefits to combining models in an ensemble.

³¹Notice that in a data rich environment, all three models would have converged on the truth, thus all would have the same behavior out-of-sample.

The many varieties of complex models in Section 3.3 is the tip of the iceberg. There are endless variations of models that one could ensemble. Beyond those we reported, we also find beneficial model differentiation when using different training windows, different shrinkages, and even with different P in the baseline KMZ model.

4 Further Responses

4.1 Nagel (2025)

The preceding sections address the central issues of Nagel (2025). In this section we address a few of his more minor comments.

4.1.1 Convergence of RFF to “Low Complexity” Kernel

Nagel spends much of Section 2 arguing that RFF behaves like a Gaussian kernel (though this is an old and well known result, e.g. [Rahimi and Recht, 2008](#), and is already discussed by KMZ). It is odd, however, for Nagel to refer to the Gaussian kernel as a low complexity model. Perhaps this is because the Gaussian kernel can be written compactly with a few parameters, giving an impression of simplicity. But this conflates algebraic simplicity and geometric simplicity. As discussed in detail in Section 2, counting parameters can be misleading, and the Gaussian kernel is a good demonstration of this. Regardless of its “nominal” parameter count, a critical aspect of well known machine learning models is their ability closely fit (in some cases, perfectly fit) any given training data set. One formal definition of parameter count (though not the only possible definition) is the following:

Definition 1 *A model has P parameters if it can perfectly match P training data observations.*

According this definition, the Gaussian kernel is an infinitely parameterized model. This is easy to see from the fact that the Gaussian kernel has the capacity to exactly fit all

observations in a training sample of arbitrary size. Another way to see that the Gaussian kernel is infinitely parameterized is simply to appreciate the [Rahimi and Recht \(2008\)](#) limit theorem. An RFF model with P parameters can exactly fit a data set of size P —and when the number of parameters P in the RFF becomes large it converges to the Gaussian kernel. Nagel’s characterizes this convergence as a “collapse” to a simple kernel model. In fact the convergence as $P \rightarrow \infty$ is smooth and it remains smooth even in the limit as $T \rightarrow \infty$ as long as $P/T \rightarrow \infty$. The Gaussian kernel is not a collapsed model—it remains an infinitely parameterized model in the limit.

Finally, the convergence of the RFF model to a Gaussian kernel is a mathematical law. This need not be investigated empirically. Above all, Nagel’s alpha comparisons of the large P RFF model versus the Gaussian kernel model is redundant. These models must “explain” each other to a close approximation by construction.

KMZ (and the statistical complexity literature more broadly, e.g. [Mei and Montanari, 2022](#)) do not use RFF in their empirical work because they believe it is a good model specification per se. They use it because it is a tractable model for analyzing complexity in which P can vary holding all else fixed (kernel regression does not allow this). RFF is just a surrogate for learning about complexity in the more interesting (but less tractable) setting of deep networks.

4.1.2 Bias Resulting From Volatility Standardization

Nagel argues that by volatility-standardizing market returns before running predictive regressions, they are introducing an upward bias in their alpha estimates. We address this in two ways.

First, Nagel’s “bias” proof does not show a bias. Instead it proves there is genuine alpha. To see this, Nagel assumes

$$r_{t+1} = \mu + \sigma_t e_{t+1}$$

where the conditional mean return μ is constant and conditional return volatility σ_t is predictable. He argues that the volatility-scaled return

$$f_{t+1} = r_{t+1} \frac{\mu}{\sigma_t}$$

is predictable only because σ_t is predictable, thus any evidence that f_{t+1} is predictable is an artifact of predictability in σ_t . While it is true that all predictability comes through σ_t in this example, this does not imply that alpha is biased. If μ is fixed and σ_t is predictable, then f_{t+1} has a higher unconditional Sharpe ratio than r_{t+1} ! This is because it takes more risk when the conditional Sharpe ratio ($\frac{\mu}{\sigma_t}$) is high and less when it is low. Alpha compares mean-variance efficiency of two trading strategies, it does not just compare their means. The alpha calculation in Nagel’s equation A.8 is not a proof of bias, but a proof that when means are constant and volatility is predictable, you earn a higher sharpe ratio from volatility-standardized returns than from raw returns.³²

Second, KMZ show that volatility standardization is not the main driver of the results in KMZ.³³ Their appendix Figure IA.6 reports VoC curves using raw market returns and shows that their conclusions are unaffected by using raw returns.

We believe that volatility standardization is the best choice for the KMZ analysis for the following reasons. Volatility-managing the market (Moreira and Muir, 2017) is a simple strategy that improves over buy-and-hold market returns. In KMZ’s sample, the volatility-managed market has a Sharpe ratio of 0.5 (versus 0.4 for the raw market) and has an alpha t -statistic of 2.6 versus the raw market. If KMZ *did not* standardize market returns, we might be worried that the complex model was just re-learning Moreira and Muir (2017). Next, heteroskedasticity is confounding to statistical analyses in general. We prefer first

³²This setup—more predictability in conditional volatility than conditional means—is essentially the mechanism behind the empirical discovery of Moreira and Muir (2017).

³³Nagel is clear about this point, noting “much of the predictability captured by the RFF is due to something else, not the above mechanical effect related to standardization.”

to understand empirical phenomena after purging data of heteroskedasticity (to the extent possible) and then move on to understanding any additional role that heteroskedasticity might be playing. In addition, volatility-standardized returns align more closely with KMZ’s theoretical environment, and the main point of the KMZ empirical analysis is to investigate their theoretical predictions. Finally, volatility standardized returns justify our use of return predictions as timing portfolio weights.³⁴

4.1.3 Critique of Complexity in Cross-Sectional Models

Nagel claims that his arguments regarding KMZ’s complexity analysis in the univariate return setting extends to the cross-sectional analysis in DKKM’s “AIPT.” This claim is speculative and Nagel presents no empirical evidence to support his speculation. Furthermore, the conditions that Nagel focuses on regarding KMZ (small T and persistent predictors) are far less relevant for DKKM. DKKM focuses on much larger T than KMZ (the training sample is 360 months in their main analysis) and uses more than 130 stock-level predictors. Many of these predictors have low persistence, including the usual suite of past-return variables as well as changes in various accounting ratios and metrics (see [Jensen et al., 2023](#), for detailed information on the predictors used by DKKM). The remaining aspects of Nagel’s DKKM critique are addressed by the same conceptual points that we lay out in Sections 2 and 3.

In the univariate time series setting of KMZ, the gains from complexity are modest relative to simpler models (see their comparison versus the linear “kitchen sink”). The picture is very different in cross-sectional models. In the cross section, complex models dominate standard simple factor models like [Fama and French \(2015\)](#) by a large margin.

³⁴In particular, the return to our timing strategy is

$$\frac{r_{t+1}}{\sigma_t} E_t \left[\frac{r_{t+1}}{\sigma_t} \right] \approx r_{t+1} \underbrace{\frac{E_t[r_{t+1}]}{\sigma_t^2}}_{\text{Markowitz}},$$

where $E_t[r_{t+1}]/\sigma_t^2$ is the Markowitz optimal timing weight given conditional mean $E_t[r_{t+1}]$ and conditional volatility σ_t^2 .

This has been demonstrated repeatedly in the literature, most recently in the transformer specification of [Kelly et al. \(2025a\)](#) which integrates a large language model architecture into a cross-sectional asset pricing model.

4.1.4 MA(2) Reversal Simulation

Nagel presents the following simulation result:

“I generate artificial return data by adding a observations from a simulated MA(2) process with strong negative autocorrelation to the actual market index return data. When I feed this artificial return series into the high-complexity ridgeless regression with RFF, it still produces weights on recent past returns that are positive, and hence effectively still a volatility-timed momentum strategy, even though the returns data now features reversals, not momentum. As a consequence, the strategy earns negative abnormal returns out-of-sample. This shows that the high-complexity ridgeless regression with RFF does not learn from the training data about the presence of momentum or reversal effects.”

Nagel’s simulations only modify the returns data; the Goyal-Welch predictors are unchanged. Because the MA(2) reversal pattern is not part of the real world data, we should not expect the model to be able to detect it with the real world Goyal-Welch predictors. In fact, it is reassuring that the model does not find reversal, as this would be evidence of RFF producing spurious predictability. It is good news that RFF does not “work” in this simulation.

The bad news is that the RFF model continues to build persistent forecasts, even though the simulated target return possesses a strong reversal effect. The ideal result from the perspective of KMZ would be that the model finds nothing—that it realizes there is no predictability to be found in the simulated returns data using the Goyal-Welch predictors. But it is not surprising that the model builds a persistent forecast. This is really all it

can build given the predictors that it has access too. And the fact that it fails to find the true model with no predictability is a consequence of limits to learning (there is not enough data for the model to find the truth). These two facts—the predictors are persistent and the training set is small—explains why RFF suffers in Nagel’s simulation. But Nagel’s conclusion that “*This shows that the high-complexity ridgeless regression with RFF does not learn from the training data about the presence of momentum or reversal effects*” is neither supported nor refuted by this analysis.

4.2 Buncic (2025): Aggregation, Intercepts, and Flimsy Logic

Buncic (2025) levies two main criticisms of KMZ. The first is that the virtue of complexity is an artifact of the way that KMZ aggregate their model performance metrics. The second is that a linear model with an intercept outperforms the complex models presented by KMZ. In this section we explain why both of these claims are wrong in fairly obvious ways.

4.2.1 Aggregation of Performance Metrics

The virtue of complexity is a statement about the average out-of-sample performance of high complexity models (those with $P \gg T$) to low complexity models ($P \ll T$) holding the all else fixed. The theory of KMZ shows that complex models outperform simple models in terms of Sharpe ratio. KMZ construct an empirical counterpart to the *expected out-of-sample Sharpe ratio* in their theory.

To show the flaw in Buncic’s argument, we first revisit some notation from KMZ. A random feature regression of R_{t+1} given feature set X_t takes the form

$$R_{t+1} = \beta' S(X_t) + \epsilon_{t+1} \tag{19}$$

where $S(X_t)$ is a P -vector of “random features” with individual elements $i = 1, \dots, P$

$$S_{i,t}(X_t) = f(\omega_i' X_t), \quad \omega_i \sim iidN(0, \gamma I).$$

Each random feature $S_{i,t}$ is a weighted sum of predictors passed through a nonlinear activation function. With a sufficiently large number P of random features one can approximate a generic conditional expectation function $E_t[R_{t+1}|X_t]$ to arbitrary precision (in population). While one could use fixed parameters ω_i to construct nonlinear basis terms, one would then need to estimate these parameters. [Rahimi and Recht \(2007\)](#) point out that one can instead randomly generate ω_i and still achieve the same limiting (large P) reconstruction of $E_t[R_{t+1}|X_t]$.

Consider evaluating the behavior of a simple model with $P = 1$ (this case is sufficient for illustrating Buncic’s error). Because the nonlinear features are random, there are infinitely many possible models with $P = 1$. In order to estimate the expected out-of-sample Sharpe ratio of a $P = 1$ model, KMZ generate 1,000 models of the form (19) with $P = 1$, each with a different seed for the randomization of ω . They calculate the out-of-sample Sharpe ratio of each model, then average these results across the 1,000 seeds. Denote an out-of-sample return forecast from a $P = 1$ model with random seed $j = 1, \dots, 1000$ as $\hat{\pi}_t^{(j)} = \hat{E}_t^{(j)}[R_{t+1}|X_t]$. The trading strategy return studied by KMZ is $\hat{\pi}_t^{(j)} \times R_{t+1}$. The estimated Sharpe ratio of this strategy is

$$SR^{(j)} = \frac{\hat{E} \left[\hat{\pi}_t^{(j)} \times R_{t+1} \right]}{\hat{\sigma} \left[\hat{\pi}_t^{(j)} \times R_{t+1} \right]} \quad (20)$$

where $\hat{E}$ and $\hat{\sigma}$ are the sample mean and standard deviation over out-of-sample realized trading strategy returns. Finally, the KMZ estimate of out-of-sample Sharpe ratio for a

representative $P = 1$ model is

$$\overline{SR}^{KMZ} = \frac{1}{1000} \sum_{j=1}^{1000} SR^{(j)}. \quad (21)$$

To sum up, KMZ produce 1,000 different $P = 1$ trading strategies and report the average of the individual strategy Sharpe ratios. As KMZ claim, this indeed estimates the average Sharpe ratio of a model with $P = 1$.

Buncic suggests a different aggregation scheme. He argues that trading strategy returns should *first be averaged* over the 1,000 seeds with $P = 1$, and then performance should be estimated as the Sharpe ratio of this average trading strategy return.

In other words, the aggregation proposed by Buncic creates a single trading strategy as an ensemble of 1,000 different $P = 1$ models:

$$\left(\hat{\pi}_t^{(1)} + \dots + \hat{\pi}_t^{(1000)} \right) \times R_{t+1}. \quad (22)$$

Obviously, this ensemble is no longer a $P = 1$ model! Instead it uses $P = 1,000$ features and estimates 1,000 regression coefficients. While Buncic calls this a $P = 1$ model, in fact he has created a highly complex ensemble whose individual sub-models are simple. His Sharpe ratio calculation is

$$\overline{SR}^{Buncic} = \frac{\hat{E} \left[\left(\sum_j \hat{\pi}_t^{(j)} \right) \times R_{t+1} \right]}{\hat{\sigma} \left[\left(\sum_j \hat{\pi}_t^{(j)} \right) \times R_{t+1} \right]}. \quad (23)$$

Buncic argues that there is no VoC because his aggregation scheme results in a Sharpe ratio of similar magnitude to a single $P = 1,000$ model. Indeed, the out-of-sample Sharpe ratio of this ensemble is high—exactly because the ensemble is a complex model.³⁵

³⁵In fact, the ensemble of Buncic is equivalent to a particular form of ridge regression with a shrunk covariance matrix, where the off-diagonal blocks across groups of random features are shrunk to zero. In other words, Buncic inadvertently affirms the main argument of KMZ. Another way to describe Buncic's

Buncic’s error has a positive spin: It demonstrates that complexity can be virtuous even if arrived at by ensembling individually weak learners (see, e.g. [Schapire, 1990](#)). This is the conceptual underpinning of boosting and bootstrap aggregation (as commonly applied in tree algorithms, see e.g. [Friedman, 2001](#); [Breiman, 2001](#)). Recently, [Belkin et al. \(2019\)](#) demonstrates that the virtue of complexity also holds in ensembles and emphasizes that the size of the ensemble is a measure of model complexity.

In summary, Buncic’s aggregation scheme compares two different types of complex models. One is a complex ensemble of many simple models, the other is a single complex model. He finds that these two models (both of which are highly complex) produce similar out-of-sample performance. His analysis has nothing to say about the comparative performance of simple and complex models. Buncic concludes that “*KMZ’s empirical VoC results thus appear to be an artifact of aggregation choice.*” The logic underlying this conclusion is flawed and his conclusion is incorrect.

4.2.2 Including an Intercept

Buncic’s second main criticism is that KMZ omit an intercept in their RFF predictive regressions, and that including an intercept overturns their findings. This is incorrect. There are two errors in Buncic’s logic that lead to his false conclusion. First, he fails to appreciate that inclusion of an intercept introduces large sampling noise which disrupts one’s ability to draw clear inferences about the behavior of complex models. In other words, he merely asserts that omission of an intercept is an error without considering the merits of this design choice. Second, he compares models in terms of standalone Sharpe ratio, rather than comparing models in terms of additivity (e.g., alpha or information ratio) to appropriate benchmarks.³⁶

error is a failure to understand Jensen’s inequality. The Sharpe ratio is a nonlinear function and he reverses the order of averaging and function evaluation, which leads to his erroneous conclusion.

³⁶As an illustration of this, he states on p.11 “*The passive buy-and- hold strategy outperforms all RFF machine learning models.... From an investors perspective, once in the market, doing nothing is the better and cheaper investment strategy than forming dynamic market timing strategies based on forecasts from the proposed ‘complex’ RFF model.*” This is an elementary error. As KMZ show, their timing strategy has large

The root of Buncic’s logical flaw can be traced to his statement on p.15: “*The intercept term should never be shrunk in penalized regression, as it captures the unconditional mean of the data.*” This claim has no theoretical or empirical grounding. Sure, some textbook descriptions of ridge regression recommend demeaning the dependent variable and then adding the mean back to the ridge forecast, so that the intercept is not subject to shrinkage. But statistics textbooks are not often written with financial returns data in mind! There are several reasons that justify KMZ’s omission of an intercept:

1. In the financial setting of return prediction, the intercept is an average return which is known to have a high degree of estimation uncertainty, especially in short samples. Indeed, mean return estimates typically have larger sampling variation than regression slope estimates (e.g. [Merton and Samuelson, 1992](#)). In other words, the return prediction intercept likely has greater need of shrinkage than the regression slopes.
2. The monthly mean return is close to zero (0.007 in the KMZ sample). It is common in the financial econometrics (especially in variance and correlation models) to constrain the mean return to zero based on this being a favorable bias-variance tradeoff.
3. Most importantly, KMZ use a rolling window estimation scheme to form their out-of-sample forecasts. Return prediction based on the rolling intercept estimate is tantamount to time-series momentum. This phenomenon is well understood and KMZ deliberately seek to understand predictive patterns *above and beyond* well-known momentum effects (see KMZ Section VII).

Motivated by these considerations, KMZ fix their intercept to zero. This is indeed a constraint on the specification, but one justified by the desire to focus on the benefits

alpha to buy-and-hold, meaning that an investor is better off combining her buy-and-hold strategy with the timing strategy. This misguided focus on standalone Sharpe ratio leads him to build a linear model with an intercept that behaves very similarly to a vanilla momentum strategy. Had he thought more about benchmarking, he would have seen that the complex KMZ model remains a highly successful model, with or without an intercept, because it learns predictive patterns above and beyond momentum (and above and beyond buy-and-hold).

of complex predictive relationships between Goyal-Welch predictors and returns without obscuring their results with noisy intercept estimates and well-known momentum effects.

Next, we demonstrate that Buncic’s analysis fails to substantiate his conclusion empirically. We fix $T = 12$ and estimate the following prediction models and their associated timing strategies:

- “Linear - Int.”: Buncic’s preferred model specification which is linear in the Goyal-Welch predictors and includes an intercept.
- “Linear - No Int.”: The same linear model but with intercept constrained to zero.
- “KMZ - No Int.”: The exact $P = 12,000$ RFF model specification used in KMZ.
- “KMZ - Int.”: The same KMZ specification but including an intercept.
- “Momentum”: The linear model specification with no predictors, only an intercept.

The associated timing strategy is thus a 12-month momentum strategy.

As in KMZ, we consider ridge shrinkage of $z \in \{10^{-3}, 10^{-2}, \dots, 10^3\}$ applied *only* to slope coefficients (for models using an intercept, we do not shrink the intercept, following Buncic).

Table 2 reports model performance statistics. Panel A reports annualized out-of-sample Sharpe ratios. The key benchmark is momentum, whose Sharpe ratio is 0.43. With sufficient shrinkage, the linear model matches the Sharpe ratio of momentum, and with insufficient shrinkage the Sharpe ratio is essentially zero.

What is more telling is Panel B, which reports the correlation of each model with momentum. Evidently, the linear-intercept model achieves a decent Sharpe ratio by exactly replicating 12-month momentum (the linear model with the highest Sharpe ratio has a correlation of 100% with momentum).

What gives rise to this pattern? When there is heavy slope shrinkage, the non-intercept portion of the model (i.e. the Goyal-Welch predictor portion) has much smaller variation

Table 2: Comparison of Models With and Without an Intercept ($T = 12$)

Panel A reports the out-of-sample Sharpe ratio of various model specifications estimated in training windows of $T = 12$ months. Panel B reports the information ratio of each model versus the 12-month momentum strategy. Panel C reports the information ratio of each model versus both the 12-month momentum strategy and the linear model with an intercept. We report t -statistics in parentheses.

	Ridge Shrinkage						
	10^{-3}	10^{-2}	10^{-1}	10^0	10^1	10^2	10^3
Panel A: Sharpe Ratio							
Momentum	0.43	0.43	0.43	0.43	0.43	0.43	0.43
Linear - Int	-0.04	-0.02	-0.02	0.02	0.27	0.41	0.43
Linear - No Int.	-0.12	-0.07	-0.04	0.03	0.29	0.43	0.46
KMZ - No Int.	0.41	0.41	0.41	0.41	0.41	0.43	0.47
KMZ - Int.	0.47	0.47	0.47	0.47	0.47	0.47	0.46
Panel B: Correlation With Momentum							
Linear - Int	0.07	0.11	0.19	0.38	0.83	1.00	1.00
Linear - No Int.	0.03	0.11	0.17	0.38	0.84	0.99	0.98
KMZ - No Int.	0.38	0.38	0.38	0.38	0.38	0.40	0.50
KMZ - Int.	0.84	0.84	0.85	0.85	0.86	0.89	0.97
Panel C: Info. Ratio vs. Momentum							
Linear - Int	-0.07 (-0.62)	-0.07 (-0.67)	-0.10 (-0.96)	-0.15 (-1.43)	-0.15 (-1.42)	-0.14 (-1.30)	-0.14 (-1.29)
Linear - No Int.	-0.13 (-1.22)	-0.11 (-1.08)	-0.11 (-1.06)	-0.15 (-1.39)	-0.14 (-1.29)	0.01 (0.09)	0.21 (1.99)
KMZ - No Int.	0.27 (2.52)	0.27 (2.52)	0.27 (2.52)	0.27 (2.52)	0.27 (2.54)	0.28 (2.61)	0.30 (2.79)
KMZ - Int.	0.20 (1.87)	0.20 (1.87)	0.20 (1.87)	0.20 (1.87)	0.20 (1.86)	0.19 (1.81)	0.18 (1.69)

than the intercept (which is exactly the momentum signal). On the other hand, with very light shrinkage, a much larger portion of the linear model’s variation comes from the Goyal-Welch predictors. But because shrinkage is small, this component is too noisy to be useful. Therefore the low shrinkage linear model has a Sharpe ratio of about zero. Buncic’s preferred model ranges from being useless (when shrinkage is small) to being identically momentum (when shrinkage is large). This fact is further corroborated by Panel C which reports each model’s information ratio versus momentum (and the associated t -statistic).³⁷ The information ratio of Buncic’s model is negative even in cases where it has a reasonably high Sharpe ratio.

Recall that KMZ make an extensive comparison to the *linear model with no intercept*. This is a much more interesting benchmark because it helps isolate the *linear* usefulness of the Goyal-Welch predictors after incorporating shrinkage. That is, the high shrinkage “Linear - No Int.” results hone in on what can be learned from the predictors without the confounding and mechanical momentum effects of an intercept. The linear model with no intercept still has a high correlation with momentum, but it has a borderline significant information ratio versus momentum (as high as 0.21 with a t -statistic of 2.0). This model is a stronger benchmark for the complex model, and this is why KMZ impose a zero intercept constraint even when building their linear benchmark model.

The table also reports the performance of the original KMZ model (the highest complexity model they study with $P = 12,000$). Again the raw Sharpe ratio is about the same as momentum. But this is clearly a different model than momentum, its correlation with momentum is less than 50%. Its information ratio versus momentum is 0.3 with a t -statistic of 2.8.

Finally, the table explores the impact of including an intercept in the KMZ model (all

³⁷In a regression of the form

$$R_{\text{model 1}} = \alpha + \beta R_{\text{model 2}} + \epsilon,$$

the information ratio is defined as α divided by the standard deviation of regression residuals.

other aspects of the KMZ model are unchanged). The results here are unsurprising. The intercept introduces the expected mechanical momentum effect, and as a result the KMZ correlation with momentum roughly doubles. However, despite the confounding effects of intercept sampling noise and mechanical momentum effect, the KMZ model can still identify useful additive predictive information in the Goyal-Welch predictors. Its information ratio versus momentum is about 0.20 and its t -statistic is around 1.8. Clearly, the intercept masks part of the non-momentum predictability captured by the original KMZ model, but the complex model’s predictability is strong enough that it shows through despite the intercept’s adverse influence.

To summarize, just like in his “aggregation” critique, Buncic’s arguments regarding regression intercepts is flawed and leads him to incorrect conclusions regarding the behavior of complex models. Buncic fails to appreciate the adverse effects of sampling noise and momentum that arise from including an intercept without shrinkage. He misses this point because he does not benchmark models appropriately. Claims like *“I show that simple models perform much better than complex ones when milder and more reasonable ridge shrinkage is applied”* are misleading because they ignore the very large overlap of his preferred model with well-known benchmarks like time series momentum.

4.2.3 Searching for the Highest Sharpe Ratio

The goal of KMZ is *not* to find the model that delivers the best out-of-sample timing strategy Sharpe ratio. Instead, it is to demonstrate the potential benefits of highly parameterized models for empirical finance applications. The bulk of their demonstration is theoretical. They use a deliberately oversimplified empirical environment to show the mechanics of their theory in real data.³⁸ A critically important aspect of their empirical analysis is that their model comparisons are apples-to-apples—their simple and complex models have access to

³⁸Their paper is clear on this point, stating “We present empirical analyses that are direct empirical analogs to the theoretical comparative statics.... To make the conclusions from this analysis as easy to digest as possible, we perform our analysis in a conventional setting with conventional data.”

identical underlying information sets, use the same function class, and differ only in the richness of their parameterizations.

Misguidedly, Buncic takes KMZ’s proof-of-concept empirical work as a cue to seek out models with higher out-of-sample Sharpe ratios than those reported in KMZ.³⁹ Without reasonable process controls,⁴⁰ such an agenda leads to out-of-sample data mining and quickly degenerates into meaninglessness. KMZ do not claim that their complex model is the best model. Nor do they claim that a simple model cannot outperform a complex model. Different model types will have different degrees of specification error and unless they are nested one cannot know which has smaller specification error (as discussed in Section 1). KMZ use nested RFF regressions that place simple and complex models on equal information footing so that apple-to-apples comparisons can be made. A linear model might be a simple model, but next to KMZ’s RFF it is an orange.

4.3 Berk (2023)

Berk (2023) has four main complaints about KMZ. The first is that *“the partial equilibrium model in the article cannot be embedded in a rational expectations general equilibrium.”* Berk states that he proves this claim, but ultimately no such proof is supplied. In any case, this claim is falsified by Kelly et al. (2025b) who analyze a complex rational expectations equilibrium.

Second, Berk argues that KMZ’s tests are not truly out-of-sample *“because the predictors were identified from the full sample.”* KMZ indeed use a set of predictors that were identified in the past few decades. The goal of KMZ is to begin from a standard set of predictors that the profession agree are not “cherry picked,” hence they turn to predictors collected in the

³⁹In his own words, he claims to demonstrate *“that substantially simpler linear models...produce significantly higher Sharpe ratios.”*

⁴⁰Such controls might include evaluating models on a “secret” centralized test set that researchers cannot mine for out-of-sample gains, as discussed in the common task framework of Hellum et al. (2025).

survey of [Goyal and Welch \(2008\)](#). In any case, the entire point of [Goyal and Welch \(2008\)](#) is that these predictors show *no evidence of out-of-sample predictability* in simple linear models.

Third, Berk claims that KMZ models are not complex because they are subject to shrinkage. This is similar to Nagel’s (2025) point regarding effective complexity and is thus addressed in Section 2 above.

Fourth, Berk “*question(s) the underlying notion in the article that predictability, by itself, is of interest for financial economists.*” His point is that one cannot econometrically identify economic mechanisms from a purely statistical prediction model. We agree. We disagree, however, with the premise that statistical prediction is of no interest to financial economists. There are many important economic discoveries that trace back to atheoretical statistical models.⁴¹ We believe that prediction has an important place in economics even when (perhaps especially when) we don’t know yet how to interpret the predictive pattern (see [Kelly and Xiu, 2023](#), for more discussion of this point).

As an aside to one of his comments, Berk (2023) states, “*As the authors acknowledge, the underlying statistical theory in this paper was derived in statistics literature.*” This statement is incorrect. KMZ are very clear about their theoretical contributions and how they build on and differ from prior literature. For interested readers, we discuss the theoretical contributions of KMZ, DKKM, and their follow-on papers in Appendix B.

Finally, we draw a connection between complex financial models and economic theory. The object of interest in most machine learning asset pricing models is the unique, tradable stochastic discount factor, or SDF (as in DKKM, as well as [Cong et al., 2021](#); [Chen et al., 2023](#); [Kelly et al., 2025a](#), among others). On the other hand, economic theories often imply a non-tradable SDF represented in terms of agents’ marginal utilities.

Received wisdom based on the principle of parsimony suggests that a simple non-

⁴¹One example is early return prediction regressions that documented time variation in expected market returns ([Fama and French, 1988](#); [Campbell and Shiller, 1988](#); [Keim and Stambaugh, 1986](#), among others). These empirical papers did not identify specific mechanisms themselves, but their findings gave birth to an enormous literature investigating economic mechanisms for dynamic risk premia.

tradable SDF naturally leads to a simple tradable SDF, hence there is no need for statistical complexity. One of the early attempts to justify this wisdom is [Berk et al. \(1999\)](#), who show that in an economy with a simple non-tradable SDF, the tradable SDF can be approximated by a simple, linear three-factor model. However, recent research from [Back and Ober \(2025\)](#) shows that this approximation is extremely crude. Even in a model as simple as that of [Berk et al. \(1999\)](#), the tradable SDF requires a highly complex representation to properly approximate the non-tradable. Thus, complex predictive relationships can arise even in very simple economic models, and we need statistical tools to study such relationships.

4.4 [Cartea et al. \(2025\)](#)

[Cartea et al. \(2025\)](#) extend the analysis of KMZ to observable signals that are noisy versions of the true unobservable signals.⁴² Their theory satisfies our Assumption 4' below with a particular covariance structure.⁴³ Thus, their theoretical results, while rigorous and new, are a special case of Theorems 2 and 3 in the appendix.

While the theory of [Cartea et al. \(2025\)](#) is potentially interesting, their data experiments are less so. Instead of using real datasets to test their theory, they create an artificial dataset by adding noise to the Goyal-Welch predictors. Obviously if you add enough noise it will accumulate and destroy any model. Furthermore, their theoretical predictions—that the virtue of complexity eventually reverses when predictors are sufficiently noisy—are counterfactual and contradicted by the *real world data* results of KMZ. In addition, [Cartea et al. \(2025\)](#) only considers R^2 and not Sharpe Ratio in their theory.

⁴²In particular, the true DGP is $R_{t+1} = \beta' F_t + \varepsilon_{t+1}$, but the econometrician only observes $\tilde{S}_t = F_t + \delta_t$ where δ_t is noise.

⁴³We can rewrite the DGP as $R_{t+1} = (\beta^{(1)})' \tilde{S}_t + (\beta^{(2)})' \delta_t + \varepsilon_{t+1}$, with $\beta^{(1)} = \beta$, $\beta^{(2)} = -\beta$.

4.5 Fallahgoul (2025)

As Nagel (2025) notes, KMZ do not use the raw random Fourier features, but first normalizes them by their trailing volatility. Unfortunately, Nagel goes on to cite Fallahgoul (2025) to support his arguments.

Fallahgoul (2025) makes a trivial observation that with volatility scaling, the limiting $P \rightarrow \infty$ random feature model does not converge to the Gaussian kernel but to a data-dependent kernel. This undergraduate exercise is then advertised in Fallahgoul (2025) as:

Theorem 3.1 resolves the fundamental puzzles in high-dimensional financial prediction by revealing that claimed theoretical properties simply do not hold in practice. Kelly et al. (2024) develop their theoretical analysis assuming RFF converge to Gaussian kernels. Their random matrix theory characterization, effective complexity bounds, and optimal shrinkage formula all depend critically on this convergence. However, their empirical implementation employs standardization, which fundamentally alters the convergence properties, creating a notable difference between theory and practice.

Every single statement in this paragraph is incorrect. It is as if Fallahgoul (2025) has not looked at the theory of KMZ at all, because the claims he makes are completely at odds with the framework laid out by KMZ. Fallahgoul (2025) also reports some standard theoretical lower bounds for the approximation capacity of RFF (essentially taken from textbooks). As he himself explains, these bounds do not apply for $P \gg T$. In fact, his bounds do not even work asymptotically (in the limit as $T, P \rightarrow \infty$) when $P < T$.⁴⁴

⁴⁴His Assumption 2.3 cannot hold. Gaussian random features are bounded and real analytic. Hence, kernel eigenvalues converge to zero very quickly and $c_z \rightarrow 0$ as $P \rightarrow \infty$.

5 Extending Our Theoretical Understanding of Complex Models

In this section we extend the theoretical analysis of KMZ in order to discuss the variety of shapes for VoC curves that emerge under different data generating processes. Comparing empirical patterns in KMZ to theoretical predictions helps us learn about the DGP underlying market returns.

Our analysis follows Assumptions 1 through 5 of KMZ with the following modifications. The first alters Assumption 2 as follows.

Assumption 2' $S_t = \Psi_P^{1/2} X_t$, where Ψ_P is a nonnegative definite matrix and $X_t = (X_{i,t})_{i=1}^P$, where $X_{i,t}$ are iid with $E[X_{i,t}] = 0$, $E[X_{i,t}^2] = 1$, $E[X_{i,t}^{4+\delta}] < \infty$ for some $\delta > 0$.

KMZ's Assumption 2 is more general, but requires new RMT results developed for DKKM. The more restrictive Assumption 2' allows us to combine certain results of [Hastie et al. \(2022\)](#) with remarkable algebraic identities discovered in KMZ. Thanks to Assumption 2', we can significantly relax Assumption 4 of KMZ regarding the distribution of β coefficients. Rather than assuming betas are iid, we now only require:

Assumption 4' $\beta = P^{-1/2} \Sigma_\beta^{1/2} b$, $b = (b_i)$ has iid coordinates with $E[b_i] = 0$, $E[b_i^2] = 1$, $E[b_i^4] < \infty$ and Σ_β are uniformly bounded. Furthermore, $\Psi = \text{diag}(\lambda_i)_{i=1}^P + A$ where $\text{tr}(A)$ is uniformly bounded and λ_i are uniformly bounded.

Under these modified assumptions, we are able to characterize more completely the limiting behavior of bias and variance in the KMZ model. These are presented in Theorem 1. In Appendix C, we derive more general results under significantly more general Assumption 1, imposing essentially no structure on β .

Following [Kelly et al. \(2024b\)](#), we pick $P_1 < P$ and define the objects

$$\beta = \begin{pmatrix} \beta^{(1)} \\ \beta^{(2)} \end{pmatrix}, \quad \beta^{(1)} \in \mathbb{R}^{P_1}, \quad \beta^{(2)} \in \mathbb{R}^{P-P_1}, \quad S_t = \begin{pmatrix} S_t^{(1)} \\ S_t^{(2)} \end{pmatrix}, \quad \Psi = \begin{pmatrix} \Psi_{1,1} & \Psi_{1,2} \\ \Psi_{2,1} & \Psi_{2,2} \end{pmatrix}.$$

The ridge estimator in the mis-specified model is

$$\hat{\beta}(z; P_1) = (zI + \hat{\Psi}_{1,1})^{-1} T^{-1} \sum_{t=1}^T S_t^{(1)} R_{t+1}, \quad (24)$$

with corresponding return predictions $\hat{\pi}_t(z; P_1) = \hat{\beta}(z; P_1)' S_t^{(1)}$. Our goal is to characterize the behavior of the timing strategy return $R_{t+1}^\pi(z; P_1) = \hat{\pi}_t(z; P_1) R_{t+1}$, which requires the following objects. First, following DKKM, implicit regularization is captured by the term⁴⁵

$$Z_*(z; c; P_1) = \frac{z}{1 - c + czm_{P_1}(-z; c)}.$$

Next, following [Chen et al. \(2025a\)](#) we require the “limits to learning” gap, defined as

$$\mathcal{L}(z; c; P_1) = Z'_*(z; c; P_1) - 1.$$

This describes the distance between the trained complex model and the infeasible true model. It reflects the inherent constraints on how much can be learned with large models in finite samples (and independent of the structure of the β coefficients).⁴⁶

Now we can state our main theoretical result. We use $X \approx Y$ to denote the fact that $X - Y \rightarrow 0$ in probability.

⁴⁵The function $m(-z; c)$ is the unique positive solution to the fixed point equation

$$m_{P_1}(-z; c) = \frac{1}{1 - c + czm(-z; c)} m_{P_1} \left(\frac{-z}{1 - c + czm_{P_1}(-z; c)} \right), \quad (25)$$

where $m_{P_1}(z)$ is the *Stieltjes transform* of the $\Psi_{1,1}$ matrix,

$$m_{P_1}(z) = P_1^{-1} \text{tr}((-zI + \Psi_{1,1})^{-1}). \quad (26)$$

⁴⁶Under the above assumptions, $\Psi_{1,2}$ has no impact on the out-of-sample performance. In the Appendix (Theorem 3) we relax the above assumptions, in which case $\Psi_{1,2}$ impacts out-of-sample performance in an explicit, elegant manner.

Theorem 1 *We have*

$$E[R_{t+1}^\pi(z; P_1)] \approx \sum_{i=1}^{P_1} \frac{\lambda_i}{1 + Z_*(z; c; P_1)/\lambda_i} \beta_i^2 \quad (27)$$

and

$$\begin{aligned} E[(R_{t+1}^\pi(z; P_1))^2] &\approx \sigma_R^2 \Xi(z; P_1) + 2(E[R_{t+1}^\pi(z; P_1)])^2 \\ \Xi(z; c; P_1) &= \mathcal{L}(z; c; P_1) \sigma_{Res}^2(P_1) + \sum_{i=1}^{P_1} \frac{(\mathcal{L}(z; c; P_1) \lambda_i^{-2} + 1)}{(1 + Z_*(z; c; P_1)/\lambda_i)^2} \lambda_i \beta_i^2 \\ \sigma_{Res}^2(P_1) &= \sigma^2 + \sum_{i=P_1+1}^P \beta_i^2 \lambda_i. \end{aligned} \quad (28)$$

This result sheds new light on the theory of KMZ. Per our discussion in Section 2, the expected return to market timing, $E[R_{t+1}^\pi]$ captures the model’s bias, while variance is captured by the second moment of the timing strategy $E[(R_{t+1}^\pi)^2]$.⁴⁷ This depends on the eigenvalues of the predictor distribution (λ_i), the predictive coefficients associated with those predictors (β_i), and the amount of implicit shrinkage (Z_* , which also determines the Limits-to-Learning Gap $\mathcal{L}$). Implicit shrinkage is pinned down by the complexity of the model. The other two drivers λ_i and β_i ($i = 1, \dots, P_1$) are properties of the DGP, and we focus our discussion on these. Together they determine how complexity translates into out-of-sample model performance, and thus determines the shape of VoC curves.

In this section, we show that two aspects of the DGP are key to understanding the role of complexity on model behavior. We refer to these as “concentration” and “alignment.” Concentration is about the distribution of eigenvalues in the predictor covariance matrix. When covariation is concentrated in a few dominant components it means that there is substantial redundancy among predictors. Eigenvalue concentration can be thought of as a form of sparsity or parsimony in the DGP, so it can be learned well with relatively few

⁴⁷The expected out-of-sample return is always smaller than that of the infeasible true model due to limits to learning. The higher $E[R_{t+1}^\pi]$, the smaller the bias.

parameters, all else equal. On the other hand, if covariation is more evenly dispersed among principal components, it means that many predictors contribute differentiated information. This is a complex setting in which low dimensional models will tend to suffer.

Alignment refers to the dependence between betas and eigenvalues. If predictors associated with large eigenvalues tend to have the largest betas, then the generating process has a high degree of alignment. If alignment is low (or perhaps negative), then even minor eigenvalues are important for prediction. Therefore, the less alignment the more prominent the virtue of complexity.

In this section, we use formulas from Theorem 1 to show how concentration and alignment drive the behavior of complex models. To do so we use the following calibration of our theory. We set $T = 100$, $P = 1000$, and we increase $P_1 = 1, 2, \dots, 1000$. We define $c = P_1/T$, and assume that the eigenvalue distribution is given by $\lambda_i = i^{-\alpha_\lambda}$, $i = 1, \dots, P$, where $\alpha_\lambda \geq 0$ is the concentration parameter. The larger α_λ the more concentrated the predictors are on a few common components. Next, we set $\beta_i = b_* \lambda_i^{\alpha_\beta}$, where $\alpha_\beta \in \mathcal{R}$ is the alignment parameter. The larger α_β , the more return predictability comes from the largest principal components.

Figure 17 plots Sharpe ratio VoC curves for various choices of α_λ and α_β .⁴⁸ The case shown in the upper left, $\alpha_\lambda = 0$ and $\alpha_\beta = 0$, is that originally reported in KMZ.⁴⁹ The remaining calibrations are available thanks to the generalization of KMZ's culminating in Theorem 1. The upper right shows the case $\alpha_\lambda = 0.1$ and $\alpha_\beta = 0.1$ corresponding to moderate concentration and alignment, and the Sharpe ratio curves differ minimally from those in KMZ. The middle row increases alignment to holding concentration fixed at $\alpha_\lambda = 0.1$. We see that the VoC curves again are largely unaffected when $\alpha_\beta = 1.0$ (middle left). When there is more concentration ($\alpha_\lambda = 1.0$) and moderate alignment, the VoC curves change shape (middle right). We see that the parsimony associated with higher concentration indeed

⁴⁸All calibrations fix $\sigma^2 = 1$ and $b_* = 0.2$. These only effect the scale of plots and not their shapes.

⁴⁹The shape of the Sharpe ratio curve in this calibration differs slightly from that of KMZ do to our modifications in model assumptions.

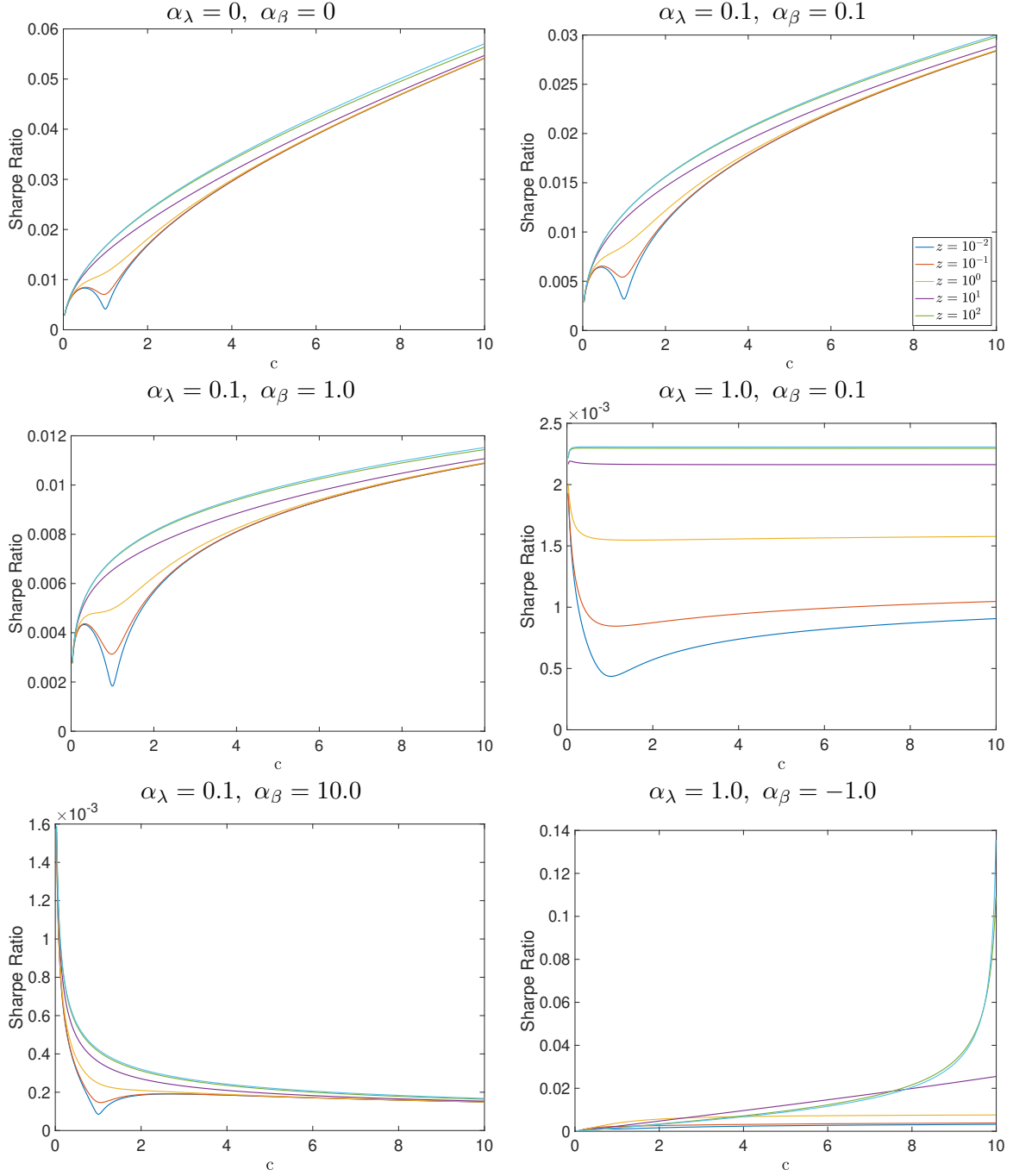


Figure 17: Concentration, Alignment, and Complexity

Note. The figure shows expected out-of-sample Sharpe ratios based on Theorem 1 under various scenarios for concentration and alignment in the return prediction data generating process.

translates into better performance of simple models. But somewhat surprisingly, even in this case complex models continue to perform well as long as appropriate shrinkage is used.⁵⁰ In the bottom left of the figure we experiment with extremely strong alignment ($\alpha_\beta = 10.0$). In this case, predictability is dominated by a small number of predictors and in essence the DGP is sparse. As a result, there is no benefit to using complex models. Finally, in the bottom right, we experiment with negative alignment ($\alpha_\beta = -1.0$). In this case, smaller components are stronger predictors, therefore there are enormous benefits to using complex models.

In summary Figure 17 demonstrates how the behavior of complex models vary with the nature of predictability in the data. Again this theory provides a complete characterization of model behavior in the high complexity regime in a more general environment than KMZ. We have focused our discussion here on Sharpe ratios, but Appendix 5 provides comprehensive theoretical results.

Lastly, the upward sloping Sharpe ratio curves in KMZ’s empirics (and also in Kelly et al., 2023; Didisheim et al., 2024) are consistent with relatively low degrees of concentration and alignment in return prediction data sets.

6 Conclusion: The Research Agenda Going Forward

Here we want to lay out what we believe are important topics for future work with complex models and machine learning in finance.

6.1 Enriching the Theory of Complex Models

While KMZ and DKKM have made early strides in characterizing the properties of complexity models, there remains enormous scope for expanding our understanding. Thus far, our theory works with high dimensional linear models. Which of these properties are preserved

⁵⁰See KMZ for a derivation of optimal ridge shrinkage in their model environment.

for more practically useful models including deep neural networks, and which properties need adjustment? These are challenging questions due to nonlinearities and the fact that complexity likely has different impact when added through network depth versus network width.

We typically interpret KMZ theory as describing parameters of a high dimensional output layer, holding the input layer fixed. An important and open theoretical question is how complexity is affected by the dimension of the inputs. As we increase the number of raw predictors in a shallow network, there can be a curse of dimensionality in that it takes many more parameters and data to learn from few inputs than from many inputs.

What are the finite sample (small- T) properties of complex models? In this paper we provide simulation evidence that KMZ’s large- T complexity theory accurately describes model behavior in small samples, but a general theoretical understanding of this phenomenon is lacking. This topic is particularly relevant in light of likely non-stationarities in financial markets data, which begs an agenda to understand how machine learning models can aid in the development of models that are robust to structural breaks and aid in their detection.

Relatedly, another gap in complexity theory is distribution theory of estimators. Recent work has started to make progress on statistical testing ([Chernov et al., 2025](#); [Liao et al., 2025](#)), but this remains an important and large area ripe for improvement.

6.2 Feature Learning

One of the conclusions in KMZ is that “Our results are not a license to add arbitrary predictors to a model. Instead, we recommend i) including all plausibly relevant predictors and ii) using rich nonlinear models rather than simple linear specifications. Doing so confers prediction and portfolio benefits, even when training data are scarce, particularly when accompanied by prudent shrinkage.” From where do these relevant predictors come? How should we design complex models if we are uncertain which predictors are useful and which

are not. Relatedly, if we have raw data that may contain useful predictor information but do not know how to design ideal predictors from this data, how can we incorporate the feature learning problem into the machine learning pipeline? If we take such an approach, how must the theory of complex models change? A fascinating example in this line of inquiry is [Liao et al. \(2023\)](#), who show that increasing model complexity by simply adding noise to the predictors set can produce counterintuitive improvements in model performance. Feature learning is likely to be a major growth area of financial machine learning in years to come, with an interesting example in this direction being [Li et al. \(2025\)](#).

6.3 Economic Mechanisms

So far the complex model research agenda has focused on statistical topics. But this is only a first step. As economists, we know division of labor is important, and this is true in scientific inquiry. Observe how the world deviates from theory, move to data collection and statistical modeling, then develop theories to explain the newly documented phenomena. Rinse and repeat. The ultimate goal is drawing economic conclusions, and approaches for economic inference in machine learning and asset pricing is perhaps the most important topic for research going forward.⁵¹

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⁵¹One bright spot outside of asset pricing is recent progress using machine learning to aid causal identification in empirical microeconomics (e.g. [Chernozhukov et al., 2018](#); [Künzel et al., 2019](#); [Athey and Wager, 2019](#), among others).

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Table 3: Comparison of Models With and Without an Intercept ($T = 120$)

Panel A reports the out-of-sample Sharpe ratio of various model specifications estimated in training windows of $T = 120$ months. Panel B reports the information ratio of each model versus the 120-month momentum strategy. Panel C reports the information ratio of each model versus both the 120-month momentum strategy and the linear model with an intercept. We report t -statistics in parentheses.

	Ridge Shrinkage						
	10^{-3}	10^{-2}	10^{-1}	10^0	10^1	10^2	10^3
Panel A: Sharpe Ratio							
Momentum	0.54	0.54	0.54	0.54	0.54	0.54	0.54
Linear - Int	0.24	0.21	0.26	0.49	0.61	0.56	0.55
Linear - No Int.	0.22	0.24	0.31	0.51	0.60	0.55	0.49
KMZ - No Int.	0.41	0.41	0.41	0.40	0.40	0.42	0.41
KMZ - Int.	0.50	0.50	0.50	0.50	0.51	0.55	0.56
Panel B: Correlation With Momentum							
Linear - Int	0.20	0.18	0.20	0.47	0.92	1.00	1.00
Linear - No Int.	0.24	0.23	0.27	0.52	0.93	1.00	0.95
KMZ - No Int.	0.27	0.27	0.27	0.27	0.28	0.34	0.38
KMZ - Int.	0.64	0.64	0.64	0.65	0.70	0.88	0.99
Panel C: Info. Ratio vs. Momentum							
Linear - Int	0.13 (1.16)	0.12 (1.05)	0.15 (1.37)	0.27 (2.39)	0.27 (2.40)	0.23 (2.08)	0.23 (2.03)
Linear - No Int.	0.09 (0.79)	0.12 (1.08)	0.18 (1.56)	0.26 (2.34)	0.25 (2.24)	0.07 (0.58)	-0.10 (-0.87)
KMZ - No Int.	0.27 (2.41)	0.27 (2.41)	0.27 (2.40)	0.27 (2.36)	0.25 (2.25)	0.25 (2.21)	0.22 (1.98)
KMZ - Int.	0.20 (1.75)	0.20 (1.75)	0.20 (1.75)	0.19 (1.71)	0.18 (1.56)	0.15 (1.33)	0.14 (1.22)
Panel D: Info. Ratio vs. Momentum and Linear - Int.							
Linear - No Int.	-0.10 (-0.90)	0.03 (0.23)	0.11 (0.99)	0.00 (0.01)	-0.02 (-0.14)	-0.07 (-0.61)	-0.12 (-1.02)
KMZ - No Int.	0.26 (2.29)	0.26 (2.30)	0.25 (2.26)	0.24 (2.13)	0.23 (2.07)	0.23 (2.05)	0.20 (1.81)
KMZ - Int.	0.20 (1.75)	0.20 (1.74)	0.19 (1.72)	0.18 (1.59)	0.15 (1.31)	0.12 (1.04)	0.10 (0.87)

A Additional Comments on Buncic (2025)

KMZ focused their analysis on $T = 12$ months (noting that this allows them to analyze very high complexity models with less computation time), but also conduct their empirical analysis with longer windows of $T = 60$ and $T = 120$ month. Buncic prefers to focus his analysis on longer training windows because here he finds higher Sharpe ratios for his preferred linear model.⁵² However, the same issues documented in Table 2 hold for larger training samples. In Table 3, we repeat our comparison of models with and without intercepts

⁵²This preference is driven by two logical errors. The first is Buncic’s claim that finding higher Sharpe ratios with larger T undermines the virtue of complexity—we dispel this argument in the previous section.

using $T = 120$. In this case, momentum is defined using the average return over the previous 120 months. This maintains the tight link between intercepts and momentum that are the key flaw in Buncic’s argument

For $T = 120$, the momentum Sharpe ratio rises to 0.54. Low and behold, the Sharpe ratio of the linear model with an intercept also rises to 0.55. But the correlation between the linear model and momentum remains close to 100% when shrinkage is large. Unlike $T = 12$, however, the linear model with an intercept has a significant alpha relative to momentum, indicating that with $T = 120$ this model is able to benefit from the predictive information in the Goyal-Welch data. In contrast, the original KMZ model has correlation of less than 40% with momentum, and its information ratio is around 0.27 and is significant. Finally, in Panel D of Table 3, I add a new analysis that compares KMZ to both momentum and Buncic’s linear-intercept model. Here we see that even after controlling for momentum and Buncic’s model, the KMZ information ratio is above 0.20 and is generally significant. Lastly, Table 3 shows that when $T = 120$, including an intercept in the KMZ model again drives the model to mimic momentum and partially obscures the complex model’s ability to leverage predictive information in the Goyal-Welch data.

B Complexity Theory Contributions

Here we provide a detailed list of theory contributions in KMZ and DKKM relative to the existing literature.

KMZ:

- Prove the VoC both for Sharpe Ratio and for R^2 . Sharpe Ratio differs from the literature in that it requires fourth-order moments of signals (regression only needs second moments). Even the VoC for R^2 result is novel. It has been known when $\Psi = I$ (Hastie et al. (2022)), but was never proved for generic eigenvalue distributions. The key is to first develop the RMT for mis-specified models (see next bullet).
- Develop RMT for the mis-specified case. It requires developing novel RMT techniques to handle generic omitted signals and generic covariance structures between observable and omitted signals.⁵³

DKKM:

- Develop RMT for the Markowitz problem and *managed portfolios*, which is an entirely new topic in the literature and requires additional techniques beyond the usual regression problem.

The second is that models should be compared in terms of Sharpe ratio levels, rather than compared based on additivity (alpha or information ratio), as discussed earlier in this section.

⁵³This generalizes the misspecified analysis of Hastie et al. (2022) who focus on the isotropic case with identity covariance.

- Introduce a novel approach to managing “three infinities:” A large number of signals, a large number of observations, and a large number of assets. Since DKKM construct factors as managed portfolios, they violate the common technical assumptions of [Hastie et al. \(2022\)](#) and [Kelly et al. \(2024b\)](#) because managed portfolios exhibit complex dependencies in higher-order moments.
- Develop novel mathematical RMT techniques to tackle the three infinities problem and compute asymptotic portfolio performance in the joint limit as $N, P, T \rightarrow \infty$. It turns out that only $c = \lim P/T$ matters! The rate at which $N \rightarrow \infty$ does not matter. This is striking and might explain why managed portfolios work so well in the high-dimensional settings.
- Prove that *managed portfolios always span the true SDF when there is enough complexity*: In the limit as $P \rightarrow \infty$, the cross-effects introduced by inverting a complex conditional covariance matrix vanish, and SDF converges to that where a stock portfolio depends only on its own signals. This is proved under the conditions in which there is no cross-stock predictability. [Kelly et al. \(2025a\)](#) extend this analysis with models that allow for cross stock predictability.

[Kelly et al. \(2024a\)](#)

- [Nagel \(2025\)](#) argues that for managed portfolios based on random features the limit is “simple” and is also given by a kernel. [Kelly et al. \(2024a\)](#) establish more than a year earlier that portfolios behave this way, but that these models are complex rather than simple. They call this the “Portfolio Kernel.” Relatedly [Nagel \(2025\)](#) calls for studying whether there is a link between KMZ, kernel methods and neural networks. [Kelly et al. \(2024a\)](#) already establish this link using a new object they have introduced called the “portfolio tangent kernel (PTK).”

C Theoretical Detail

This appendix includes detailed derivations for the results discussed in Section 5.

Assumption 1 *We have*

$$R_{t+1} = \beta_P' S_t + \varepsilon_{t+1}, \quad (29)$$

where both $S_t \in \mathbb{R}^P$ and ε_{t+1} are independent and identically distributed over time, and the processes $(S_t)_{t \geq 0}$ and $(\varepsilon_t)_{t \geq 0}$ are independent.

Furthermore, we assume that

- $E[\varepsilon_t] = 0$, $E[\varepsilon_t^2] = \sigma_\varepsilon^2$, and $E[\varepsilon_t^4] < \infty$.
- $S_t = \Psi_P^{1/2} X_t$, where Ψ_P is a nonnegative definite matrix and $X_t = (X_{i,t})_{i=1}^P$, where $X_{i,t}$ are iid with $E[X_{i,t}] = 0$, $E[X_{i,t}^2] = 1$, $E[X_{i,t}^{4+\delta}] < \infty$ for some $\delta > 0$.

- The sequence of matrices Ψ_P is uniformly bounded, $\|\Psi_P\| < K$ for some $K > 0$.
- $\|\beta_P\| \leq K$ for some $K > 0$.
- Either $E[X_{i,t}^4] = 3$ or the vector $\Psi_P^{1/2}\beta_P$ has all of its coordinates converging to zero: $\sup_i |(\Psi_P^{1/2}\beta_P)_i| = o(1)$ as $P \rightarrow \infty$.

Following [Kelly et al. \(2024b\)](#), we pick $P_1 < P$ and define the decomposition

$$\beta = \begin{pmatrix} \beta^{(1)} \\ \beta^{(2)} \end{pmatrix}, \quad \beta^{(1)} \in \mathbb{R}^{P_1}, \quad \beta^{(2)} \in \mathbb{R}^{P-P_1}, \quad (30)$$

as well as the corresponding decomposition for the covariance matrix Ψ :

$$\Psi = \begin{pmatrix} \Psi_{1,1} & \Psi_{1,2} \\ \Psi_{2,1} & \Psi_{2,2} \end{pmatrix} \quad (31)$$

We also define $S_t = \begin{pmatrix} S_t^{(1)} \\ S_t^{(2)} \end{pmatrix}$ and the ridge estimator in the mis-specified model, given by

$$\hat{\beta}(z; P_1) = (zI + \hat{\Psi}_{1,1})^{-1} T^{-1} \sum_{t=1}^T S_t^{(1)} R_{t+1}, \quad (32)$$

as well as the corresponding predictions

$$\hat{\pi}_t(z; P_1) = \hat{\beta}(z; P_1)' S_t^{(1)}. \quad (33)$$

Our goal here is to characterize the behavior of

$$R_{t+1}^\pi = \hat{\pi}_t(z; P_1) R_{t+1}. \quad (34)$$

We will need the following auxilliary object:

$$\tilde{\beta}_{P_1} = \Psi_{1,1}\beta^{(1)} + \Psi_{1,2}\beta^{(2)} = \text{Cov}(S_t^{(1)}, R_{t+1}). \quad (35)$$

Finally, following [Didisheim et al. \(2024\)](#), we introduce the implicit regularization parameter

$$Z_*(z; c; P_1) = \frac{z}{1 - c + czm_{P_1}(-z; c)}, \quad (36)$$

where $m(-z; c)$ is the unique positive solution to the fixed point equation

$$m_{P_1}(-z; c) = \frac{1}{1 - c + czm(-z; c)} m_{P_1} \left(\frac{-z}{1 - c + czm_{P_1}(-z; c)} \right), \quad (37)$$

where $m_{P_1}(z)$ is the *Stieltjes transform* of the $\Psi_{1,1}$ matrix,

$$m_{P_1}(z) = P_1^{-1} \text{tr}((-zI + \Psi_{1,1})^{-1}). \quad (38)$$

Theorem 2 (First Moment) *In the limit as $P_1, T \rightarrow \infty, P_1/T \rightarrow c$, we have*

$$E[R_{t+1}^\pi] - \tilde{\beta}'_{P_1}(Z_*(z; c; P_1)I + \Psi_{1,1})^{-1}\tilde{\beta}_{P_1} \rightarrow 0 \quad (39)$$

is probability.

Similarly to [Didisheim et al. \(2024\)](#), the behavior of managed returns is pinned down by the implicit regularization parameter Z_* that depends only on the eigenvalue distribution of $\Psi_{1,1}$. Strikingly, $E[R_{t+1}^\pi]$ coincides with that in the “classical” ($c = 0$) case, but with the true ridge penalty z replaced by the “effective” shrinkage Z_* . In the case when $\Psi = I$, this expression is particularly simple,

$$E[R_{t+1}^\pi] \approx \underbrace{\frac{1}{Z_*(z; c) + 1}}_{\text{implicit regularization}} \|\beta^{(1)}\|^2. \quad (40)$$

Mechanically, $\|\beta^{(1)}\|^2$ is always monotone increasing in P_1 : As we keep adding predictors, we capture more predictive power. At the same time, $Z_*(z; c)$ is also monotone increasing in c . These two competing mechanisms become particularly explicit when $z = 0$. In this case,

$$Z_*(0; c) = \max\{c - 1, 0\} \quad (41)$$

and we get, for $c > 1$, that

$$E[R_{t+1}^\pi] \approx c^{-1} \|\tilde{\beta}_{P_1}\|^2. \quad (42)$$

In the setting of [Kelly et al. \(2024b\)](#), $\|\tilde{\beta}_{P_1}\| \approx b_* P_1/P$ and, hence, the two channels exactly offset each other: $E[R_{t+1}^\pi | \hat{\beta}(z; P_1)] \approx (T/P)b_*$ is independent of P_1 .

We now derive the expression for the second moment. We will need the following quantities:

$$\begin{aligned} \sigma_R^2 &= \text{Var}[R] = \sigma^2 + \beta' \Psi \beta \\ \mathcal{V}_*(z; P_1) &= \tilde{\beta}' \Psi_{1,1} (\Psi_{1,1} + zI)^{-2} \tilde{\beta} \\ \sigma_{Res}^2(P_1) &= \sigma^2 + (\beta^{(2)})' (\Psi_{2,2} - \Psi_{2,1} \Psi_{1,1}^{-1} \Psi_{1,2}) \beta^{(2)} \\ \mathcal{V}_{*,Res}(z; P_1) &= \tilde{\beta}' \Psi_{1,1}^{-1} (\Psi_{1,1} + zI)^{-2} \tilde{\beta} \end{aligned} \quad (43)$$

Following [Chen et al. \(2025a\)](#), we introduce the Limits-to-Learning Gap (LLG) defined as

$$\mathcal{L}(z; c; P_1) = Z'_*(z; c; P_1) - 1. \quad (44)$$

As [Chen et al. \(2025a\)](#) explain, $\mathcal{L}(z; c; P_1)$ imposes inherent constraints on what can be learned in finite samples from the data independent of the structure of the β coefficients.

Theorem 3 (Second Moment) *In the limit as $P_1, T \rightarrow \infty, P_1/T \rightarrow c$, we have, in the sense of convergence in probability, that*

$$\begin{aligned} E[(R_{t+1}^\pi)^2] - \mathcal{V}(z; c; P_1) &\rightarrow 0 \\ \mathcal{V}(z; c; P_1) &= \sigma_R^2 \Xi(z; P_1) + 2(E[R_{t+1}^\pi])^2, \end{aligned} \quad (45)$$

where $Z_* = Z_*(z; c; P_1)$ and

$$\Xi(z; c; P_1) = \mathcal{L}(z; c; P_1) \left(\sigma_{Res}^2(P_1) + \mathcal{V}_{*,Res}(Z_*; P_1) \right) + \mathcal{V}_*(Z_*; P_1) \quad (46)$$

C.1 Preliminaries on Random Matrix Theory

We will need the Stieltjes transform and its derivative:

$$\begin{aligned} m(-z) &= \lim_{P \rightarrow \infty} P^{-1} \text{tr}((zI + \Psi_P)^{-1}) \\ m'_\Psi(-z) &= \lim_{P \rightarrow \infty} P^{-1} \text{tr}((zI + \Psi_P)^{-2}) \end{aligned} \quad (47)$$

and

$$\begin{aligned} \hat{m}(-z) &= P^{-1} \text{tr}((zI + \hat{\Psi})^{-1}) \\ \hat{m}'(-z) &= P^{-1} \text{tr}((zI + \hat{\Psi})^{-2}) \end{aligned} \quad (48)$$

Let also $m(-z; c)$ be the unique, positive solution to the fixed point equation⁵⁴

$$m(-z; c) = \frac{1}{1 - c + czm(-z; c)} m_\Psi \left(\frac{-z}{1 - c + czm(-z; c)} \right). \quad (49)$$

Many proofs in this paper are based on the classic “concentration of quadratic forms” lemma (see, e.g., [Kelly et al. \(2024b\)](#)).

Lemma 1 *Suppose that $u = (u_i)_{i=1}^P$ where u_i are iid, with $E[u_i] = 0$, $E[u_i^2] = \sigma^2$, $E[u_i^4] < \infty$. Suppose also A_P is a sequence of symmetric random matrices that is independent of u and is such that $E[(P^{-1} \text{tr}(A_P))^2] < K$ and $\lim P^{-2} E[\text{tr}(A_P^2)] = 0$. Then,*

$$\frac{1}{P} u' A u - P^{-1} \sigma^2 \text{tr}(A) \rightarrow 0 \quad (50)$$

⁵⁴See, for example, [Chernov et al. \(2025\)](#) for a proof that this equation indeed has a unique solution.

in L_2 and, hence, in probability.

We will now need the following result from [Kelly et al. \(2024b\)](#).

Proposition 4 *We have*

$$\lim_{T \rightarrow \infty} \frac{1}{T} \text{tr}((zI + \hat{\Psi})^{-1} \Psi) \rightarrow \xi(z; c) \quad (51)$$

almost surely, where

$$\xi(z; c) = \frac{1 - zm(-z; c)}{c^{-1} - 1 + zm(-z; c)}.$$

Similarly,

$$\lim_{T \rightarrow \infty} \frac{1}{T} \text{tr}((zI + \hat{\Psi})^{-2} \Psi) \rightarrow -\xi'(z; c) \quad (52)$$

almost surely, where

$$\begin{aligned} \xi'(z; c) &= \frac{d}{dz} \left(\frac{1 - zm(-z; c)}{c^{-1} - 1 + zm(-z; c)} \right) \\ &= \frac{d}{dz} \left(-1 + \frac{1}{1 - c + czm(-z; c)} \right) \\ &= - \frac{c(m(-z; c) - zm'(-z; c))}{(1 - c + czm(-z; c))^2} \end{aligned} \quad (53)$$

We will also define

$$Z_*(z; c) = \frac{z}{1 - c + czm(-z; c)} = z(1 + \xi(z; c)), \quad Z'_* = 1 + \xi + z\xi'. \quad (54)$$

to be the implicit shrinkage and

$$\underline{m}(z; c) = 1/Z_*(z; c). \quad (55)$$

Then, the results of [Bai and Saranadasa \(1996\)](#); [Li et al. \(2020\)](#); [Liu et al. \(2015\)](#); [Knowles and Yin \(2017\)](#); [Hastie et al. \(2022\)](#) imply that the following is true:

Theorem 5 *Suppose that $S_t = \Psi^{1/2}X_t$, where $X_t = (X_{i,t})$ where $E[X_{i,t}] = 0$, $E[X_{i,t}^2] = 1$, $E[X_{i,t}^4] < \infty$ are independent and identically distributed, and $\Psi = E[S_t S_t']$ is a positive semi-definite, uniformly bounded signal covariance matrix. Then, in the limit as $P, T \rightarrow$*

$\infty, P/T \rightarrow c$, for any uniformly bounded sequence of non-random matrices A_P , we have

$$P^{-1} z \operatorname{tr}(A_P(zI + \underbrace{\hat{\Psi}}_{\text{random}})^{-1}) - P^{-1} Z_* \operatorname{tr}(A_P(Z_*I + \underbrace{\Psi}_{\text{deterministic}})^{-1}) \rightarrow 0 \quad (56)$$

almost surely. Similarly, for any sequences of uniformly bounded vectors β_1, β_2 we have

$$z\beta'_1(zI + \underbrace{\hat{\Psi}}_{\text{random}})^{-1}\beta_2 - Z_*\beta'_1(A_P(Z_*I + \underbrace{\Psi}_{\text{deterministic}})^{-1})\beta_2 \rightarrow 0 \quad (57)$$

almost surely. Furthermore, the same result holds for $\tilde{\Psi} = \hat{\Psi} - \bar{S}\bar{S}'$, where $\bar{S} = T^{-1} \sum_t S_t$:

$$\begin{aligned} P^{-1} z \operatorname{tr}(A_P(zI + \tilde{\Psi})^{-1}) - P^{-1} Z_* \operatorname{tr}(A_P(Z_*I + \Psi)^{-1}) &\rightarrow 0 \\ z\beta'(zI + \tilde{\Psi})^{-1}\beta - Z_*\beta'(Z_*I + \Psi)^{-1}\beta &\rightarrow 0 \end{aligned} \quad (58)$$

almost surely.

Informally, we can rewrite this theorem as

$$\begin{aligned} z(zI + \hat{\Psi})^{-1} &\approx Z_*(Z_*I + \Psi)^{-1} \\ z(zI + \tilde{\Psi})^{-1} &\approx Z_*(Z_*I + \Psi)^{-1}. \end{aligned} \quad (59)$$

We will need the following result that follows from a combination of results in [Hastie et al. \(2022\)](#) with the algebraic identities in the proofs of [Kelly et al. \(2024b\)](#) (see [Chen et al. \(2025a\)](#) for details).

Theorem 6 *We have*

$$z^2\beta'_1(zI + \hat{\Psi})^{-1}\Psi(zI + \hat{\Psi})^{-1}\beta_2 - Z'_*(z; c) Z_*(z; c)^2\beta'_1\Psi(\Psi + Z_*I)^{-2}\beta_2 \rightarrow 0 \quad (60)$$

almost surely, for any bounded β_1, β_2 .

We will also need the following result from [Kelly et al. \(2024b\)](#) (Lemma IA14):

Lemma 2 *We have*

$$\lim T^{-1} E[\operatorname{tr}(\Psi_{1,1}(zI + \hat{\Psi}_{1,1})^{-1}\Psi_{1,1}(zI + \hat{\Psi}_{1,1})^{-1})] = (\xi + z\xi')(1 + \xi)^2. \quad (61)$$

C.2 Proof of Theorem 2

Throughout this appendix, we use the notation $\Psi(1) = \Psi_{1,1}$, $\Psi(2) = \Psi_{2,2}$, $\Psi(1,2) = \Psi_{1,2}$, $\Psi(2,1) = \Psi_{2,1}$. Similarly, we use the notation $\beta^{(i)} = \beta(i)$.

We have

$$R_{t+1} = \beta(1)'S_t(1) + \beta(2)'S_t(2) + \varepsilon_t, \quad (62)$$

or, in vector form,

$$y = S(1)\beta(1) + S(2)\beta(2) + \varepsilon = S(1)\beta(1) + \varepsilon(1), \quad (63)$$

where we have defined

$$\varepsilon(1) = S(2)\beta(2) + \varepsilon. \quad (64)$$

Note that

$$E[S(1)'\varepsilon(1)] = E[S(1)'S(2)]\beta(2) = \Psi_{1,2}\beta(2) = \tilde{\beta}(2) \in \mathbb{R}^{P(1)}. \quad (65)$$

We will also define

$$\tilde{\beta}(1) = \Psi(1)\beta(1), \quad (66)$$

and we build the ridge prediction

$$\hat{\pi}_T = \hat{\beta}_1(z)'S_T(1), \quad \hat{\beta}_1(z) = (zI + T^{-1}S(1)'S(1))^{-1}T^{-1}S(1)'y. \quad (67)$$

Now, we define managed returns

$$R_{T+1}^\pi = \hat{\pi}_T R_{T+1}, \quad (68)$$

and we need to compute

$$\begin{aligned} E[R_{T+1}^\pi] &= E[(\varepsilon_{T+1}(1) + \beta(1)'S_T(1))S_T(1)'(zI + T^{-1}S(1)'S(1))^{-1}T^{-1}S(1)'y] \\ &= E[(\varepsilon_{T+1}(1) + \beta(1)'S_T(1))S_T(1)'(zI + T^{-1}S(1)'S(1))^{-1}T^{-1}S(1)'y] \\ &= \tilde{\beta}(2)'E[(zI + T^{-1}S(1)'S(1))^{-1}T^{-1}S(1)'y] \\ &\quad + \beta(1)'E[\Psi(1)(zI + T^{-1}S(1)'S(1))^{-1}T^{-1}S(1)'y] \\ &= \tilde{\beta}(2)'E[(zI + T^{-1}S(1)'S(1))^{-1}T^{-1}S(1)'(S(1)\beta(1) + \varepsilon(1))] \\ &\quad + \beta(1)'E[\Psi(1)(zI + T^{-1}S(1)'S(1))^{-1}T^{-1}S(1)'(S(1)\beta(1) + \varepsilon(1))] \\ &= \tilde{\beta}(2)'E[(zI + T^{-1}S(1)'S(1))^{-1}\Psi(1)]\beta(1) \\ &\quad + \tilde{\beta}(2)'E[(zI + T^{-1}S(1)'S(1))^{-1}T^{-1}S(1)'\varepsilon(1)] \\ &\quad + \beta(1)'E[\Psi(1)(zI + T^{-1}S(1)'S(1))^{-1}\Psi(1)]\beta(1) \\ &\quad + \beta(1)'E[\Psi(1)(zI + T^{-1}S(1)'S(1))^{-1}T^{-1}S(1)'\varepsilon(1)]. \end{aligned} \quad (69)$$

Let us compute the asymptotics of this expectation. We will need

$$P^{-1}S_t'(zI + \hat{\Psi}_{T,t})^{-1}S_t - P^{-1}\text{tr}(\Psi(zI + \hat{\Psi}_{T,t})^{-1}) \rightarrow 0 \quad (70)$$

in probability. At the same time, standard concentration results imply that

$$P^{-1} \text{tr}(\Psi(zI + \hat{\Psi}_{T,t})^{-1}) - E[P^{-1} \text{tr}(\Psi(zI + \hat{\Psi}_{T,t})^{-1})] \rightarrow 0$$

almost surely. In particular,

$$\hat{\xi}_t = T^{-1} S_t' (zI + \hat{\Psi}_{T,t})^{-1} S_t \quad (71)$$

converges to the $\xi(z; c)$ function from [Kelly et al. \(2024b\)](#) in probability.

Lemma 3 *We have*

$$\lim E[R_{T+1}^\pi] = \lim (\tilde{\beta}(1) + \tilde{\beta}(2))' (Z_* I + \Psi_{1,1})^{-1} (\tilde{\beta}(1) + \tilde{\beta}(2)) \quad (72)$$

Proof of Lemma 3. Let $\Psi_{T,t} = \frac{1}{T} \sum_{\tau \neq t} S_\tau(1) S_\tau(1)'$. By the Sherman-Morrison formula (see [Bartlett \(1951\)](#)),

$$(zI + \hat{\Psi}_T)^{-1} = (zI + \hat{\Psi}_{T,t})^{-1} - \frac{1}{T} (zI + \hat{\Psi}_{T,t})^{-1} S_t S_t' (zI + \hat{\Psi}_{T,t})^{-1} \frac{1}{1 + (T)^{-1} S_t' (zI + \hat{\Psi}_{T,t})^{-1} S_t}. \quad (73)$$

and

$$(zI + \hat{\Psi}_T)^{-1} S_t = (zI + \hat{\Psi}_{T,t})^{-1} S_t - \frac{1}{T} (zI + \hat{\Psi}_{T,t})^{-1} S_t S_t' (zI + \hat{\Psi}_{T,t})^{-1} \frac{1}{1 + (T)^{-1} S_t' (zI + \hat{\Psi}_{T,t})^{-1} S_t}. \quad (74)$$

so that

$$\begin{aligned} & \beta(1)' E[\Psi(1)(zI + T^{-1} S(1)' S(1))^{-1} T^{-1} S(1)' \varepsilon(1)] \\ &= T^{-1} \sum_t \beta(1)' E[\Psi(1)(zI + T^{-1} S(1)' S(1))^{-1} S_t(1) \varepsilon_t(1)] \\ & \quad \underbrace{=}_{\text{Sherman-Morrison}} T^{-1} \sum_t \beta(1)' E[\Psi(1)(zI + \Psi_{T,t})^{-1} S_t(1) \varepsilon_t(1) (1 + \hat{\xi}_t)^{-1}] \\ & \approx (1 + \xi(z; c))^{-1} \sum_t \beta(1)' E[\Psi(1)(zI + \Psi_{T,t})^{-1} S_t(1) \varepsilon_t(1)] \\ & \quad \underbrace{=}_{\text{symmetry}} \beta(1)' E[\Psi(1)(zI + \Psi_{T,t})^{-1} S_t(1) \varepsilon_t(1)] \\ &= (1 + \xi(z; c))^{-1} \beta(1)' E[\Psi(1)(zI + \Psi_{T,t})^{-1}] \tilde{\beta}(2) \\ &= \frac{z}{Z_*} \beta(1)' E[\Psi(1)(zI + \Psi_{T,t})^{-1}] \tilde{\beta}(2) \\ &\approx \beta(1)' E[\Psi(1)(Z_* I + \Psi_{T,t})^{-1}] \tilde{\beta}(2) \end{aligned} \quad (75)$$

where the last claim follows from [Theorem 5](#).

A similar argument applies to other terms, and the proof is complete. $\square$

C.3 Proof of Theorem 3

We now come to the second moment. For simplicity, we omit the (1) index for S, ε . Then,

$$E[(R_{T+1}^\pi)^2] = E[(\varepsilon_{T+1}(1) + \beta(1)'S_T)(S_T'\hat{\beta})^2]. \quad (76)$$

First, we integrate over S_T using the following lemma.

Lemma 4 *Let $\check{\beta} \in \mathbb{R}^{P_1}$ be uniformly bounded and $\tilde{\beta} = \Psi^{1/2}\beta$ and $\hat{\beta} = \Psi^{1/2}\begin{pmatrix} \check{\beta} \\ 0 \end{pmatrix}$ and suppose that either $E[X_{i,t}^4] - 3 = 0$ or $\tilde{\beta}_i^2 = o(1)$. Then,*

$$\begin{aligned} & E[(\varepsilon_{t+1} + \beta'S_t)^2(\check{\beta}'S_t(1))^2] \\ &= \sigma^2\check{\beta}'\Psi(1)\check{\beta} + \check{\beta}'\Psi(1)\check{\beta}\beta'\Psi\beta + 2(\beta'\Psi\begin{pmatrix} \check{\beta} \\ 0 \end{pmatrix})^2 + o(1). \end{aligned} \quad (77)$$

Proof of Lemma 4. We have

$$\begin{aligned} & E[(\varepsilon_{t+1} + \beta'S_t)^2(\hat{\beta}(1)'S_t(1))^2] \\ &= \sigma^2\hat{\beta}'\Psi(1)\hat{\beta} + E[(\beta'S_t)^2(\check{\beta}'S_t(1))^2]. \end{aligned} \quad (78)$$

Then, substituting $S_t = \Psi^{1/2}X_t$, we get

$$\begin{aligned} & E[(\beta'S_t)^2(\check{\beta}'S_t(1))^2] \\ &= E[(\tilde{\beta}'X_t)^2(\hat{\beta}'X_t)^2] = E\left[\sum_{i_1, i_2, i_3, i_4} \tilde{\beta}_{i_1}\tilde{\beta}_{i_2}\hat{\beta}_{i_3}\hat{\beta}_{i_4}X_{i_1,t}X_{i_2,t}X_{i_3,t}X_{i_4,t}\right] \\ &= \sum_{i_1 \neq i_3} \tilde{\beta}_{i_1}^2\hat{\beta}_{i_4}^2 \\ &+ 2\sum_{i_1 \neq i_3} \tilde{\beta}_{i_1}\hat{\beta}_{i_1}\tilde{\beta}_{i_3}\hat{\beta}_{i_3} + \sum_i \tilde{\beta}_i^2\hat{\beta}_i^2 E[X_{i,t}^4] \\ &= \|\tilde{\beta}\|^2\|\hat{\beta}\|^2 + 2(\tilde{\beta}'\hat{\beta})^2 \\ &+ \sum_i \tilde{\beta}_i^2\hat{\beta}_i^2(E[X_{i,t}^4] - 3) \end{aligned} \quad (79)$$

and the claim follows. $\square$

Thus, it remains to compute

$$\begin{aligned} \beta'\Psi\begin{pmatrix} \check{\beta} \\ 0 \end{pmatrix} &= \beta(1)'\Psi(1)\check{\beta} + \beta(2)'\Psi_{2,1}\check{\beta} \\ &= \tilde{\beta}'\check{\beta}. \end{aligned} \quad (80)$$

Thus, we have proved the following result

Lemma 5 *Let $\tilde{\beta} = \Psi(1)\beta(1) + \Psi(1, 2)\beta(2)$. We have*

$$\begin{aligned} E[(R_{T+1}^\pi)^2] &= (\sigma^2 + \beta' \Psi \beta) E \left[\hat{\beta}(z; P_1)' \Psi(1) \hat{\beta}(z; P_1) \right] \\ &+ 2E \left[\left(\tilde{\beta}' \hat{\beta}(z; P_1) \right)^2 \right] + o(1). \end{aligned} \quad (81)$$

C.4 Computing $E \left[\hat{\beta}(z; P_1)' \Psi(1) \hat{\beta}(z; P_1) \right]$

Now, setting $\tilde{\beta} = \hat{\beta}(z)$, we get

$$\begin{aligned} E[\tilde{\beta}' \Psi(1) \tilde{\beta}] &= T^{-2} E[y' S(1) (zI + T^{-1} S(1)' S(1))^{-1} \Psi(1) (zI + T^{-1} S(1)' S(1))^{-1} S(1)' y] \\ &= T^{-2} \text{tr} E[(zI + T^{-1} S(1)' S(1))^{-1} \Psi(1) (zI + T^{-1} S(1)' S(1))^{-1} S(1)' y y' S(1)] \\ &= T^{-2} \text{tr} E[(zI + T^{-1} S(1)' S(1))^{-1} \Psi(1) (zI + T^{-1} S(1)' S(1))^{-1} S(1)' (\sigma^2 I + S \beta \beta' S') S(1)] \\ &= \sigma^2 T^{-2} \text{tr} E[(zI + T^{-1} S(1)' S(1))^{-1} \Psi(1) (zI + T^{-1} S(1)' S(1))^{-1} S(1)' S(1)] \\ &+ T^{-2} E[(zI + T^{-1} S(1)' S(1))^{-1} \Psi(1) (zI + T^{-1} S(1)' S(1))^{-1} S(1)' S \beta \beta' S' S(1)] \\ &= \sigma^2 T^{-1} \text{tr} E[(zI + \hat{\Psi}(1))^{-1} \Psi(1) (zI + \hat{\Psi}(1))^{-1} \hat{\Psi}(1)] \\ &+ \beta' T^{-2} E[S' S(1) (zI + T^{-1} S(1)' S(1))^{-1} \Psi(1) (zI + T^{-1} S(1)' S(1))^{-1} S(1)' S] \beta \\ &= \mathcal{T}_1 + \mathcal{T}_2. \end{aligned} \quad (82)$$

For the first term, we have

$$\begin{aligned} \mathcal{T}_1 &= \sigma^2 T^{-1} \text{tr} E[(zI + \hat{\Psi}(1))^{-1} \Psi(1) (zI + \hat{\Psi}(1))^{-1} \hat{\Psi}(1)] \\ &= \sigma^2 T^{-1} \text{tr} E[(zI + \hat{\Psi}(1))^{-1} \Psi(1) (zI + \hat{\Psi}(1))^{-1} (zI + \hat{\Psi}(1) - zI)] \\ &= \sigma^2 T^{-1} \text{tr} E[(zI + \hat{\Psi}(1))^{-1} \Psi(1)] - \sigma^2 T^{-1} z \text{tr} E[(zI + \hat{\Psi}(1))^{-2} \Psi(1)] \\ &\approx \sigma^2 (\xi + z \xi'). \end{aligned} \quad (83)$$

Let us now deal with the second term. We have

$$\begin{aligned} \mathcal{T}_2 &= \beta' T^{-2} E[S' S(1) (zI + T^{-1} S(1)' S(1))^{-1} \Psi(1) (zI + T^{-1} S(1)' S(1))^{-1} S(1)' S] \beta \\ &= \beta(1)' E[\hat{\Psi}(1) (zI + \hat{\Psi}(1))^{-1} \Psi(1) (zI + \hat{\Psi}(1))^{-1} \hat{\Psi}(1)] \beta(1) \\ &+ 2\beta(1)' E[\hat{\Psi}(1) (zI + \hat{\Psi}(1))^{-1} \Psi(1) (zI + \hat{\Psi}(1))^{-1} \hat{\Psi}(1, 2)] \beta(2) \\ &+ \beta(2)' E[\hat{\Psi}(2, 1) (zI + \hat{\Psi}(1))^{-1} \Psi(1) (zI + \hat{\Psi}(1))^{-1} \hat{\Psi}(1, 2)] \beta(2) \\ &= \mathcal{T}_{2,1} + 2\mathcal{T}_{2,2} + \mathcal{T}_{2,3} \end{aligned} \quad (84)$$

where we have defined

$$\hat{\Psi}(1) = T^{-1}S(1)'S(1), \quad \hat{\Psi}(2, 1) = T^{-1}S(2)'S(1), \quad \hat{\Psi}(1, 2) = T^{-1}S(1)'S(2). \quad (85)$$

Now,

$$\begin{aligned} \mathcal{T}_{2,1} &= \beta(1)'E[\hat{\Psi}(1)(zI + \hat{\Psi}(1))^{-1}\Psi(1)(zI + \hat{\Psi}(1))^{-1}\hat{\Psi}(1)]\beta(1) \\ &= \beta(1)'E[(-zI + zI + \hat{\Psi}(1))(zI + \hat{\Psi}(1))^{-1}\Psi(1)(zI + \hat{\Psi}(1))^{-1}\hat{\Psi}(1)]\beta(1) \\ &= \beta(1)'E[\Psi(1)(zI + \hat{\Psi}(1))^{-1}\hat{\Psi}(1)]\beta(1) \\ &\quad - z\beta(1)'E[(zI + \hat{\Psi}(1))^{-1}\Psi(1)(zI + \hat{\Psi}(1))^{-1}(-zI + zI + \hat{\Psi}(1))]\beta(1) \\ &= \beta(1)'E[\Psi(1)(zI + \hat{\Psi}(1))^{-1}(-zI + zI + \hat{\Psi}(1))]\beta(1) \\ &\quad - z\beta(1)'E[(zI + \hat{\Psi}(1))^{-1}\Psi(1)(zI + \hat{\Psi}(1))^{-1}(-zI + zI + \hat{\Psi}(1))]\beta(1) \\ &= \beta(1)'\Psi(1)\beta(1) - 2z\beta(1)'E[(zI + \hat{\Psi}(1))^{-1}\Psi(1)]\beta(1) \\ &\quad + z^2\beta(1)'E[(zI + \hat{\Psi}(1))^{-1}\Psi(1)(zI + \hat{\Psi}(1))^{-1}]\beta(1) \\ &= \beta(1)'\Psi(1)\beta(1) - 2Z_*\beta(1)'E[(Z_*I + \Psi(1))^{-1}\Psi(1)]\beta(1) \\ &\quad + Z'_*Z_*^2\beta(1)'(Z_*I + \Psi(1))^{-2}\Psi(1)\beta(1) \\ &= \beta(1)'\Psi(1)^3(Z_*I + \Psi(1))^{-2}\beta(1) + (Z'_* - 1)Z_*^2\beta(1)'(Z_*I + \Psi(1))^{-2}\Psi(1)\beta(1), \end{aligned} \quad (86)$$

where we have used (57) and (60).

Now, for the first cross-term, we have

$$\begin{aligned} \mathcal{T}_{2,2} &= \beta(1)'E[\hat{\Psi}(1)(zI + \hat{\Psi}(1))^{-1}\Psi(1)(zI + \hat{\Psi}(1))^{-1}\hat{\Psi}(1, 2)]\beta(2) \\ &= \beta(1)'E[(-zI + zI + \hat{\Psi}(1))(zI + \hat{\Psi}(1))^{-1}\Psi(1)(zI + \hat{\Psi}(1))^{-1}\hat{\Psi}(1, 2)]\beta(2) \\ &= \beta(1)'E[(-zI + zI + \hat{\Psi}(1))(zI + \hat{\Psi}(1))^{-1}\Psi(1)(zI + \hat{\Psi}(1))^{-1}\hat{\Psi}(1, 2)]\beta(2) \\ &= -z\beta(1)'E[(zI + \hat{\Psi}(1))^{-1}\Psi(1)(zI + \hat{\Psi}(1))^{-1}\hat{\Psi}(1, 2)]\beta(2) \\ &\quad + \beta(1)'E[\Psi(1)(zI + \hat{\Psi}(1))^{-1}\hat{\Psi}(1, 2)]\beta(2) \\ &= -z\mathcal{T}_{2,2,1} + \mathcal{T}_{2,2,2}. \end{aligned} \quad (87)$$

Now, by Theorem 6,

$$\begin{aligned}
\mathcal{T}_{2,2,1} &= \beta(1)' E[(zI + \hat{\Psi}(1))^{-1} \Psi(1)(zI + \hat{\Psi}(1))^{-1} \hat{\Psi}(1, 2)] \beta(2) \\
&\approx T^{-1} \sum_t \beta(1)' E[(zI + \hat{\Psi}_{T,t})^{-1} - T^{-1}(1 + \hat{\xi}_t)^{-1}(zI + \hat{\Psi}_{T,t})^{-1} S_t(1) S_t(1)' (zI + \hat{\Psi}_{T,t})^{-1}) \\
&\quad \times \Psi(1)(zI + \hat{\Psi}_{T,t})^{-1} (1 + \hat{\xi}_t)^{-1} S_t(1) S_t(2)'] \beta(2) \\
&\stackrel{\text{symmetry}}{=} \beta(1)' E[(1 + \hat{\xi}_t)^{-1} (zI + \hat{\Psi}_{T,t})^{-1} \Psi(1)(zI + \hat{\Psi}_{T,t})^{-1}] \Psi(1, 2) \beta(2) \\
&\quad - T^{-1} \beta(1)' E[(1 + \hat{\xi}_t)^{-1} (zI + \hat{\Psi}_{T,t})^{-1} S_t(1) S_t(1)' (zI + \hat{\Psi}_{T,t})^{-1} \\
&\quad \times \Psi(1)(zI + \hat{\Psi}_{T,t})^{-1} (1 + \hat{\xi}_t)^{-1} S_t(1) S_t(2)'] \beta(2) \\
&\approx (1 + \xi)^{-1} z^{-2} Z_*' Z_*^2 \beta(1)' (Z_* I + \Psi(1))^{-2} \Psi(1) \Psi(1, 2) \beta(2) \\
&\quad - T^{-1} (1 + \xi)^{-2} \beta(1)' E[(zI + \hat{\Psi}_{T,t})^{-1} S_t(1) S_t(1)' (zI + \hat{\Psi}_{T,t})^{-1} \\
&\quad \times \Psi(1)(zI + \hat{\Psi}_{T,t})^{-1} S_t(1) S_t(2)'] \beta(2) \\
&= \mathcal{T}_{2,2,1,1} - \mathcal{T}_{2,2,1,2}.
\end{aligned} \tag{88}$$

We will need

Lemma 6 *We have, for any vectors β_1, β_2 and matrix A independent of S_t that*

$$E[\beta_1' S_t S_t' A S_t S_t' \beta_2] = E[\beta_1' \Psi \beta_2 \text{tr}(\Psi A)] + O(\|A\| \|\tilde{\beta}_1\| \|\tilde{\beta}_2\|) \tag{89}$$

Proof of Lemma 6. Let $\tilde{\beta}_1 = \Psi^{1/2} \beta_1$, $\tilde{\beta}_2 = \Psi^{1/2} \beta_2$, $\tilde{A} = \Psi^{1/2} A \Psi^{1/2}$. Then,

$$\begin{aligned}
E[\beta_1' S_t S_t' A S_t S_t' \beta_2] &= E[\tilde{\beta}_1' X_t X_t' \tilde{A} X_t X_t' \tilde{\beta}_2] \\
&= E\left[\sum_{i_1, i_2, i_3, i_4} X_{i_1, t} X_{i_2, t} X_{i_3, t} X_{i_4, t} \tilde{\beta}_{1, i_1} \tilde{\beta}_{2, i_4} \tilde{A}_{i_2, i_3} \tilde{\beta}_2\right] \\
&= 2 \sum_{i_1 \neq i_3} \tilde{\beta}_{1, i_1} \tilde{\beta}_{2, i_3} \tilde{A}_{i_1, i_3} \\
&\quad + \sum_{i_1 \neq i_2} \tilde{\beta}_{1, i_1} \tilde{\beta}_{2, i_1} \tilde{A}_{i_2, i_2} \\
&\quad + \sum_i E[X_{i, t}^4] \tilde{\beta}_{1, i} \tilde{\beta}_{2, i} \tilde{A}_{i, i} \\
&= 2 \tilde{\beta}_1' \tilde{A} \tilde{\beta}_2 + \tilde{\beta}_1' \tilde{\beta}_2 \text{tr}(\tilde{A}) + O(\|A\| \|\tilde{\beta}_1\| \|\tilde{\beta}_2\|) \\
&= 2 \tilde{\beta}_1' \tilde{A} \tilde{\beta}_2 + \tilde{\beta}_1' \tilde{\beta}_2 \text{tr}(\tilde{A}) + \tilde{\beta}_1' \text{diag}(\tilde{A}) \tilde{\beta}_2 \\
&= \tilde{\beta}_1' \tilde{\beta}_2 \text{tr}(\tilde{A}) + O(\|A\| \|\tilde{\beta}_1\| \|\tilde{\beta}_2\|)
\end{aligned} \tag{90}$$

□

Lemma 6 implies that

$$\begin{aligned}
\mathcal{T}_{2,2,1,2} &= T^{-1}(1+\xi)^{-2}\beta(1)'E[(zI + \hat{\Psi}_{T,t})^{-1}S_t(1)S_t(1)'(zI + \hat{\Psi}_{T,t})^{-1} \\
&\times \Psi(1)(zI + \hat{\Psi}_{T,t})^{-1}S_t(1)S_t(2)']\beta(2) \\
&\stackrel{\text{Lemma 6}}{\approx} (1+\xi)^{-2}T^{-1}E[\beta(1)'(zI + \hat{\Psi}_{T,t})^{-1}\Psi\begin{pmatrix} 0 \\ \beta(2) \end{pmatrix} \text{tr}((zI + \hat{\Psi}_{T,t})^{-1}\Psi(1)(zI + \hat{\Psi}_{T,t})^{-1}\Psi(1))] \\
&\approx (1+\xi)^{-2}\beta(1)'E[(zI + \hat{\Psi}_{T,t})^{-1}\Psi(1,2)\beta(2)] \\
&\times T^{-1}E[\text{tr}((zI + \hat{\Psi}_{T,t})^{-1}\Psi(1)(zI + \hat{\Psi}_{T,t})^{-1}\Psi(1))] \\
&\approx (1+\xi)^{-2}\frac{Z_*}{z}\beta(1)'(Z_*I + \Psi_{1,1})^{-1}\Psi(1,2)\beta(2)(\xi + z\xi')(1+\xi)^2 \\
&= \frac{Z_*}{z}\beta(1)'(Z_*I + \Psi_{1,1})^{-1}\Psi(1,2)\beta(2)(\xi + z\xi')
\end{aligned} \tag{91}$$

bu Lemma 2 and (57). Summarizing, we get since $Z'_* = (z(1+\xi))' = 1+\xi + z\xi'$ that

$$\begin{aligned}
\mathcal{T}_{2,2,1} &\approx \mathcal{T}_{2,2,1,1} - \mathcal{T}_{2,2,1,2} \\
&= (1+\xi)^{-1}z^{-2}Z'_*Z_*^2\beta(1)'(Z_*I + \Psi(1))^{-2}\Psi(1)\Psi(1,2)\beta(2) \\
&- \frac{Z_*}{z}\beta(1)'(Z_*I + \Psi_{1,1})^{-1}\Psi(1,2)\beta(2)(\xi + z\xi') \\
&= (1+\xi)(1+\xi + z\xi')\beta(1)'(Z_*I + \Psi(1))^{-2}\Psi(1)\Psi(1,2)\beta(2) \\
&- (1+\xi)\beta(1)'(Z_*I + \Psi_{1,1})^{-1}\Psi(1,2)\beta(2)(\xi + z\xi') \\
&= -(\xi + z\xi')(1+\xi)Z_*\left(\beta(1)'(Z_*I + \Psi(1))^{-2}\Psi(1,2)\beta(2)\right) \\
&+ (1+\xi)\beta(1)'(Z_*I + \Psi(1))^{-2}\Psi(1)\Psi(1,2)\beta(2)
\end{aligned} \tag{92}$$

Similarly, by (57),

$$\begin{aligned}
\mathcal{T}_{2,2,2} &= \beta(1)'E[\Psi(1)(zI + \hat{\Psi}(1))^{-1}\hat{\Psi}(1,2)]\beta(2) \\
&\approx (1+\xi)^{-1}\beta(1)'E[\Psi(1)(zI + \hat{\Psi}_{T,t})^{-1}S_t(1)S_t(2)']\beta(2) \\
&= (1+\xi)^{-1}\beta(1)'E[\Psi(1)(zI + \hat{\Psi}_{T,t})^{-1}\Psi(1,2)]\beta(2) \\
&= \frac{Z_*}{z}\beta(1)'E[\Psi(1)(zI + \hat{\Psi}_{T,t})^{-1}\Psi(1,2)]\beta(2) \\
&\approx \beta(1)'\Psi(1)(Z_*I + \Psi_{1,1})^{-1}\Psi(1,2)\beta(2)
\end{aligned} \tag{93}$$

Thus,

$$\begin{aligned}
\mathcal{T}_{2,2} &= -z\mathcal{T}_{2,2,1} + \mathcal{T}_{2,2,2} \\
&\approx -z \left(-(\xi + z\xi')(1 + \xi)Z_* \left(\beta(1)'(Z_*I + \Psi(1))^{-2}\Psi(1,2)\beta(2) \right) \right. \\
&\quad \left. + (1 + \xi)\beta(1)'(Z_*I + \Psi(1))^{-2}\Psi(1)\Psi(1,2)\beta(2) \right) \\
&\quad + \beta(1)'\Psi(1)(Z_*I + \Psi_{1,1})^{-1}\Psi(1,2)\beta(2) \\
&= - \left(-(\xi + z\xi')Z_*^2 \left(\beta(1)'(Z_*I + \Psi(1))^{-2}\Psi(1,2)\beta(2) \right) \right. \\
&\quad \left. + \beta(1)'Z_*(Z_*I + \Psi(1))^{-2}\Psi(1)\Psi(1,2)\beta(2) \right) \\
&\quad + \beta(1)'\Psi(1)(Z_*I + \Psi_{1,1})^{-1}\Psi(1,2)\beta(2) \\
&= (\xi + z\xi')Z_*^2 \left(\beta(1)'(Z_*I + \Psi(1))^{-2}\Psi(1,2)\beta(2) \right) + \beta(1)'\Psi(1)^2(Z_*I + \Psi)^{-2}\Psi(1,2)\beta(2)
\end{aligned} \tag{94}$$

We now deal with the last term $\mathcal{T}_{2,3}$ in (84). We have, in complete analogy with the derivations above, that

$$\begin{aligned}
\mathcal{T}_{2,3} &= \beta(2)'E[\hat{\Psi}(2,1)(zI + \hat{\Psi}(1))^{-1}\Psi(1)(zI + \hat{\Psi}(1))^{-1}\hat{\Psi}(1,2)]\beta(2) \\
&\approx T^{-2}(1 + \xi)^{-2} \sum_{t_1, t_2} \beta(2)'E[S_{t_1}(2)S_{t_1}(1)'(zI + \hat{\Psi}_{T,t_1})^{-1}\Psi(1)(zI + \hat{\Psi}_{T,t_2})^{-1}S_{t_2}(1)S_{t_2}(2)']\beta(2) \\
&\quad \underbrace{=}_{\text{symmetry}} (1 + \xi)^{-2}T^{-1}\beta(2)'E[S_t(2)S_t(1)'(zI + \hat{\Psi}_{T,t})^{-1}\Psi(1)(zI + \hat{\Psi}_{T,t})^{-1}S_t(1)S_t(2)']\beta(2) \\
&\quad + (1 + \xi)^{-2}T^{-2}T(T - 1)\beta(2)'E[S_{t_1}(2)S_{t_1}(1)'(zI + \hat{\Psi}_{T,t_1})^{-1}\Psi(1)(zI + \hat{\Psi}_{T,t_2})^{-1}S_{t_2}(1)S_{t_2}(2)']\beta(2) \\
&= \mathcal{T}_{2,3,1} + \mathcal{T}_{2,3,2}.
\end{aligned} \tag{95}$$

The first term in (95) can be evaluated using Lemma 6 to get

$$\begin{aligned}
\mathcal{T}_{2,3,1} &= (1 + \xi)^{-2}T^{-1}\beta(2)'E[S_t(2)S_t(1)'(zI + \hat{\Psi}_{T,t})^{-1}\Psi(1)(zI + \hat{\Psi}_{T,t})^{-1}S_t(1)S_t(2)']\beta(2) \\
&\approx (1 + \xi)^{-2}\beta(2)'\Psi(2,2)\beta(2)T^{-1} \text{tr} E[(zI + \hat{\Psi}_{T,t})^{-1}\Psi(1)(zI + \hat{\Psi}_{T,t})^{-1}\Psi(1)] \\
&\approx \beta(2)'\Psi(2,2)\beta(2)(\xi + z\xi').
\end{aligned} \tag{96}$$

by Lemma 2.

It remains to evaluate the second term in (95). We have

$$\begin{aligned}
(1 + \xi)^2 \mathcal{T}_{2,3,2} &= \beta(2)' E[S_{t_1}(2)S_{t_1}(1)'(zI + \hat{\Psi}_{T,t_1})^{-1}\Psi(1)(zI + \hat{\Psi}_{T,t_2})^{-1}S_{t_2}(1)S_{t_2}(2)']\beta(2) \\
&= \beta(2)' E[S_{t_1}(2)S_{t_1}(1)'] \left((zI + \hat{\Psi}_{T,t_1,t_2})^{-1} - T^{-1}(1 + \hat{\xi})^{-1}(zI + \hat{\Psi}_{T,t_1,t_2})^{-1}S_{t_2}(1)S_{t_2}(1)'(zI + \hat{\Psi}_{T,t_1,t_2})^{-1} \right) \\
&\quad \times \Psi(1) \left((zI + \hat{\Psi}_{T,t_1,t_2})^{-1} - T^{-1}(1 + \hat{\xi})^{-1}(zI + \hat{\Psi}_{T,t_1,t_2})^{-1}S_{t_1}(1)S_{t_1}(1)'(zI + \hat{\Psi}_{T,t_1,t_2})^{-1} \right) S_{t_2}(1)S_{t_2}(2)']\beta(2) \\
&= \beta(2)' E[S_{t_1}(2)S_{t_1}(1)'(zI + \hat{\Psi}_{T,t_1,t_2})^{-1}\Psi(1)(zI + \hat{\Psi}_{T,t_1,t_2})^{-1}S_{t_2}(1)S_{t_2}(2)']\beta(2) \\
&\quad - 2\beta(2)' E[S_{t_1}(2)S_{t_1}(1)'T^{-1}(1 + \hat{\xi})^{-1}(zI + \hat{\Psi}_{T,t_1,t_2})^{-1}S_{t_2}(1)S_{t_2}(1)'(zI + \hat{\Psi}_{T,t_1,t_2})^{-1} \\
&\quad \times \Psi(1)(zI + \hat{\Psi}_{T,t_1,t_2})^{-1}S_{t_2}(1)S_{t_2}(2)']\beta(2) \\
&\quad + \beta(2)' E[S_{t_1}(2)S_{t_1}(1)'T^{-1}(1 + \hat{\xi})^{-1}(zI + \hat{\Psi}_{T,t_1,t_2})^{-1}S_{t_2}(1)S_{t_2}(1)'(zI + \hat{\Psi}_{T,t_1,t_2})^{-1} \\
&\quad \Psi(1)T^{-1}(1 + \hat{\xi})^{-1}(zI + \hat{\Psi}_{T,t_1,t_2})^{-1}S_{t_1}(1)S_{t_1}(1)'(zI + \hat{\Psi}_{T,t_1,t_2})^{-1}S_{t_2}(1)S_{t_2}(2)']\beta(2) \\
&= Term1 - 2Term2 + Term3.
\end{aligned} \tag{97}$$

We have

$$\begin{aligned}
Term1 &= \beta(2)' E[S_{t_1}(2)S_{t_1}(1)'(zI + \hat{\Psi}_{T,t_1,t_2})^{-1}\Psi(1)(zI + \hat{\Psi}_{T,t_1,t_2})^{-1}S_{t_2}(1)S_{t_2}(2)']\beta(2) \\
&= \beta(2)' E[\Psi(2, 1)(zI + \hat{\Psi}_{T,t_1,t_2})^{-1}\Psi(1)(zI + \hat{\Psi}_{T,t_1,t_2})^{-1}\Psi(1, 2)]\beta(2) \\
&\approx z^{-2} Z_*'(z; c) Z_*(z; c)^2 \beta(2)' \Psi(2, 1) \Psi(1) (\Psi(1) + Z_* I)^{-2} \Psi(1, 2) \beta(2)
\end{aligned} \tag{98}$$

by Theorem 6.

Then, by Lemma 6,

$$\begin{aligned}
Term2 &= \beta(2)' E[S_{t_1}(2)S_{t_1}(1)'T^{-1}(1 + \hat{\xi})^{-1}(zI + \hat{\Psi}_{T,t_1,t_2})^{-1}S_{t_2}(1)S_{t_2}(1)'(zI + \hat{\Psi}_{T,t_1,t_2})^{-1} \\
&\quad \times \Psi(1)(zI + \hat{\Psi}_{T,t_1,t_2})^{-1}S_{t_2}(1)S_{t_2}(2)']\beta(2) \\
&\approx (1 + \xi)^{-1} \beta(2)' E[\Psi(2, 1)T^{-1}(zI + \hat{\Psi}_{T,t_1,t_2})^{-1}S_{t_2}(1)S_{t_2}(1)'(zI + \hat{\Psi}_{T,t_1,t_2})^{-1} \\
&\quad \times \Psi(1)(zI + \hat{\Psi}_{T,t_1,t_2})^{-1}S_{t_2}(1)S_{t_2}(2)']\beta(2) \\
&\quad \underbrace{\approx}_{\text{Lemma 6}} (1 + \xi)^{-1} \beta(2)' E[\Psi(2, 1)(zI + \hat{\Psi}_{T,t_1,t_2})^{-1}\Psi(1, 2)]\beta(2) \\
&= T^{-1} \text{tr} E[(zI + \hat{\Psi}_{T,t_1,t_2})^{-1}\Psi(1)(zI + \hat{\Psi}_{T,t_1,t_2})^{-1}\Psi(1)] \\
&\approx \beta(2)' \Psi(2, 1)(Z_* I + \Psi_{1,1})^{-1}\Psi(1, 2)\beta(2)(\xi + z\xi')(1 + \xi)^2
\end{aligned} \tag{99}$$

and

$$\begin{aligned}
Term3 &= \beta(2)'E[S_{t_1}(2)S_{t_1}(1)'T^{-1}(1+\hat{\xi})^{-1}(zI + \hat{\Psi}_{T,t_1,t_2})^{-1}S_{t_2}(1)S_{t_2}(1)'(zI + \hat{\Psi}_{T,t_1,t_2})^{-1} \\
&\quad \Psi(1)T^{-1}(1+\hat{\xi})^{-1}(zI + \hat{\Psi}_{T,t_1,t_2})^{-1}S_{t_1}(1)S_{t_1}(1)'(zI + \hat{\Psi}_{T,t_1,t_2})^{-1}S_{t_2}(1)S_{t_2}(2)']\beta(2) \\
&\approx (1+\xi)^{-2}T^{-2}\beta(2)'E[S_{t_1}(2)S_{t_1}(1)'(zI + \hat{\Psi}_{T,t_1,t_2})^{-1}S_{t_2}(1)S_{t_2}(1)'(zI + \hat{\Psi}_{T,t_1,t_2})^{-1} \\
&\quad \Psi(1)(zI + \hat{\Psi}_{T,t_1,t_2})^{-1}S_{t_1}(1)S_{t_1}(1)'(zI + \hat{\Psi}_{T,t_1,t_2})^{-1}S_{t_2}(1)S_{t_2}(2)']\beta(2)
\end{aligned} \tag{100}$$

We can now use formula (90) to integrate over S_{t_1} to get

$$\begin{aligned}
Term3 &= (1+\xi)^{-2}T^{-2}E\left[\left(\beta(2)'\Psi(2,1)\left((zI + \hat{\Psi}_{T,t_1,t_2})^{-1}S_{t_2}(1)S_{t_2}(2)'\beta(2)\right)\right)\right. \\
&\quad \times \text{tr}\left((zI + \hat{\Psi}_{T,t_1,t_2})^{-1}S_{t_2}(1)S_{t_2}(1)'(zI + \hat{\Psi}_{T,t_1,t_2})^{-1}\right. \\
&\quad \left.\left.\Psi(1)(zI + \hat{\Psi}_{T,t_1,t_2})^{-1}\Psi(1)\right)\right] + o(1) \\
&= (1+\xi)^{-2}T^{-2}E\left[\left(\beta(2)'\Psi(2,1)\left((zI + \hat{\Psi}_{T,t_1,t_2})^{-1}S_{t_2}(1)S_{t_2}(2)'\beta(2)\right)\right)\right. \\
&\quad \times \left(S_{t_2}(1)'(zI + \hat{\Psi}_{T,t_1,t_2})^{-1}\right. \\
&\quad \left.\left.\Psi(1)(zI + \hat{\Psi}_{T,t_1,t_2})^{-1}\Psi(1)(zI + \hat{\Psi}_{T,t_1,t_2})^{-1}S_{t_2}(1)\right)\right]
\end{aligned} \tag{101}$$

where the fact that the other terms from (90) are negligible follows by a direct but tedious calculation.

Using (90) once again to integrate over S_{t_2} , we get that

$$\begin{aligned}
Term3 &= (1+\xi)^{-2}T^{-2}E\left[\left(\beta(2)'\Psi(2,1)\left((zI + \hat{\Psi}_{T,t_1,t_2})^{-1}\Psi(1,2)\beta(2)\right)\right)\right. \\
&\quad \times \text{tr}\left((zI + \hat{\Psi}_{T,t_1,t_2})^{-1}\right. \\
&\quad \left.\left.\Psi(1)(zI + \hat{\Psi}_{T,t_1,t_2})^{-1}\Psi(1)(zI + \hat{\Psi}_{T,t_1,t_2})^{-1}\Psi(1)\right)\right]
\end{aligned} \tag{102}$$

Since Ψ and β are bounded, we get that $Term3 = O(T^{-1})$ and is therefore negligible.

Thus,

$$\begin{aligned}
\mathcal{T}_{2,3,2} &\approx (1 + \xi)^2 \left(Term1 - 2Term2 + Term3 \right) \\
&= (1 + \xi)^{-2} \left(z^{-2} Z'_*(z; c) Z_*(z; c)^2 \beta(2)' \Psi(2, 1) \Psi(1) (\Psi(1) + Z_* I)^{-2} \Psi(1, 2) \beta(2) \right. \\
&\quad \left. - 2\beta(2)' \Psi(2, 1) (Z_* I + \Psi_{1,1})^{-1} \Psi(1, 2) \beta(2) (\xi + z\xi') (1 + \xi)^2 \right) \\
&= Z'_*(z; c) \beta(2)' \Psi(2, 1) \Psi(1) (\Psi(1) + Z_* I)^{-2} \Psi(1, 2) \beta(2) \\
&\quad - 2\beta(2)' \Psi(2, 1) (Z_* I + \Psi_{1,1})^{-1} \Psi(1, 2) \beta(2) (\xi + z\xi') \\
&= \beta(2)' \Psi(2, 1) \Psi(1) (\Psi(1) + Z_* I)^{-2} \Psi(1, 2) \beta(2) \\
&\quad - (\xi + z\xi') \beta(2)' \Psi(2, 1) (2Z_* I + \Psi(1)) (\Psi(1) + Z_* I)^{-2} \Psi(1, 2) \beta(2)
\end{aligned} \tag{103}$$

and, hence,

$$\begin{aligned}
\mathcal{T}_{2,3} &= \mathcal{T}_{2,3,1} + \mathcal{T}_{2,3,2} \\
&= \left(\beta(2)' \Psi(2, 2) \beta(2) - \beta(2)' \Psi(2, 1) (2Z_* I + \Psi(1)) (\Psi(1) + Z_* I)^{-2} \Psi(1, 2) \beta(2) \right) (\xi + z\xi') \\
&\quad + \beta(2)' \Psi(2, 1) \Psi(1) (\Psi(1) + Z_* I)^{-2} \Psi(1, 2) \beta(2)
\end{aligned} \tag{104}$$

and, hence,

$$\begin{aligned}
\mathcal{T}_2 &= \mathcal{T}_{2,1} + 2\mathcal{T}_{2,2} + \mathcal{T}_{2,3} \\
&= \beta(1)' \Psi(1)^3 (Z_* I + \Psi(1))^{-2} \beta(1) + (Z'_* - 1) Z_*^2 \beta(1)' (Z_* I + \Psi(1))^{-2} \Psi(1) \beta(1) \\
&\quad + 2 \left((\xi + z\xi') Z_*^2 \left(\beta(1)' (Z_* I + \Psi(1))^{-2} \Psi(1, 2) \beta(2) \right) + \beta(1)' \Psi(1)^2 (Z_* I + \Psi)^{-2} \Psi(1, 2) \beta(2) \right) \\
&\quad + \left(\beta(2)' \Psi(2, 2) \beta(2) - \beta(2)' \Psi(2, 1) (2Z_* I + \Psi(1)) (\Psi(1) + Z_* I)^{-2} \Psi(1, 2) \beta(2) \right) (\xi + z\xi') \\
&\quad + \beta(2)' \Psi(2, 1) \Psi(1) (\Psi(1) + Z_* I)^{-2} \Psi(1, 2) \beta(2)
\end{aligned} \tag{105}$$

and, hence,

$$\begin{aligned}
& E[\hat{\beta}(z)' \Psi(1) \hat{\beta}(z)] \\
&= \mathcal{T}_1 + \mathcal{T}_2 \\
&= \sigma^2(\xi + z\xi') + \beta(1)' \Psi(1)^3 (Z_* I + \Psi(1))^{-2} \beta(1) + (Z'_* - 1) Z_*^2 \beta(1)' (Z_* I + \Psi(1))^{-2} \Psi(1) \beta(1) \\
&+ 2 \left((\xi + z\xi') Z_*^2 \left(\beta(1)' (Z_* I + \Psi(1))^{-2} \Psi(1, 2) \beta(2) \right) + \beta(1)' \Psi(1)^2 (Z_* I + \Psi)^{-2} \Psi(1, 2) \beta(2) \right) \\
&+ \left(\beta(2)' \Psi(2, 2) \beta(2) - \beta(2)' \Psi(2, 1) (2Z_* I + \Psi(1)) (\Psi(1) + Z_* I)^{-2} \Psi(1, 2) \beta(2) \right) (\xi + z\xi') \\
&+ \beta(2)' \Psi(2, 1) \Psi(1) (\Psi(1) + Z_* I)^{-2} \Psi(1, 2) \beta(2) \\
&= (\xi + z\xi') \left(\sigma^2 + Z_*^2 \beta(1)' (Z_* I + \Psi(1))^{-2} \Psi(1) \beta(1) \right. \\
&+ 2 Z_*^2 \beta(1)' (Z_* I + \Psi(1))^{-2} \Psi(1, 2) \beta(2) \\
&+ \beta(2)' \Psi(2, 2) \beta(2) - \beta(2)' \Psi(2, 1) (2Z_* I + \Psi(1)) (\Psi(1) + Z_* I)^{-2} \Psi(1, 2) \beta(2) \left. \right) \\
&+ \beta(1)' \Psi(1)^3 (Z_* I + \Psi(1))^{-2} \beta(1) + 2 \beta(1)' \Psi(1)^2 (Z_* I + \Psi)^{-2} \Psi(1, 2) \beta(2) \\
&+ \beta(2)' \Psi(2, 1) \Psi(1) (\Psi(1) + Z_* I)^{-2} \Psi(1, 2) \beta(2)
\end{aligned} \tag{106}$$

Note that $\Psi(2, 2) \geq \Psi(2, 1) \Psi(1, 1)^{-1} \Psi(1, 2)$, which ensures that the sum of respective terms is positive.

Finally,

$$\begin{aligned}
& \beta(1)' \Psi(1)^3 (Z_* I + \Psi(1))^{-2} \beta(1) + 2 \beta(1)' \Psi(1)^2 (Z_* I + \Psi)^{-2} \Psi(1, 2) \beta(2) \\
&+ \beta(2)' \Psi(2, 1) \Psi(1) (\Psi(1) + Z_* I)^{-2} \Psi(1, 2) \beta(2) \\
&= \tilde{\beta}' \Psi(1) (Z_* I + \Psi(1))^{-2} \tilde{\beta}
\end{aligned} \tag{107}$$

whereas

$$\begin{aligned}
& \sigma^2 + Z_*^2 \beta(1)'(Z_* I + \Psi(1))^{-2} \Psi(1) \beta(1) \\
& + 2Z_*^2 \beta(1)'(Z_* I + \Psi(1))^{-2} \Psi(1, 2) \beta(2) \\
& + \beta(2)' \Psi(2, 2) \beta(2) - \beta(2)' \Psi(2, 1) (2Z_* I + \Psi(1)) (\Psi(1) + Z_* I)^{-2} \Psi(1, 2) \beta(2) \\
& + \beta(1)' \Psi(1)^3 (Z_* I + \Psi(1))^{-2} \beta(1) + 2\beta(1)' \Psi(1)^2 (Z_* I + \Psi)^{-2} \Psi(1, 2) \beta(2) \\
& + \beta(2)' \Psi(2, 1) \Psi(1) (\Psi(1) + Z_* I)^{-2} \Psi(1, 2) \beta(2) \\
& = \sigma^2 + \beta(2)' (\Psi(2, 2) - \Psi(2, 1) \Psi(1)^{-1} \Psi(1, 2)) \beta(2) \\
& + \tilde{\beta}' \Psi(1)^{-1} (Z_* I + \Psi(1))^{-2} \tilde{\beta}
\end{aligned} \tag{108}$$

$$\text{C.5} \quad E \left[\left(\tilde{\beta}' \hat{\beta}(z; P_1) \right)^2 \right]$$

The goal of this section is to show that

$$\begin{aligned}
\tilde{\beta}' \hat{\beta}(z; P_1) &= \tilde{\beta}' (zI + \hat{\Psi}(1))^{-1} (\hat{\Psi}(1) \beta(1) + \hat{\Psi}(1, 2) \beta(2) + T^{-1} S(1)' \varepsilon) \\
&\approx \tilde{\beta}' (Z_* I + \Psi_{1,1})^{-1} \tilde{\beta}.
\end{aligned} \tag{109}$$

We know that this holds in expectation, so it is enough to show that the variance vanishes. We have

$$\begin{aligned}
& E \left[\left(\tilde{\beta}' \hat{\beta}(z; P_1) \right)^2 \right] \\
&= T^{-2} E[\tilde{\beta}' (zI + \hat{\Psi}(1))^{-1} S(1)' (S(1) \beta(1) + S(2) \beta(2) + \varepsilon) \\
&\times ((S(1) \beta(1) + S(2) \beta(2) + \varepsilon)' S(1) (zI + \hat{\Psi}(1))^{-1}) \tilde{\beta}] \\
&= E[\tilde{\beta}' (zI + \hat{\Psi}(1))^{-1} (\hat{\Psi}(1) \beta(1) + \hat{\Psi}(2) \beta(2)) (\hat{\Psi}(1) \beta(1) + \hat{\Psi}(2) \beta(2))' (zI + \hat{\Psi}(1))^{-1}] \tilde{\beta} \\
&+ \sigma^2 T^{-1} E[\tilde{\beta}' (zI + \hat{\Psi}(1))^{-2} \hat{\Psi}(1) \tilde{\beta}] \\
&\approx E[\tilde{\beta}' (zI + \hat{\Psi}(1))^{-1} (\hat{\Psi}(1) \beta(1) + \hat{\Psi}(1, 2) \beta(2)) (\hat{\Psi}(1) \beta(1) + \hat{\Psi}(1, 2) \beta(2))' (zI + \hat{\Psi}(1))^{-1}] \tilde{\beta}.
\end{aligned} \tag{110}$$

We know from (57) that

$$\begin{aligned}
& \tilde{\beta}' (zI + \hat{\Psi}(1))^{-1} \hat{\Psi}(1) \beta(1) \\
&= \tilde{\beta}' \beta(1) - z \tilde{\beta}' (zI + \hat{\Psi}(1))^{-1} \beta(1) \\
&\approx \tilde{\beta}' \beta(1) - Z_* \tilde{\beta}' (Z_* I + \Psi(1))^{-1} \beta(1) \\
&= \tilde{\beta}' (Z_* I + \Psi(1))^{-1} \Psi(1) \beta(1).
\end{aligned} \tag{111}$$

The only term that requires special care is

$$E[\tilde{\beta}'(zI + \hat{\Psi}(1))^{-1}\hat{\Psi}(1,2)\beta(2)\beta(2)'\hat{\Psi}(1,2)'(zI + \hat{\Psi}(1))^{-1}\tilde{\beta}]. \quad (112)$$

Our goal is to show that, in fact, $\tilde{\beta}'(zI + \hat{\Psi}(1))^{-1}\hat{\Psi}(1,2)\beta(2)$ has a deterministic equivalent and becomes approximately non-random in high dimensions. To this end, we proceed as follows

$$\begin{aligned} & E[\tilde{\beta}'(zI + \hat{\Psi}(1))^{-1}\hat{\Psi}(1,2)\beta(2)\beta(2)'\hat{\Psi}(1,2)'(zI + \hat{\Psi}(1))^{-1}\tilde{\beta}] \\ &= T^{-2} \sum_{t_1, t_2} E[\tilde{\beta}'(zI + \hat{\Psi}(1))^{-1}S_{t_1}(1)S_{t_1}(2)'\beta(2)\beta(2)'S_{t_2}(2)S_{t_2}(1)'(zI + \hat{\Psi}(1))^{-1}\tilde{\beta}] \\ &= T^{-2} \sum_{t_1, t_2} E[\tilde{\beta}'(zI + \hat{\Psi}(1))^{-1}S_{t_1}(1)S_{t_1}(2)'\beta(2)\beta(2)'S_{t_2}(2)S_{t_2}(1)'(zI + \hat{\Psi}(1))^{-1}\tilde{\beta}] \\ &\approx (1 + \xi)^{-2}T^{-2} \sum_{t_1, t_2} E[\tilde{\beta}'(zI + \hat{\Psi}_{T,t_1})^{-1}S_{t_1}(1)S_{t_1}(2)'\beta(2)\beta(2)'S_{t_2}(2)S_{t_2}(1)'(zI + \hat{\Psi}_{T,t_2})^{-1}\tilde{\beta}] \\ &= (1 + \xi)^{-2}T^{-1}E[\tilde{\beta}'(zI + \hat{\Psi}_{T,t})^{-1}S_t(1)S_t(2)'\beta(2)\beta(2)'S_t(2)S_t(1)'(zI + \hat{\Psi}_{T,t})^{-1}\tilde{\beta}] \\ &+ T^{-2}T(T-1)E[\tilde{\beta}'(zI + \hat{\Psi}_{T,t_1})^{-1}S_{t_1}(1)S_{t_1}(2)'\beta(2)\beta(2)'S_{t_2}(2)S_{t_2}(1)'(zI + \hat{\Psi}_{T,t_2})^{-1}\tilde{\beta}] \\ &\approx (1 + \xi)^{-2}T^{-1}E[\tilde{\beta}'(zI + \hat{\Psi}_{T,t})^{-1}S_t(1)S_t(2)'\beta(2)\beta(2)'S_t(2)S_t(1)'(zI + \hat{\Psi}_{T,t})^{-1}\tilde{\beta}] \\ &+ E[\tilde{\beta}'(zI + \hat{\Psi}_{T,t_1})^{-1}S_{t_1}(1)S_{t_1}(2)'\beta(2)\beta(2)'S_{t_2}(2)S_{t_2}(1)'(zI + \hat{\Psi}_{T,t_2})^{-1}\tilde{\beta}] \end{aligned} \quad (113)$$

Then, the same argument as above implies that the first term is negligible, while

$$\begin{aligned} & E[\tilde{\beta}'(zI + \hat{\Psi}_{T,t_1})^{-1}S_{t_1}(1)S_{t_1}(2)'\beta(2)\beta(2)'S_{t_2}(2)S_{t_2}(1)'(zI + \hat{\Psi}_{T,t_2})^{-1}\tilde{\beta}] \\ &= E[S_{t_1}(2)'\beta(2)\tilde{\beta}'(zI + \hat{\Psi}_{T,t_1})^{-1}S_{t_1}(1)S_{t_2}(1)'(zI + \hat{\Psi}_{T,t_2})^{-1}\tilde{\beta}\beta(2)'S_{t_2}(2)] \end{aligned} \quad (114)$$

Now, standard concentration of quadratic forms implies that

$$S_{t_1}(2)'\beta(2)\tilde{\beta}'(zI + \hat{\Psi}_{T,t_1})^{-1}S_{t_1}(1) \approx \text{tr}(\beta(2)\tilde{\beta}'(zI + \hat{\Psi}_{T,t_1})^{-1}\Psi(1,2)) = \tilde{\beta}'(zI + \hat{\Psi}_{T,t_1})^{-1}\tilde{\beta}(2) \quad (115)$$

and hence

$$E\left[\left(\tilde{\beta}'\hat{\beta}(z; P_1)\right)^2\right] \approx (\tilde{\beta}'(Z_*I + \Psi(1))^{-1}\tilde{\beta})^2. \quad (116)$$

The proof of Theorem 3 is complete.

C.6 Monotonicity of Z_*

We know from [Chernov et al. \(2025\)](#) that $Z_* = 1/\tilde{m}$ with $\tilde{m}$ being the unique solution to

$$c - 1 + z\tilde{m} - c \int \frac{dH(x)}{(1 + \tilde{m}x)} = 0. \quad (117)$$

Differentiating this with respect to c , we get

$$\int \frac{\tilde{m}xdH(x)}{(1 + \tilde{m}x)} + z\tilde{m}_c + \tilde{m}_cc \int \frac{xdH(x)}{(1 + \tilde{m}x)^2} = 0, \quad (118)$$

so that

$$\tilde{m}_c = \frac{-\int \frac{\tilde{m}xdH(x)}{(1 + \tilde{m}x)}}{z + c \int \frac{xdH(x)}{(1 + \tilde{m}x)^2}}. \quad (119)$$

Thus, $\tilde{m}$ is decreasing in c and, hence, Z_* is increasing in c . Furthermore,

$$(d/dc)Z_*/Z_* = \frac{\int \frac{\tilde{m}xdH(x)}{(1 + \tilde{m}x)}}{z\tilde{m} + c \int \frac{\tilde{m}xdH(x)}{(1 + \tilde{m}x)^2}} \quad (120)$$

We have

$$\begin{aligned} & z\tilde{m} + c \int \frac{\tilde{m}xdH(x)}{(1 + \tilde{m}x)^2} - \int \frac{\tilde{m}xdH(x)}{(1 + \tilde{m}x)} \\ &= 1 - \int \frac{\tilde{m}xdH(x)}{(1 + \tilde{m}x)} - c \left(\int \frac{\tilde{m}xdH(x)}{(1 + \tilde{m}x)} - \int \frac{\tilde{m}xdH(x)}{(1 + \tilde{m}x)^2} \right) \\ &= 1 - \int \frac{\tilde{m}xdH(x)}{(1 + \tilde{m}x)} - c \int \frac{(\tilde{m}x)^2 dH(x)}{(1 + \tilde{m}x)^2} \\ &= \int \frac{(1 + \tilde{m}x - c(\tilde{m}x)^2) dH(x)}{(1 + \tilde{m}x)^2} \end{aligned} \quad (121)$$

Now, differentiating the same expression with respect to z , we get

$$\tilde{m} + z\tilde{m}' + c\tilde{m}' \int \frac{xdH(x)}{(1 + \tilde{m}x)^2} = 0, \quad (122)$$

so that

$$Z'_* = -\frac{\tilde{m}'}{\tilde{m}^2} = \tilde{m}^{-2} \frac{\tilde{m}}{z + c \int \frac{xdH(x)}{(1 + \tilde{m}x)^2}} = \frac{1}{\tilde{m}z + c \int \frac{\tilde{m}xdH(x)}{(1 + \tilde{m}x)^2}} \quad (123)$$

Rewriting (117) as

$$z\tilde{m} + c \int \frac{\tilde{m}xdH(x)}{(1+\tilde{m}x)} = 1, \quad (124)$$

we see that $Z'_* > 1$ and

$$\begin{aligned} Z'_* - 1 &= \frac{1}{\tilde{m}z + c \int \frac{\tilde{m}xdH(x)}{(1+\tilde{m}x)^2}} - 1 = \frac{1 - \tilde{m}z - c \int \frac{\tilde{m}xdH(x)}{(1+\tilde{m}x)^2}}{\tilde{m}z + c \int \frac{\tilde{m}xdH(x)}{(1+\tilde{m}x)^2}} \\ &= \frac{c \int \frac{\tilde{m}xdH(x)}{(1+\tilde{m}x)} - c \int \frac{\tilde{m}xdH(x)}{(1+\tilde{m}x)^2}}{\tilde{m}z + c \int \frac{\tilde{m}xdH(x)}{(1+\tilde{m}x)^2}} \\ &= \frac{c \int \frac{(\tilde{m}x)^2 dH(x)}{(1+\tilde{m}x)^2}}{\tilde{m}z + c \int \frac{\tilde{m}xdH(x)}{(1+\tilde{m}x)^2}} \\ &\leq \frac{c}{\tilde{m}z} = cz^{-1}Z_* \end{aligned} \quad (125)$$

Now,

$$\frac{d}{dc} Z'_* = - \frac{\tilde{m}_c z + \int \frac{\tilde{m}_c x dH(x)}{(1+\tilde{m}_c x)^2} + c \int \frac{\tilde{m}_c x dH(x)}{(1+\tilde{m}_c x)^2} - 2\tilde{m}_c c \int \frac{\tilde{m}_c x^2 dH(x)}{(1+\tilde{m}_c x)^3}}{(\tilde{m}z + c \int \frac{\tilde{m}xdH(x)}{(1+\tilde{m}x)^2})^2}. \quad (126)$$

The negative of the numerator is

$$\begin{aligned} &\int \frac{\tilde{m}xdH(x)}{(1+\tilde{m}x)^2} \\ &+ \tilde{m}_c \left(z + c \int \frac{xdH(x)}{(1+\tilde{m}x)^2} - 2c \int \frac{\tilde{m}x^2 dH(x)}{(1+\tilde{m}x)^3} \right) \\ &= \int \frac{\tilde{m}xdH(x)}{(1+\tilde{m}x)^2} \\ &- \frac{\int \frac{\tilde{m}xdH(x)}{(1+\tilde{m}x)}}{z + c \int \frac{xdH(x)}{(1+\tilde{m}x)^2}} \left(z + c \int \frac{(x - \tilde{m}x^2)dH(x)}{(1+\tilde{m}x)^3} \right) \end{aligned} \quad (127)$$

Our goal is to show that this is positive, which is equivalent to

$$\int \frac{xdH(x)}{(1+\tilde{m}x)^2} \left(z + c \int \frac{xdH(x)}{(1+\tilde{m}x)^2} \right) > \int \frac{xdH(x)}{(1+\tilde{m}x)} \left(z + c \int \frac{(x - \tilde{m}x^2)dH(x)}{(1+\tilde{m}x)^3} \right). \quad (128)$$

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